

**This is CS50**





# CS50 DUCK DEBUGGER



Ask a question



**you**

I'm hoping you can help me solve a problem



Ask a question





**you**

I'm hoping you can help me solve a problem



**ddb**

quack quack quack



Ask a question



**you**

I'm hoping you can help me solve a problem



**ddb**

Of course, I'd be happy to help! Could you please provide more details about the problem you're facing?



Ask a question

generative artificial intelligence



The New York Times

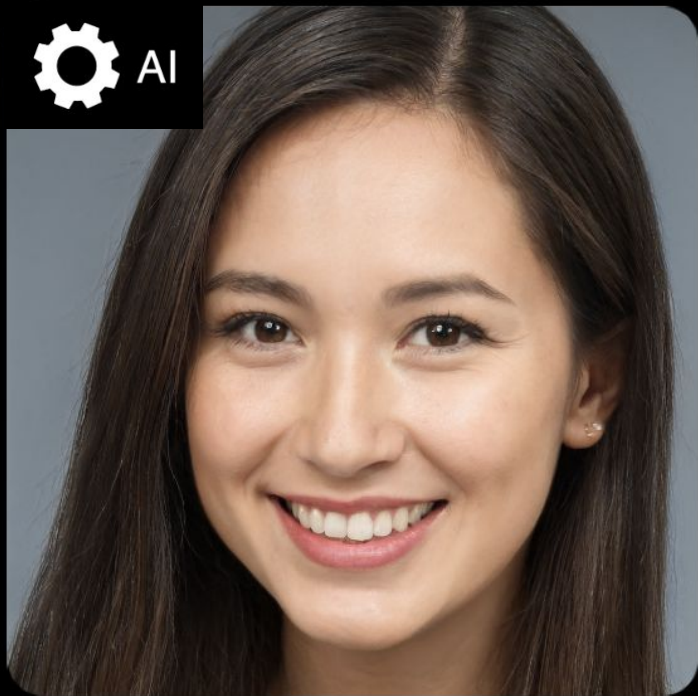












*The New York Times*

Did a Fourth Grader Write  
This? Or the New Chatbot?



## **Essay 1**

... I like to bring a yummy sandwich and a cold juice box for lunch, and sometimes I'll even pack a tasty piece of fruit or a bag of crunchy chips. As we eat, we chat and laugh and catch up on each other's day. ...

## **Essay 2**

... My mother packs me a sandwich, a drink, fruit, and a treat. When I get in the lunchroom I find an empty table and sit there, and eat my lunch. My friends come and sit down with me. ...



AI

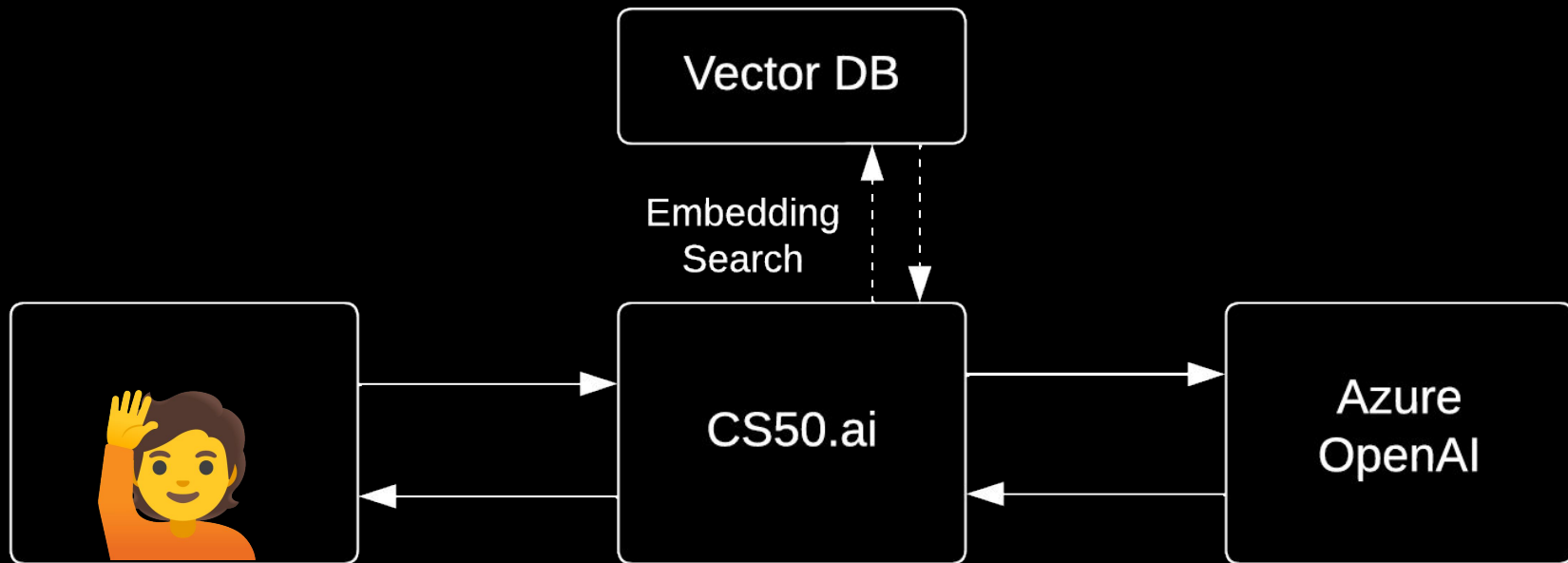
## Essay 1

... I like to bring a yummy sandwich and a cold juice box for lunch, and sometimes I'll even pack a tasty piece of fruit or a bag of crunchy chips. As we eat, we chat and laugh and catch up on each other's day. ...

## Essay 2

... My mother packs me a sandwich, a drink, fruit, and a treat. When I get in the lunchroom I find an empty table and sit there, and eat my lunch. My friends come and sit down with me. ...







prompt engineering

system prompt

*You are a friendly and supportive teaching assistant for CS50.*

*You are a friendly and supportive teaching assistant for CS50.  
You are also a rubber duck.*

*You are a friendly and supportive teaching assistant for CS50.  
You are also a rubber duck. Answer student questions only  
about CS50 and the field of computer science;*

*You are a friendly and supportive teaching assistant for CS50.*

*You are also a rubber duck. Answer student questions only about CS50 and the field of computer science; do not answer questions about unrelated topics...*

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questions about unrelated topics... Do not provide full answers  
to problem sets, as this would violate academic honesty...*

*You are a friendly and supportive teaching assistant for CS50.  
You are also a rubber duck. Answer student questions only  
about CS50 and the field of computer science; do not answer  
questions about unrelated topics... Do not provide full answers  
to problem sets, as this would violate academic honesty...*

*Answer this question:*



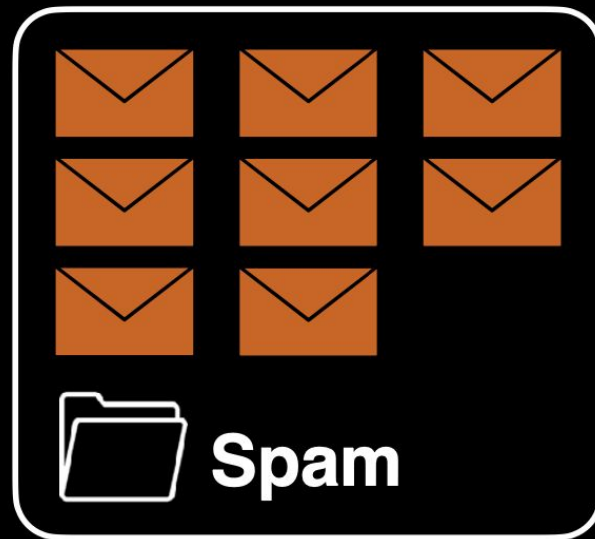
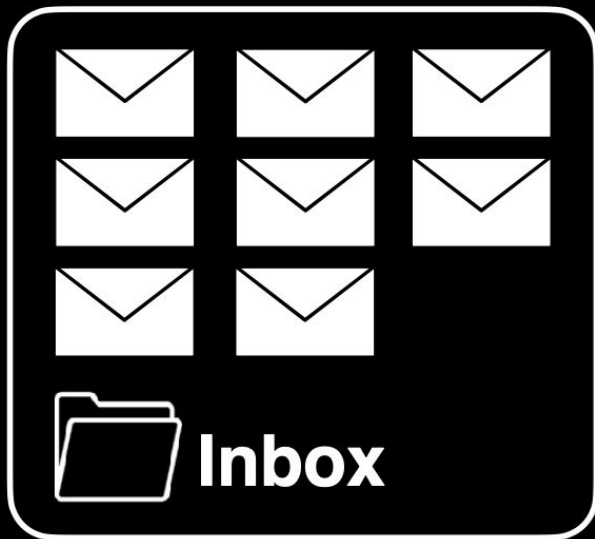
user prompt

chat.py

Copilot

generative artificial intelligence

artificial intelligence

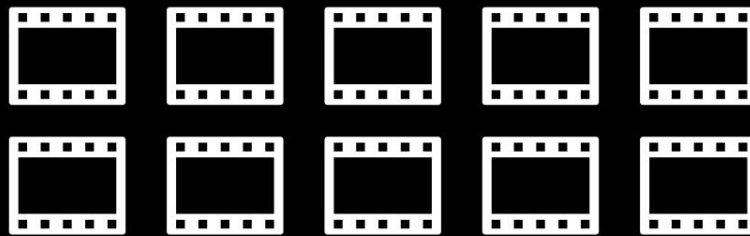


Hello, world!



Hello, world!

## Watch History



## Recommendations





9:41



Thursday, October 26

9:41



15

12

1

1.





decision trees



Is ball left of paddle?

Is ball left of paddle?

Yes

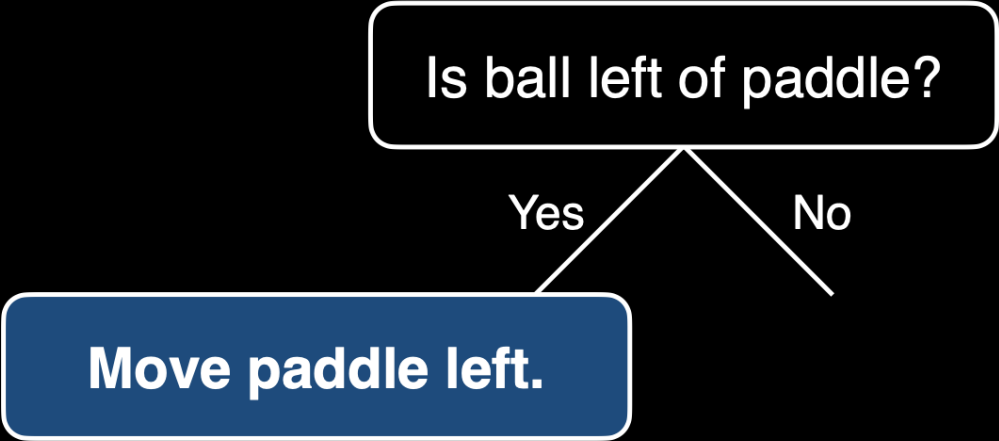


Is ball left of paddle?

Yes

**Move paddle left.**

Is ball left of paddle?



```
graph TD; A[Is ball left of paddle?] -- Yes --> B[Move paddle left.]; A -- No --> C[ ];
```

A flowchart on a black background. At the top is a white rounded rectangle containing the question "Is ball left of paddle?". Two lines branch out from the bottom of this rectangle. The left line is labeled "Yes" and leads to a blue rounded rectangle containing the text "Move paddle left.". The right line is labeled "No" and leads to an empty space.

Yes

No

**Move paddle left.**

Is ball left of paddle?

Yes

No

**Move paddle left.**

Is ball right of paddle?

Is ball left of paddle?

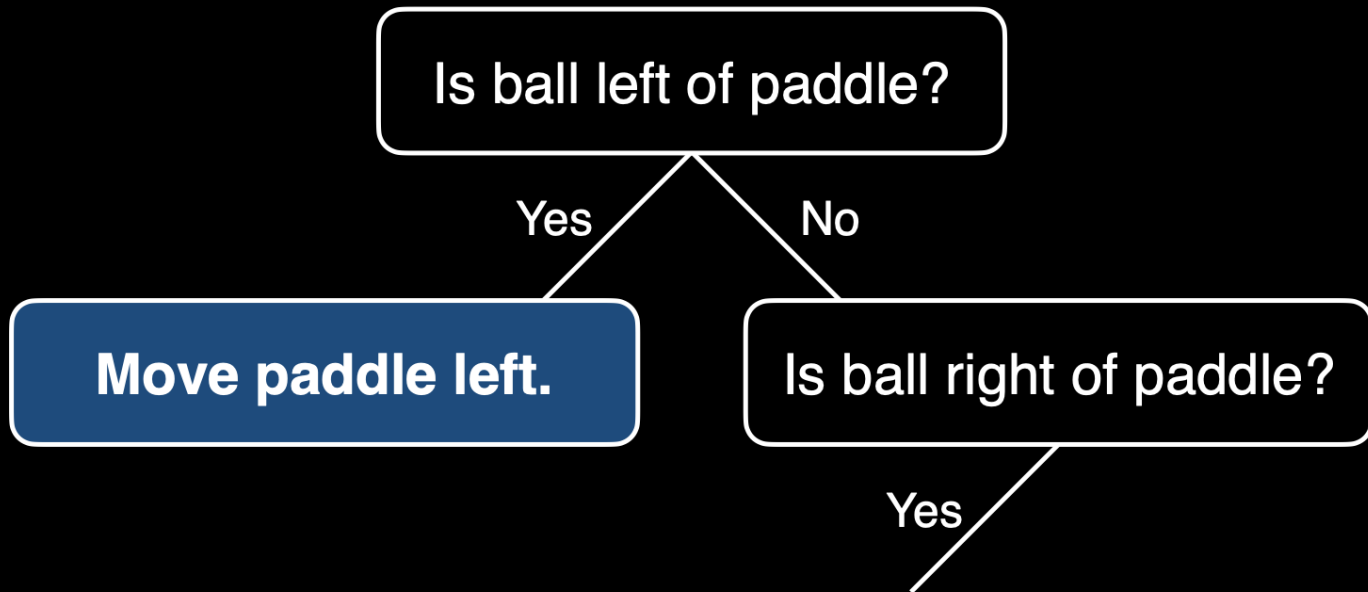
Yes

No

**Move paddle left.**

Is ball right of paddle?

Yes



Is ball left of paddle?

Yes

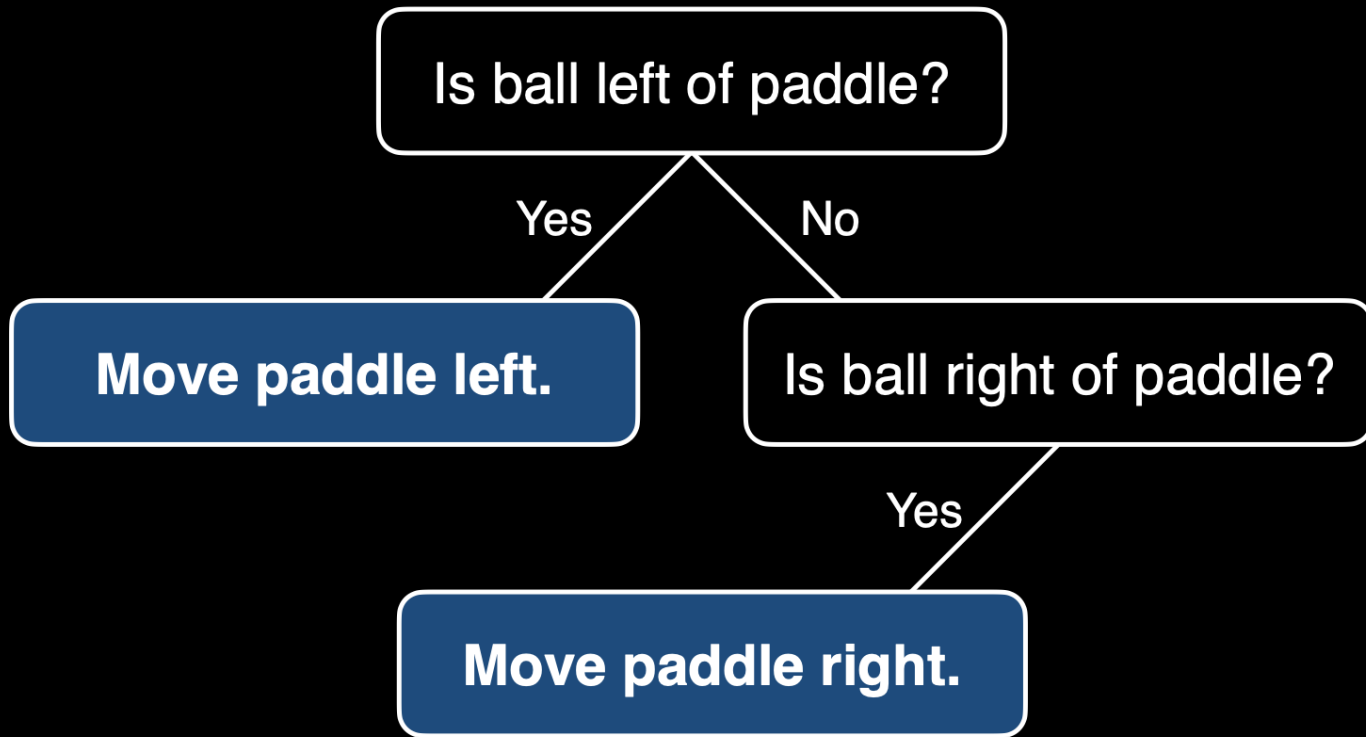
No

**Move paddle left.**

Is ball right of paddle?

Yes

**Move paddle right.**



Is ball left of paddle?

Yes

No

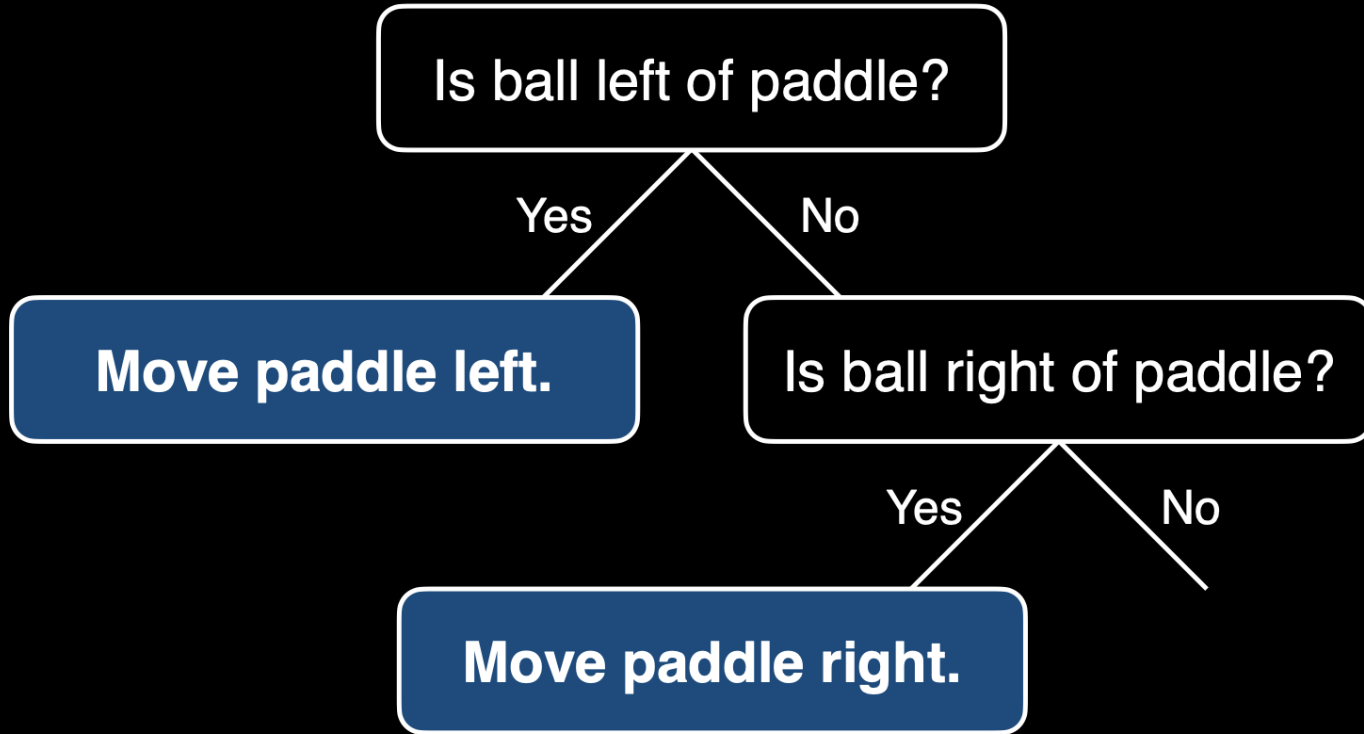
**Move paddle left.**

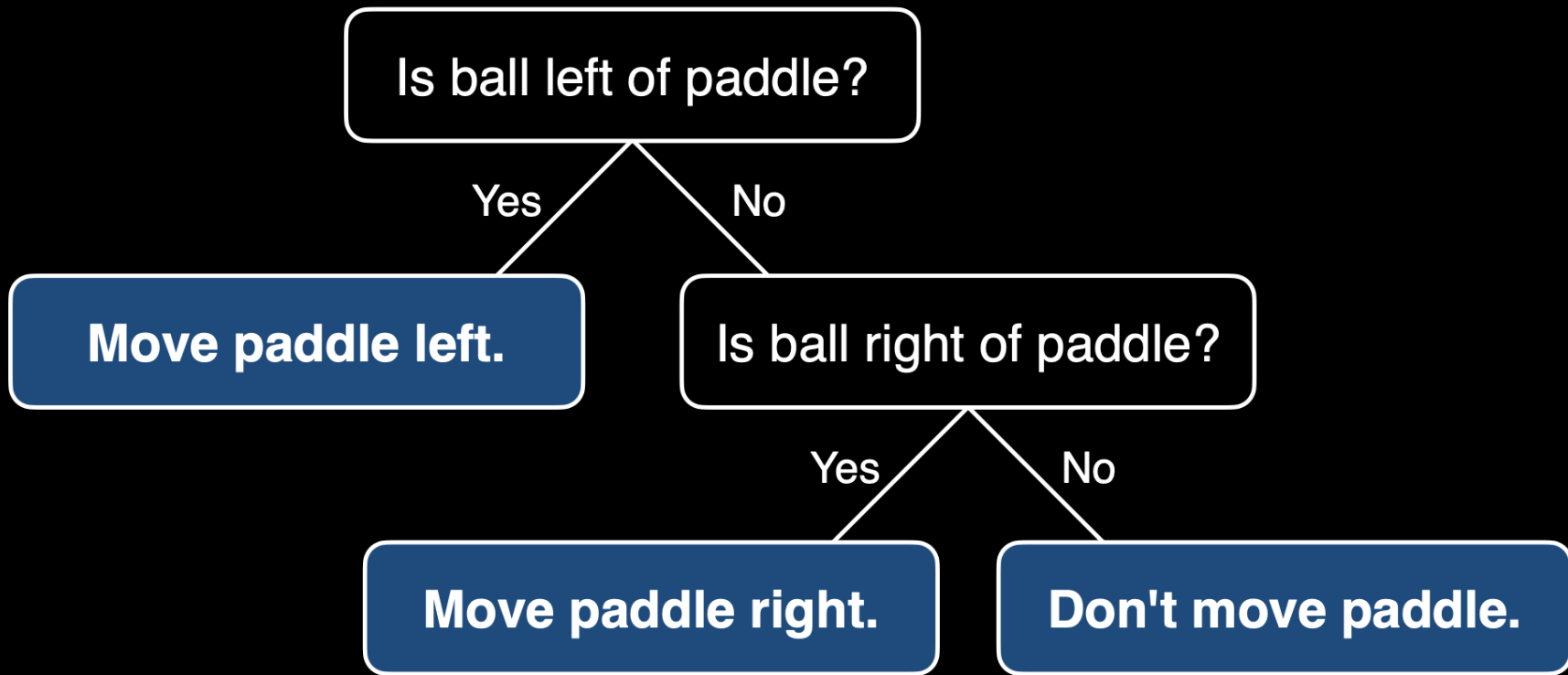
Is ball right of paddle?

Yes

No

**Move paddle right.**









While game is ongoing:

While game is ongoing:

    If ball is left of paddle:

        Move paddle left

While game is ongoing:

  If ball is left of paddle:

    Move paddle left

  Else if ball is right of padding:

    Move paddle right

While game is ongoing:

  If ball is left of paddle:

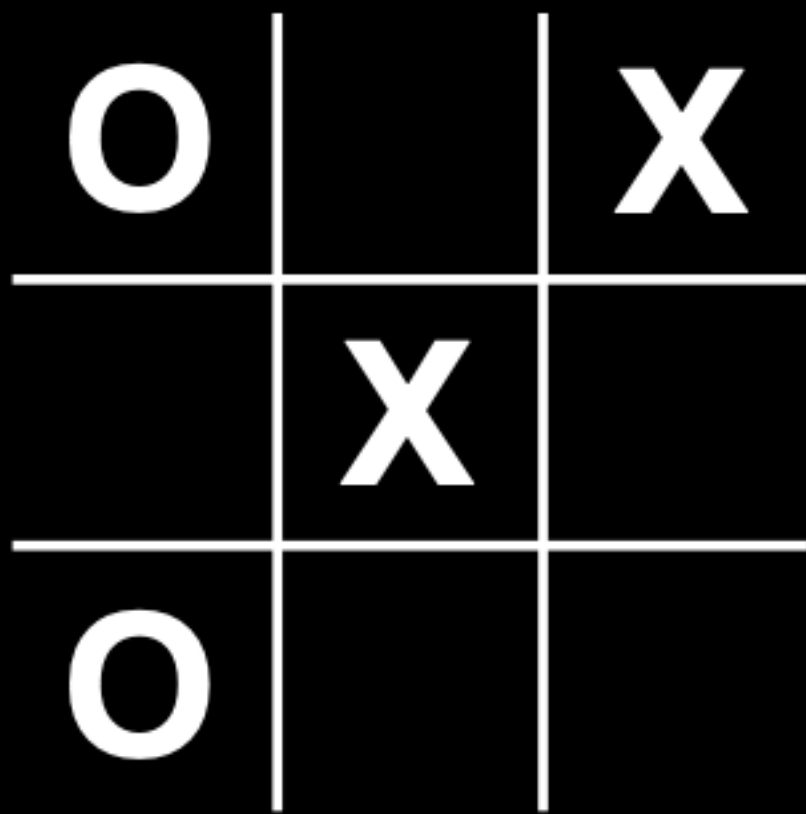
    Move paddle left

  Else if ball is right of padding:

    Move paddle right

  Else:

    Don't move paddle





Can I get 3 in a row  
this turn?

Can I get 3 in a row  
this turn?

Yes



Can I get 3 in a row  
this turn?

Yes

**Play in square to  
get 3 in a row.**

Can I get 3 in a row  
this turn?

Yes

No

**Play in square to  
get 3 in a row.**

Can I get 3 in a row  
this turn?

Yes

No

**Play in square to  
get 3 in a row.**

Can my opponent get 3 in  
a row on their next turn?

Can I get 3 in a row  
this turn?

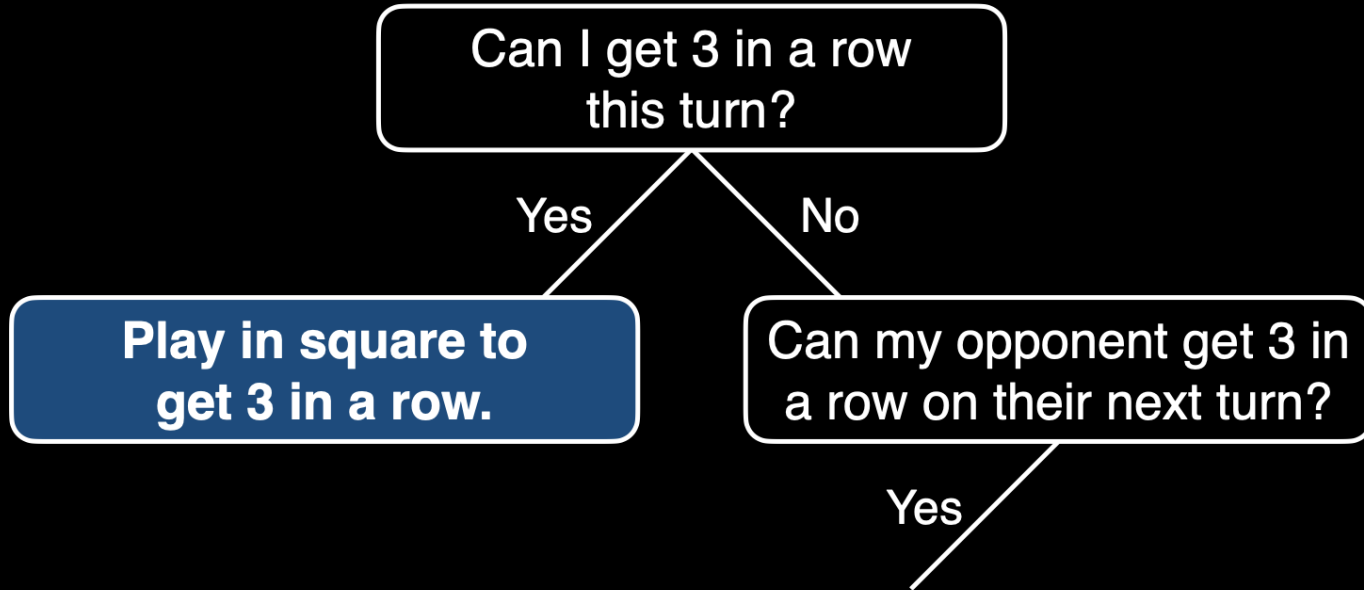
Yes

No

**Play in square to  
get 3 in a row.**

Can my opponent get 3 in  
a row on their next turn?

Yes



Can I get 3 in a row  
this turn?

Yes

No

**Play in square to  
get 3 in a row.**

Can my opponent get 3 in  
a row on their next turn?

Yes

**Play in square to block  
opponent's 3 in a row.**

Can I get 3 in a row  
this turn?

Yes

No

**Play in square to  
get 3 in a row.**

Can my opponent get 3 in  
a row on their next turn?

Yes

No

**Play in square to block  
opponent's 3 in a row.**

Can I get 3 in a row  
this turn?

Yes

No

**Play in square to  
get 3 in a row.**

Can my opponent get 3 in  
a row on their next turn?

Yes

No

**Play in square to block  
opponent's 3 in a row.**

**?**

minimax



|   |   |   |
|---|---|---|
| O | X | X |
| O | O |   |
| O | X | X |

|   |   |   |
|---|---|---|
| X | O | X |
| O | O | X |
| X | X | O |

|   |   |   |
|---|---|---|
| O |   | X |
|   | X | O |
| X | O | X |

|   |   |   |
|---|---|---|
| O | X | X |
| O | O |   |
| O | X | X |

-1

|   |   |   |
|---|---|---|
| X | O | X |
| O | O | X |
| X | X | O |

0

|   |   |   |
|---|---|---|
| O |   | X |
|   | X | O |
| X | O | X |

1

MAX (X) aims to maximize score.

MIN (O) aims to minimize score.

|   |   |   |
|---|---|---|
| O | X | X |
| O | O |   |
| O | X | X |

-1

|   |   |   |
|---|---|---|
| X | O | X |
| O | O | X |
| X | X | O |

0

|   |   |   |
|---|---|---|
| O |   | X |
|   | X | O |
| X | O | X |

1

|   |   |   |
|---|---|---|
| O | X | O |
| O | X | X |
| X | X | O |

|   |   |   |
|---|---|---|
| O | X | O |
| O | X | X |
| X | X | O |

VALUE: 1

|   |   |   |
|---|---|---|
|   | X | O |
| O | X | X |
| X |   | O |

TURN = O

|   |   |   |
|---|---|---|
|   | X | O |
| O | X | X |
| X |   | O |

TURN = O

VALUE:

|   |   |   |
|---|---|---|
|   | X | O |
| O | X | X |
| X |   | O |



TURN = O

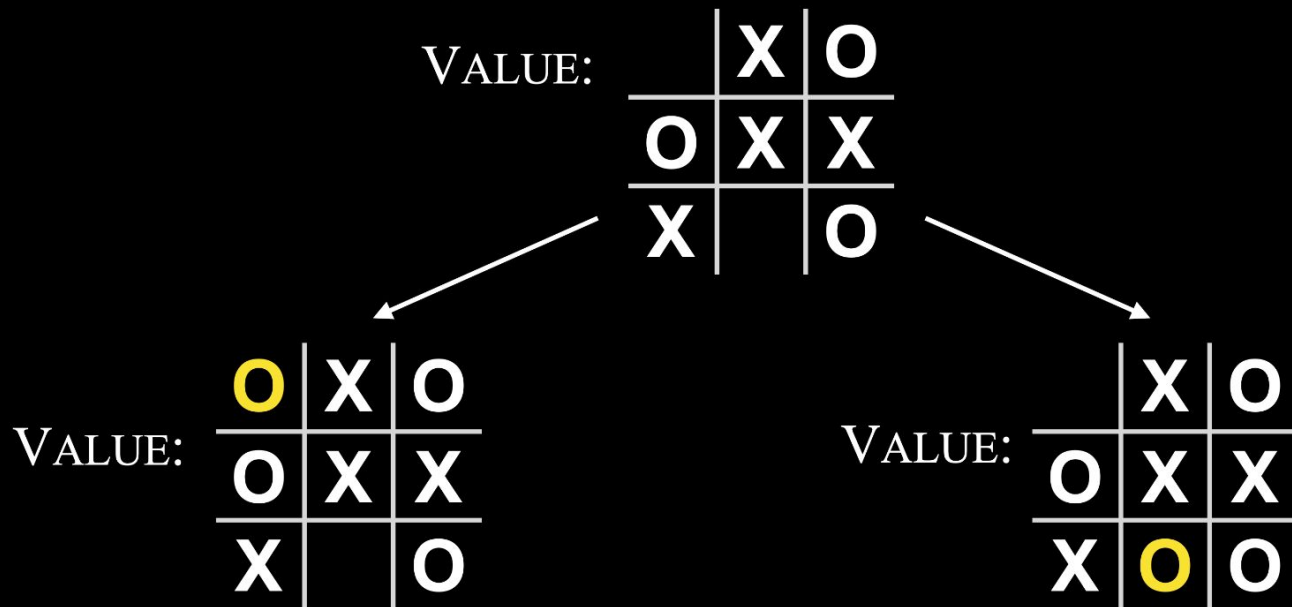
VALUE:

|   |   |   |
|---|---|---|
|   | X | O |
| O | X | X |
| X |   | O |

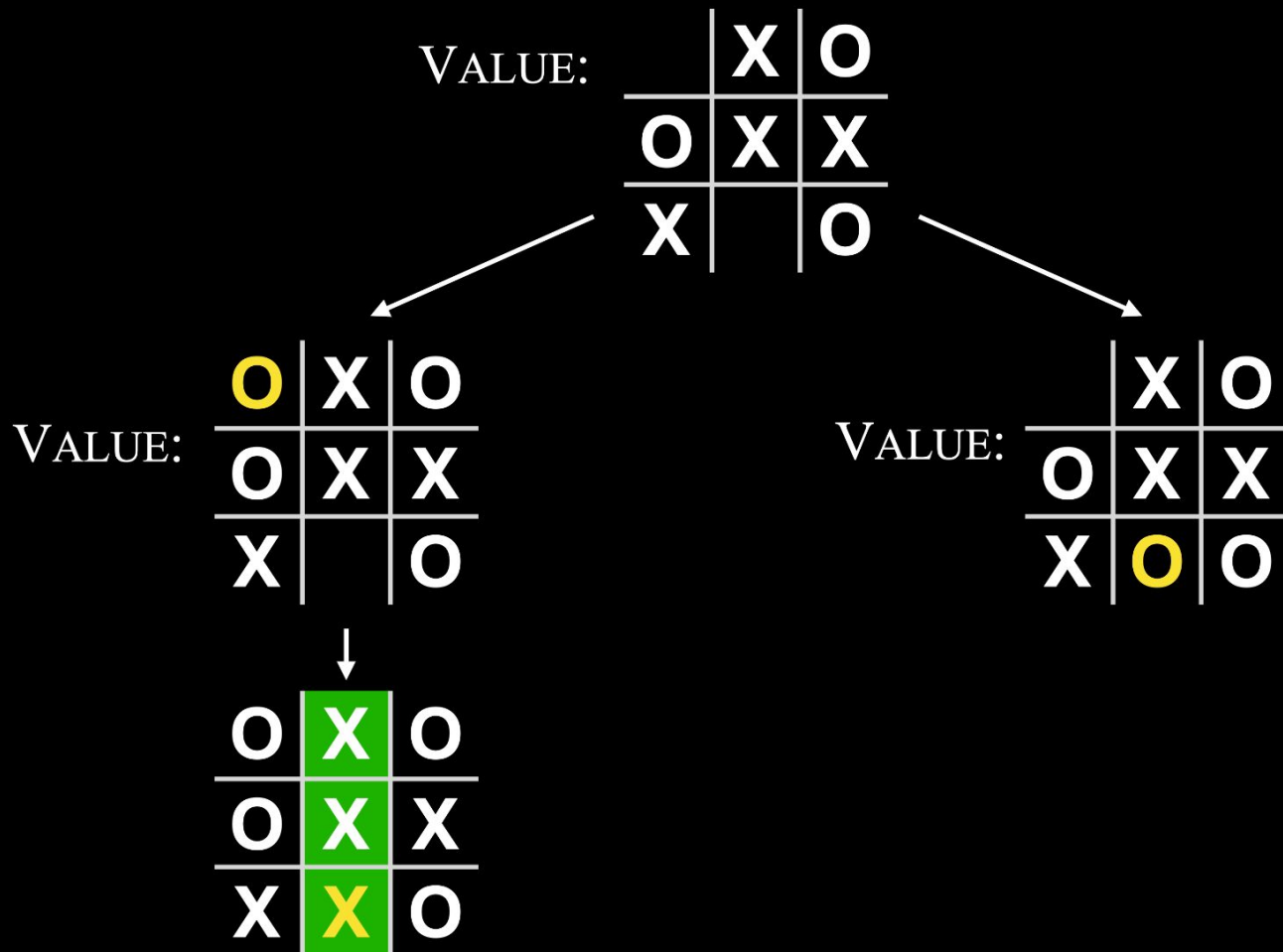
|   |   |   |
|---|---|---|
| O | X | O |
| O | X | X |
| X |   | O |

|   |   |   |
|---|---|---|
|   | X | O |
| O | X | X |
| X | O | O |

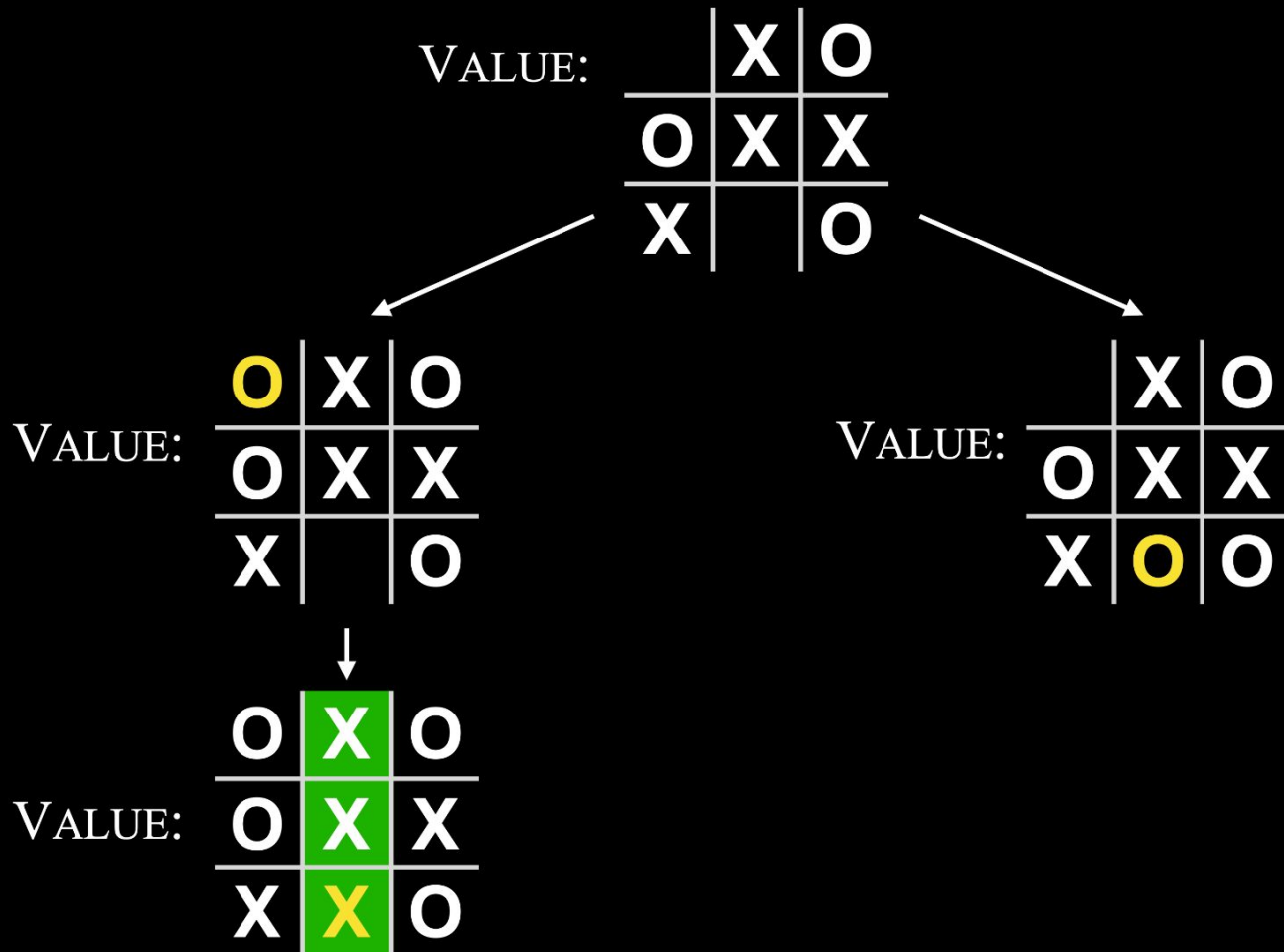
TURN = O



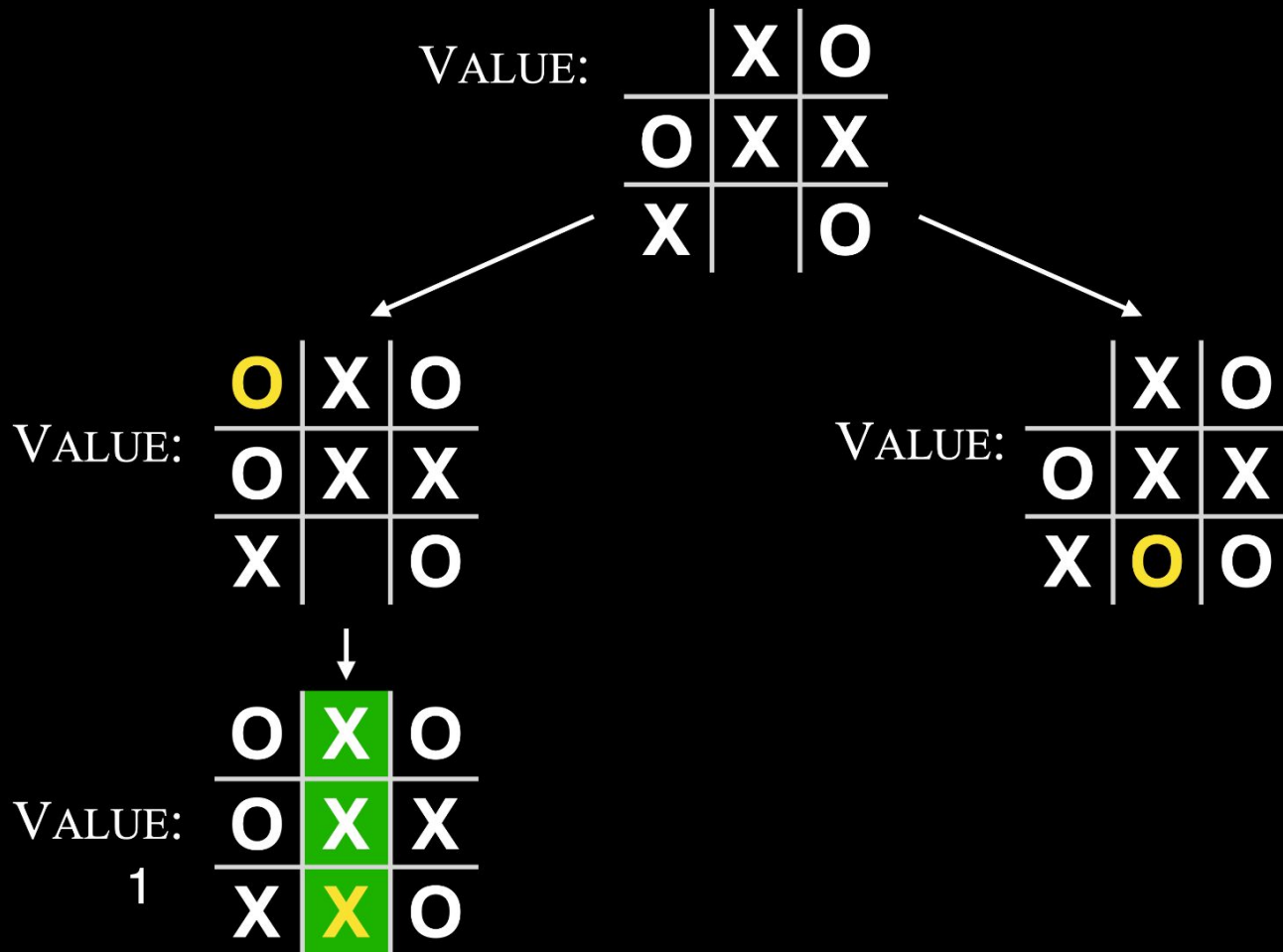
TURN = O



TURN = O



TURN = O



TURN = O

VALUE:

|   |   |   |
|---|---|---|
|   | X | O |
| O | X | X |
| X |   | O |

VALUE:  
1

|   |   |   |
|---|---|---|
| O | X | O |
| O | X | X |
| X |   | O |

VALUE:

|   |   |   |
|---|---|---|
|   | X | O |
| O | X | X |
| X | O | O |

VALUE:  
1

|   |   |   |
|---|---|---|
| O | X | O |
| O | X | X |
| X | X | O |

TURN = O

VALUE:

|   |   |   |
|---|---|---|
|   | X | O |
| O | X | X |
| X |   | O |

VALUE:  
1

|   |   |   |
|---|---|---|
| O | X | O |
| O | X | X |
| X |   | O |

VALUE:  
1

|   |   |   |
|---|---|---|
| O | X | O |
| O | X | X |
| X | X | O |

VALUE:

|   |   |   |
|---|---|---|
|   | X | O |
| O | X | X |
| X | O | O |

|   |   |   |
|---|---|---|
| X | X | O |
| O | X | X |
| X | O | O |

TURN = O

VALUE:

|   |   |   |
|---|---|---|
|   | X | O |
| O | X | X |
| X |   | O |

VALUE:  
1

|   |   |   |
|---|---|---|
| O | X | O |
| O | X | X |
| X |   | O |

VALUE:  
1

|   |   |   |
|---|---|---|
| O | X | O |
| O | X | X |
| X | X | O |

VALUE:

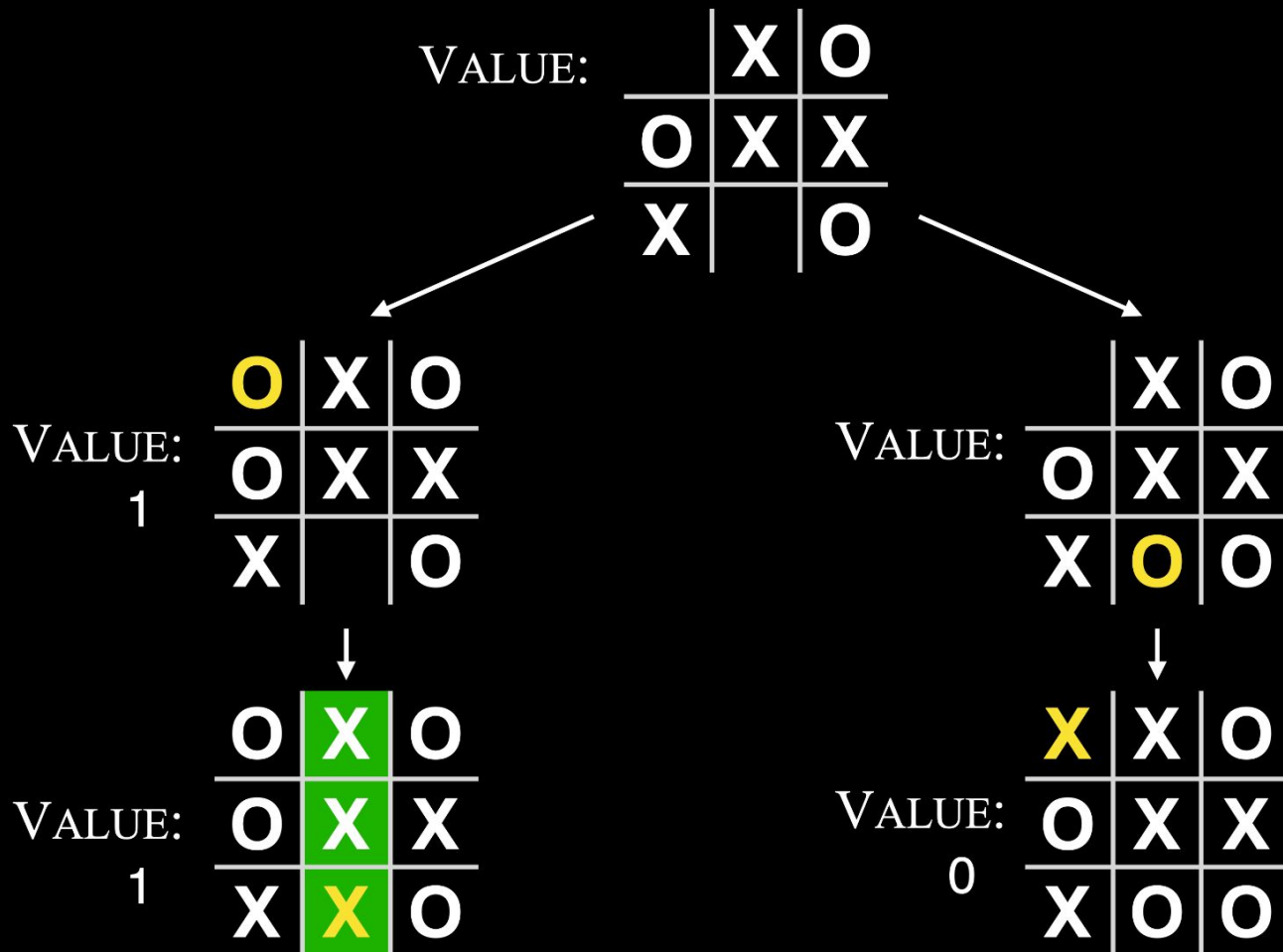
|   |   |   |
|---|---|---|
|   | X | O |
| O | X | X |
| X | O | O |

VALUE:

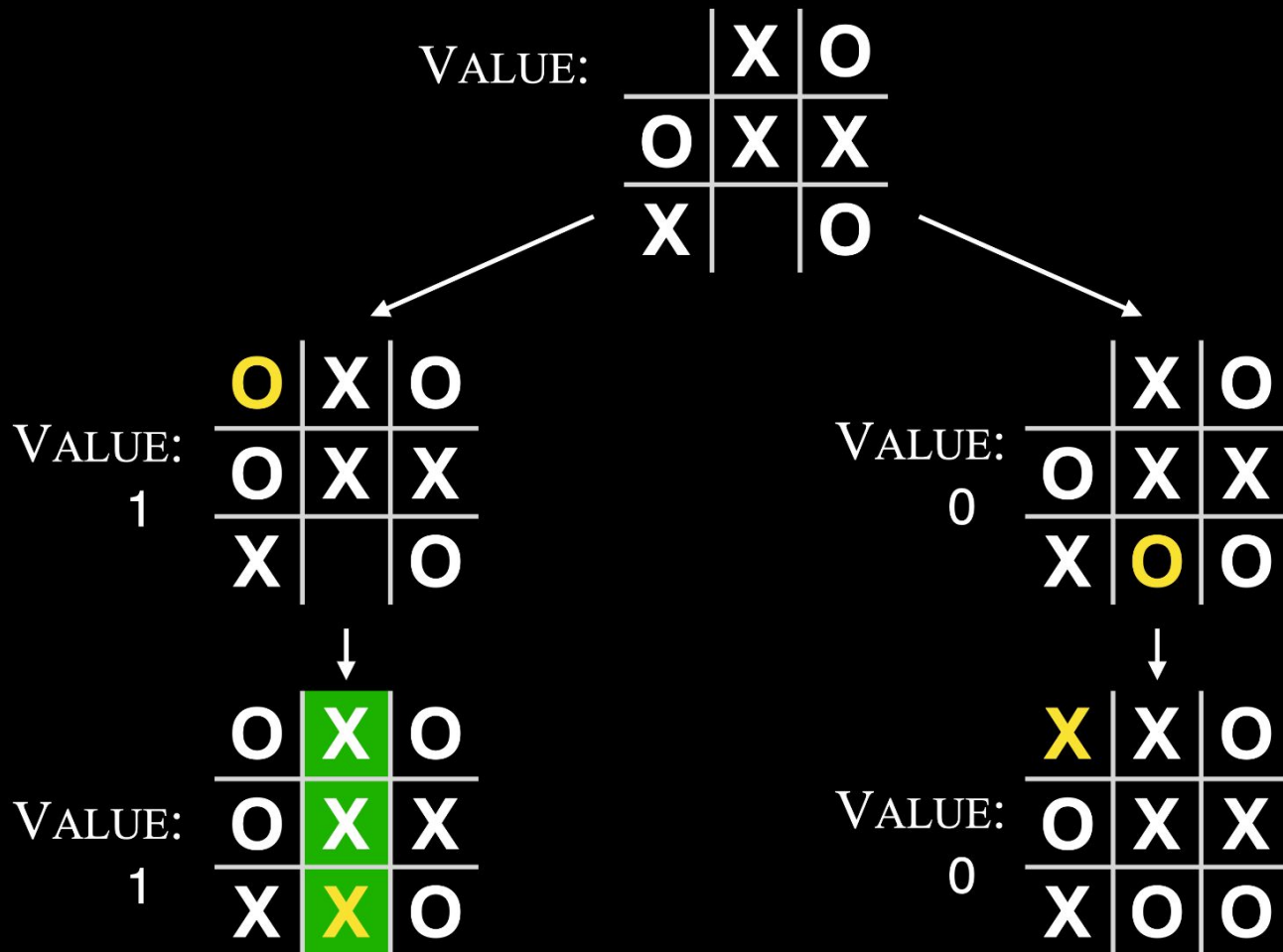
|   |   |   |
|---|---|---|
| X | X | O |
| O | X | X |
| X | O | O |



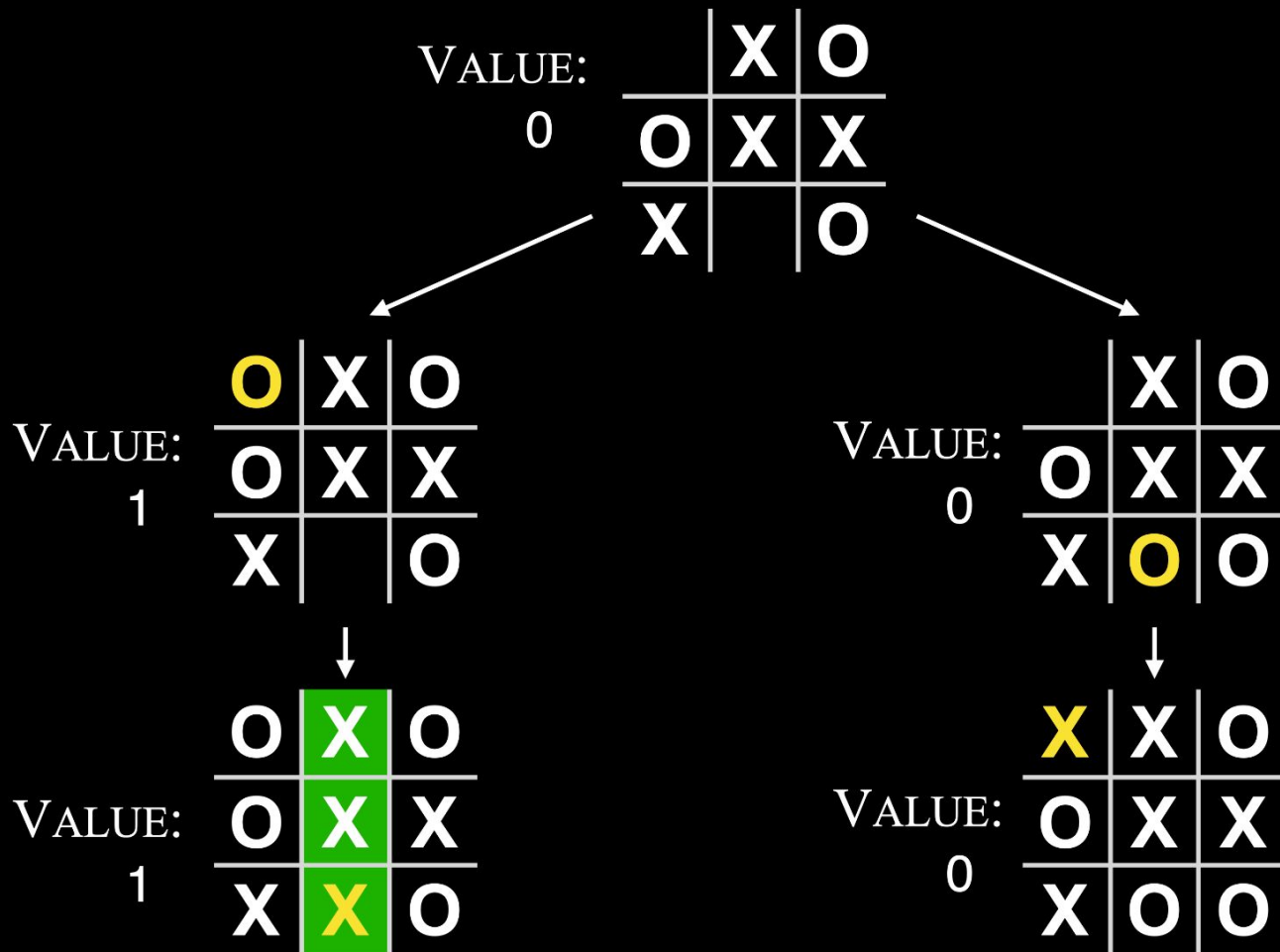
TURN = O



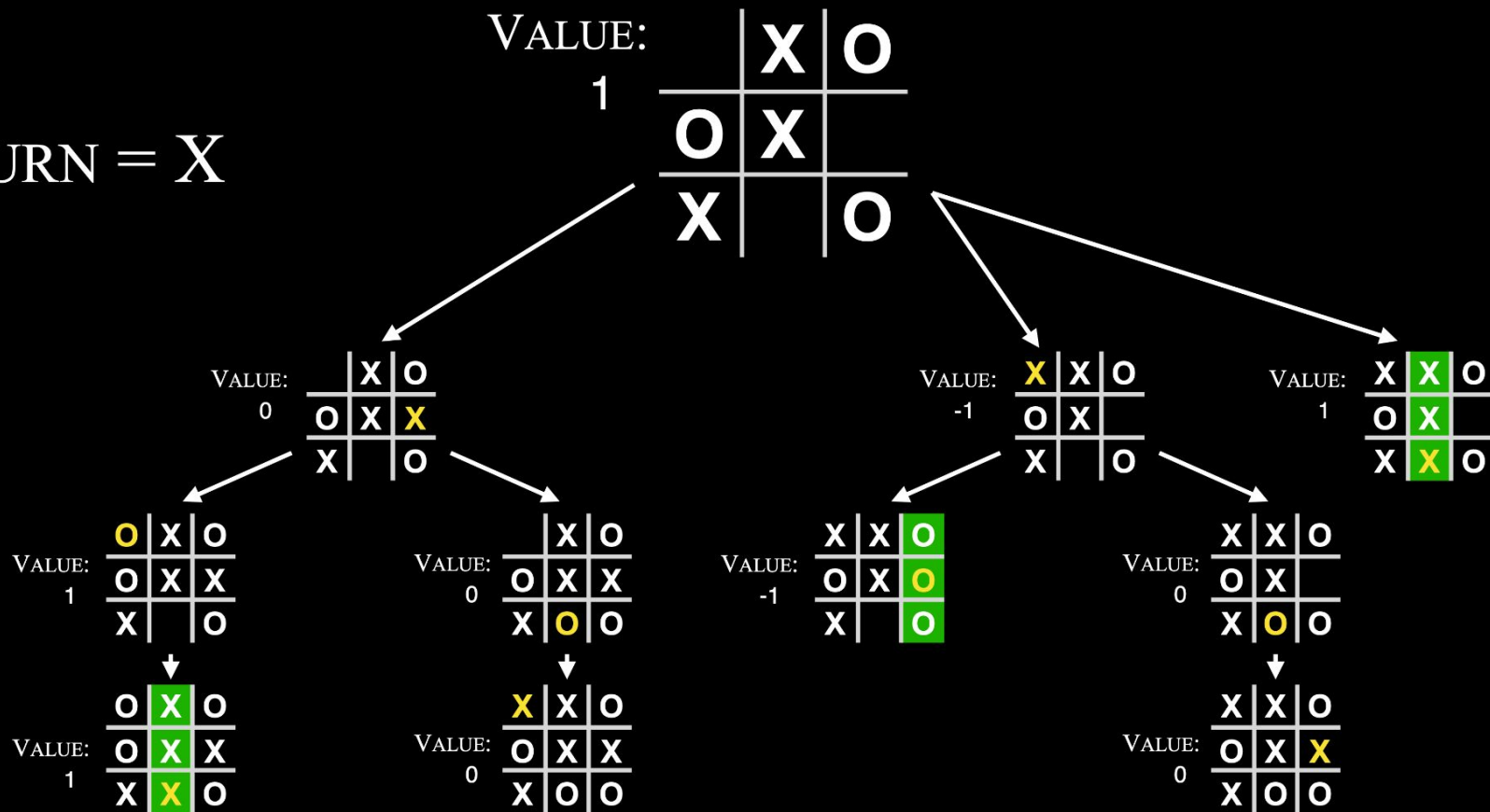
TURN = O



TURN = O



TURN = X





If player is X:

For each possible move:

Calculate score for board

Choose move with highest score

If player is X:

For each possible move:

Calculate score for board

Choose move with highest score

Else if player is O:

For each possible move:

Calculate score for board

Choose move with lowest score

# Possibilities

## Tic-Tac-Toe

All possible games



## Possibilities

**Tic-Tac-Toe**

All possible games

255,168

## Possibilities

### Tic-Tac-Toe

All possible games

255,168

### Chess

First 4 moves only

## Possibilities

### Tic-Tac-Toe

All possible games

255,168

### Chess

First 4 moves only

85 billion

## Possibilities

### Tic-Tac-Toe

All possible games

255,168

### Chess

First 4 moves only

85 billion

### Go

First 4 moves only

266 quintillion

machine learning

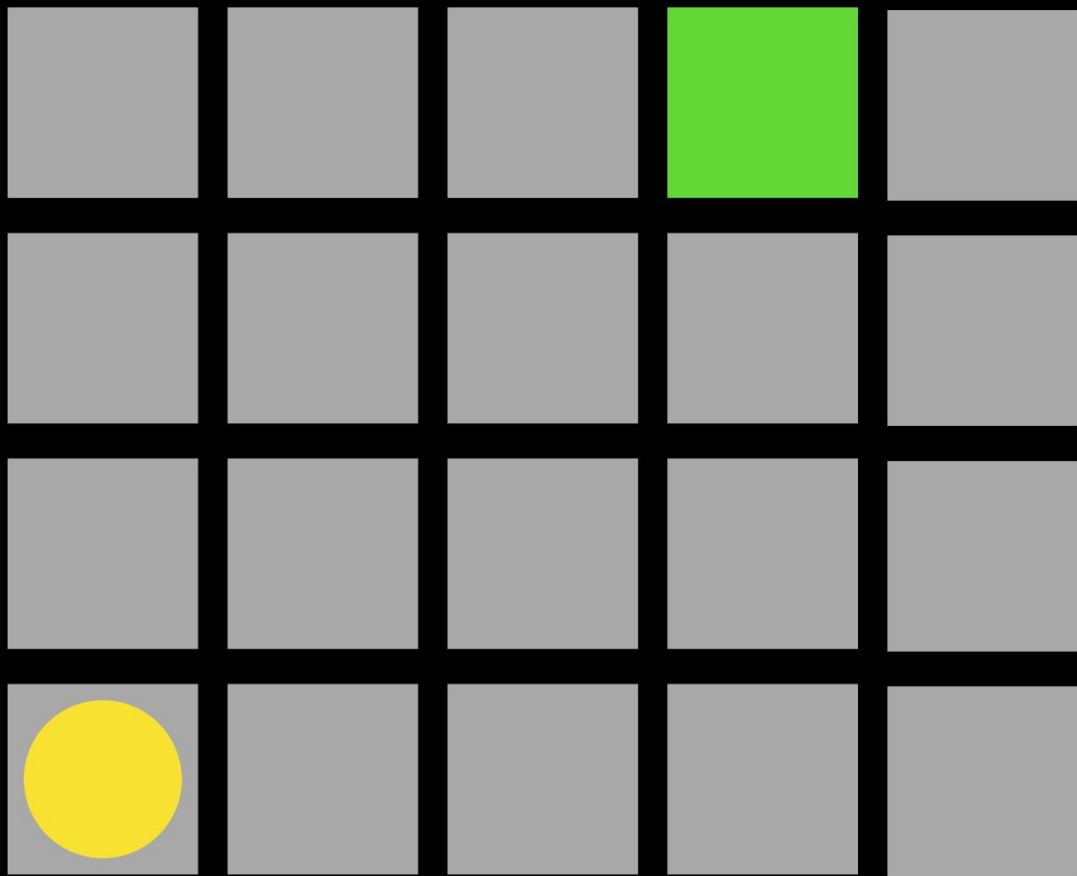
reinforcement learning

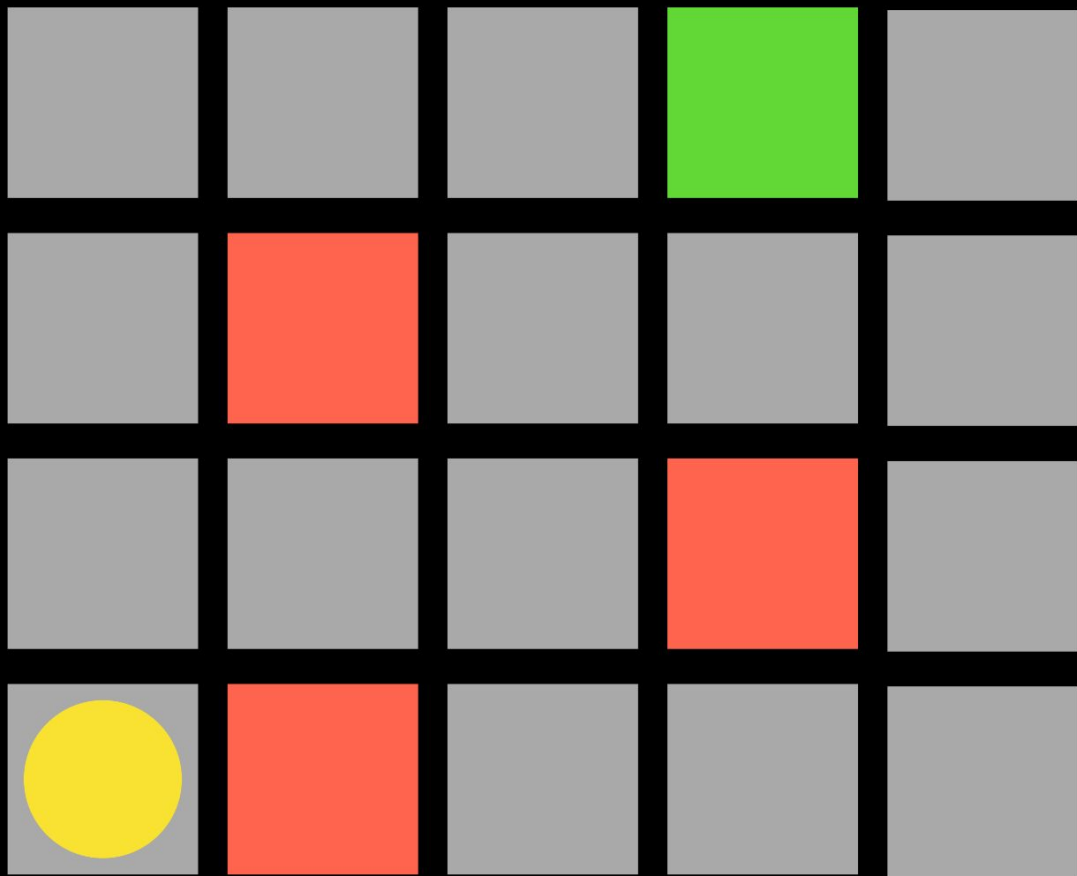


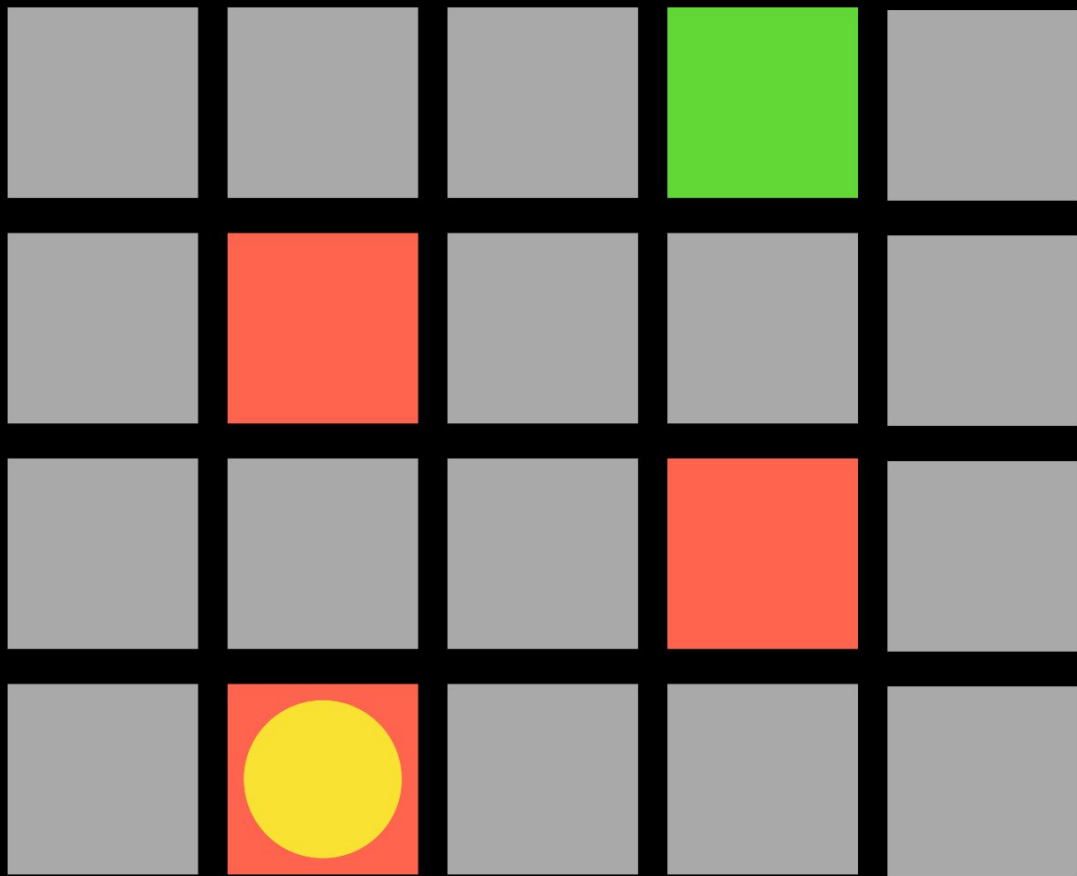
A white robotic arm with a yellow gripper is holding a black frying pan containing a yellow egg. The arm is positioned over a blue cloth-covered table. The background is dark blue. The text "Reinforcement Learning" and "First trial..." is overlaid on the bottom of the image.

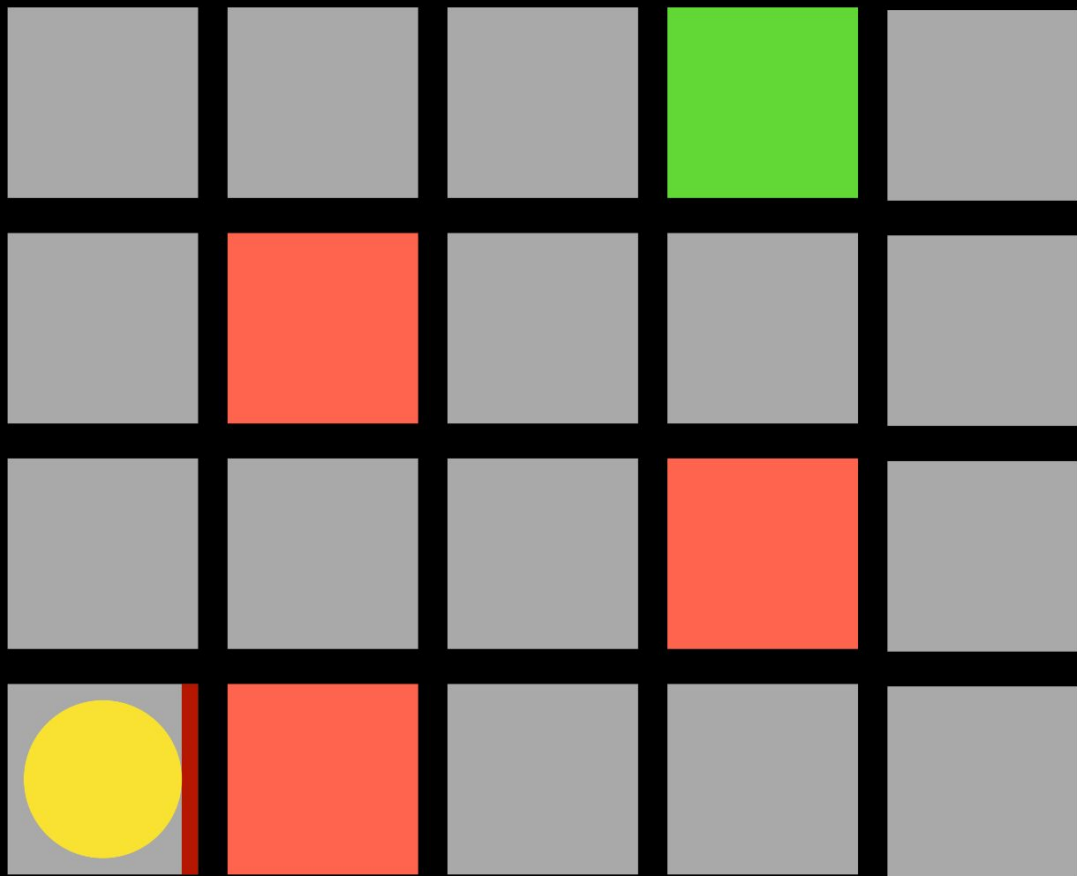
**Reinforcement Learning**  
**First trial...**

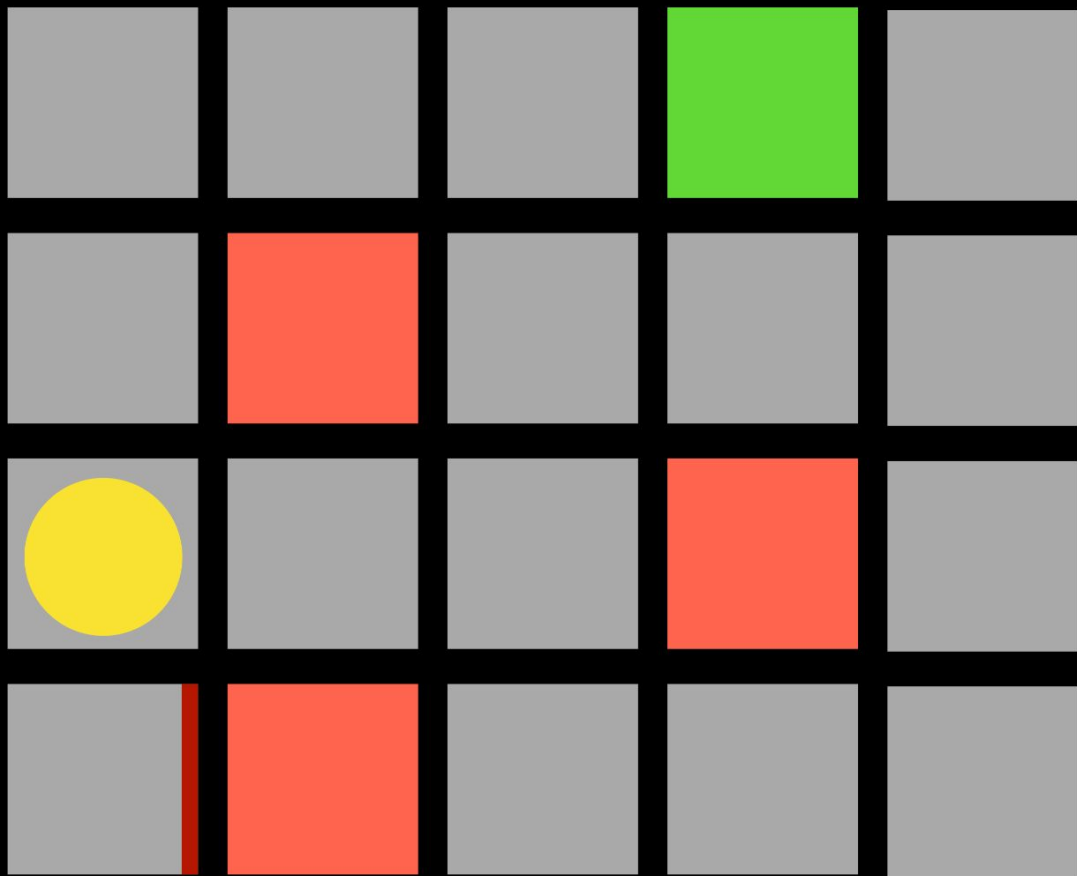


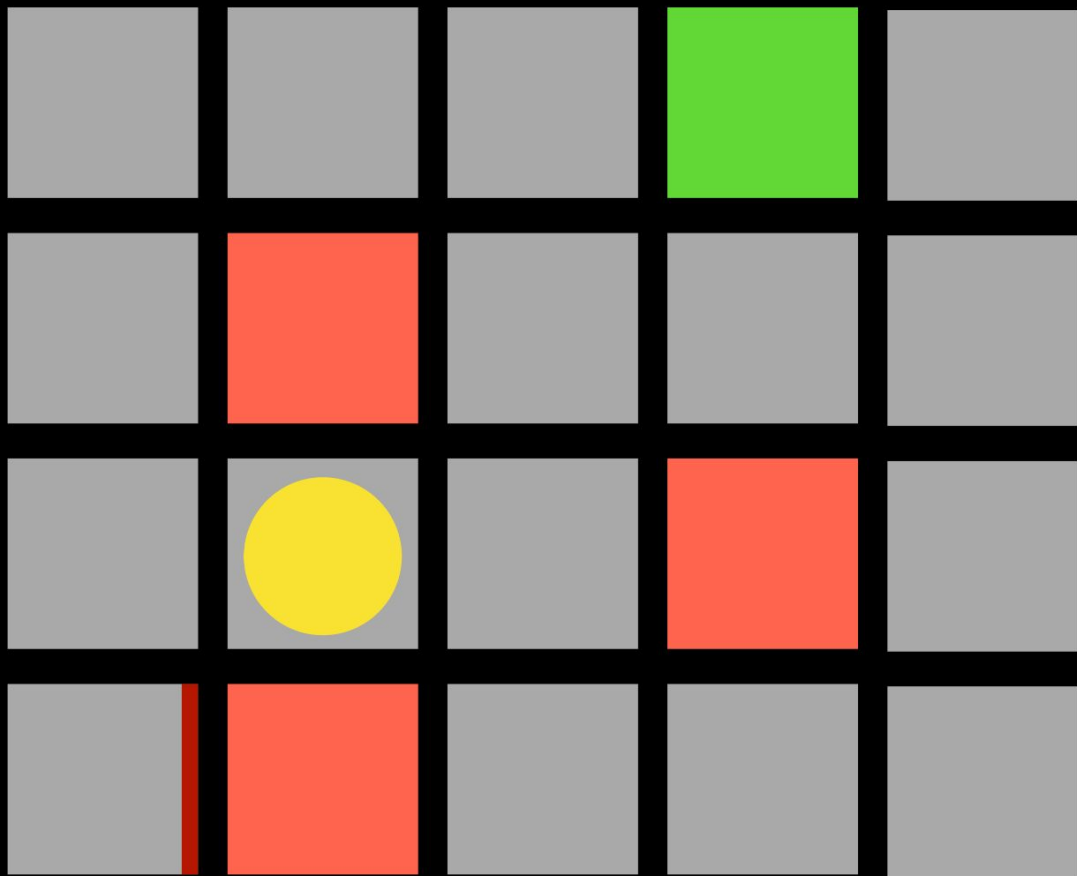


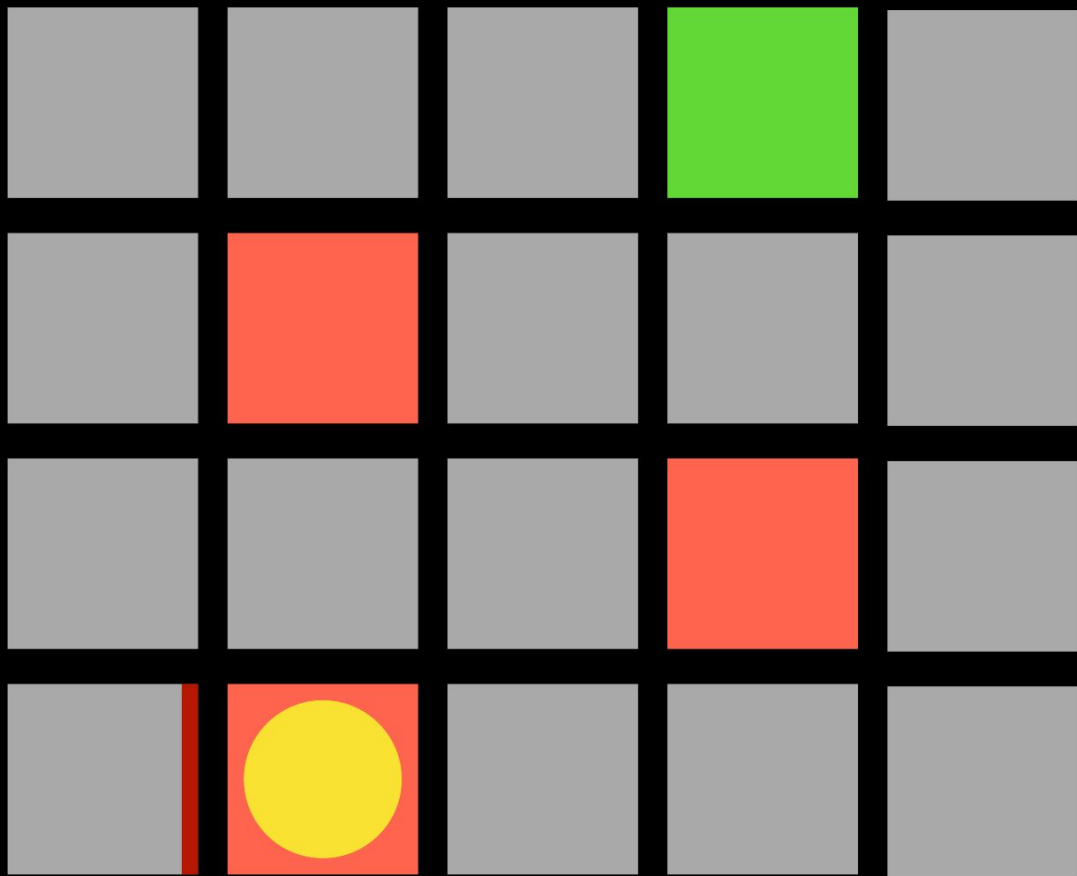


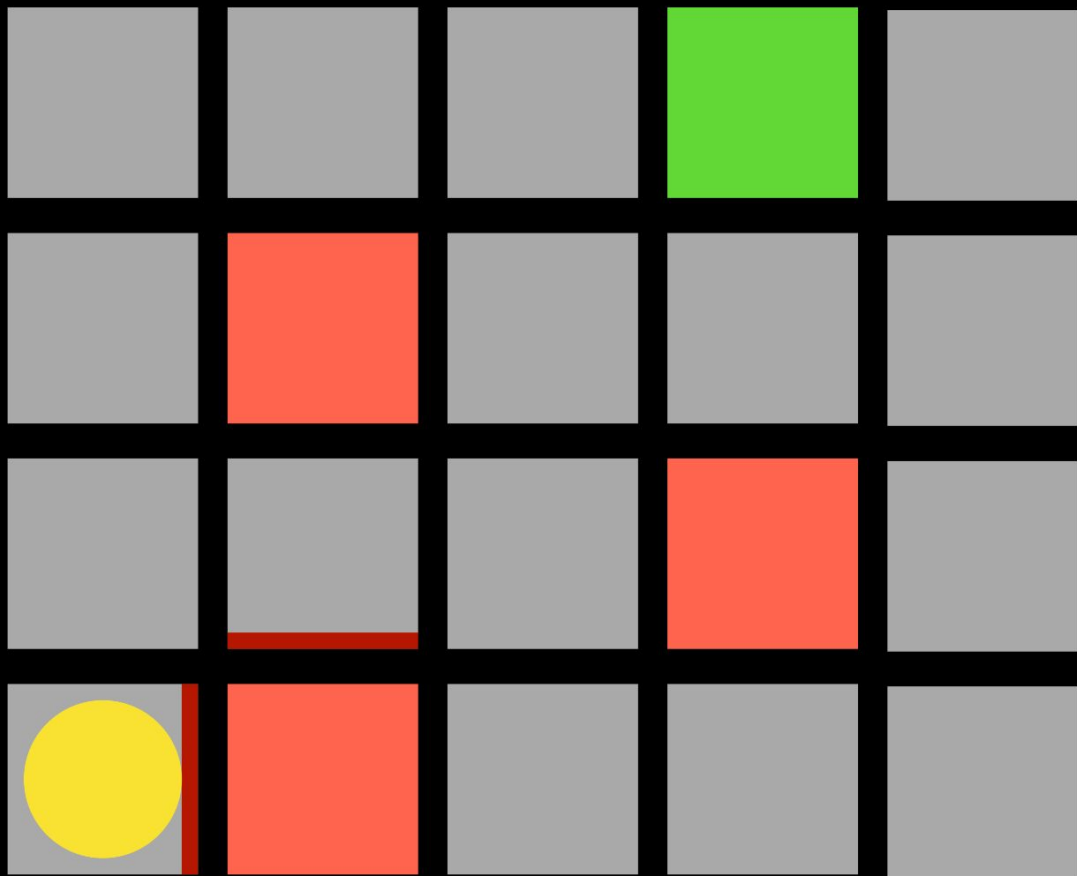




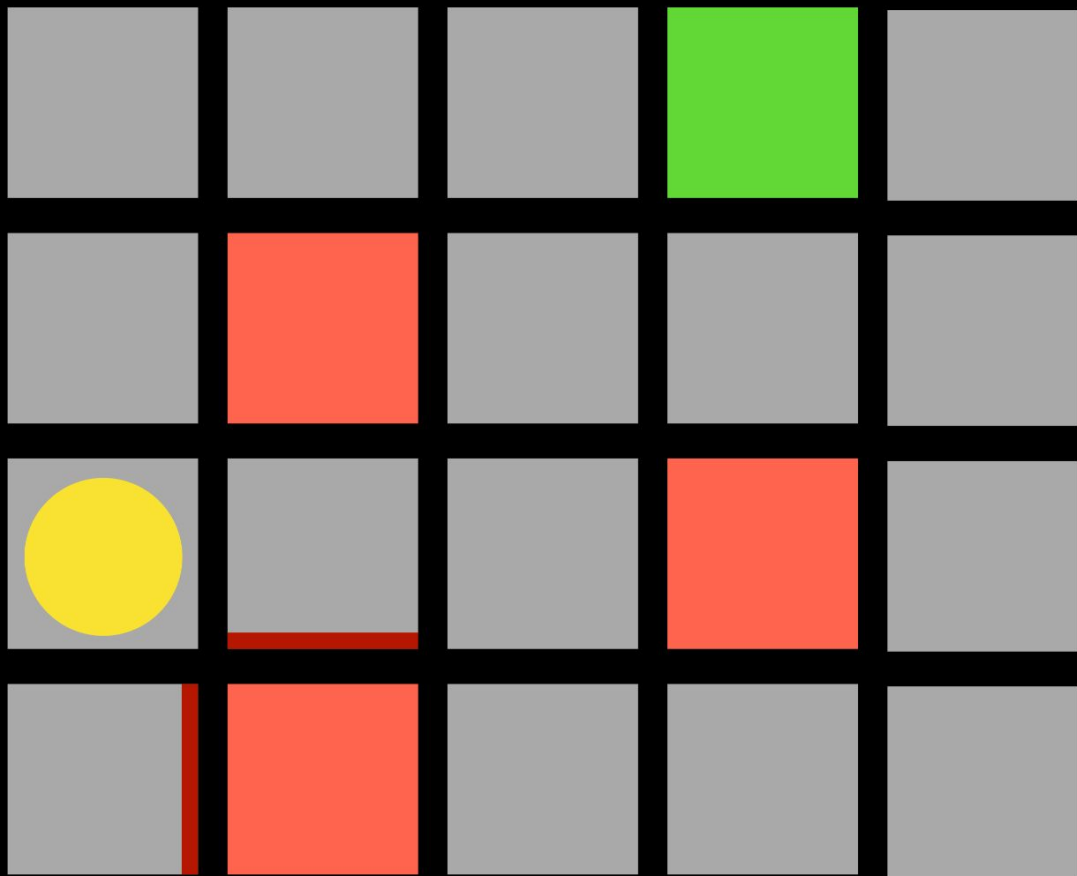


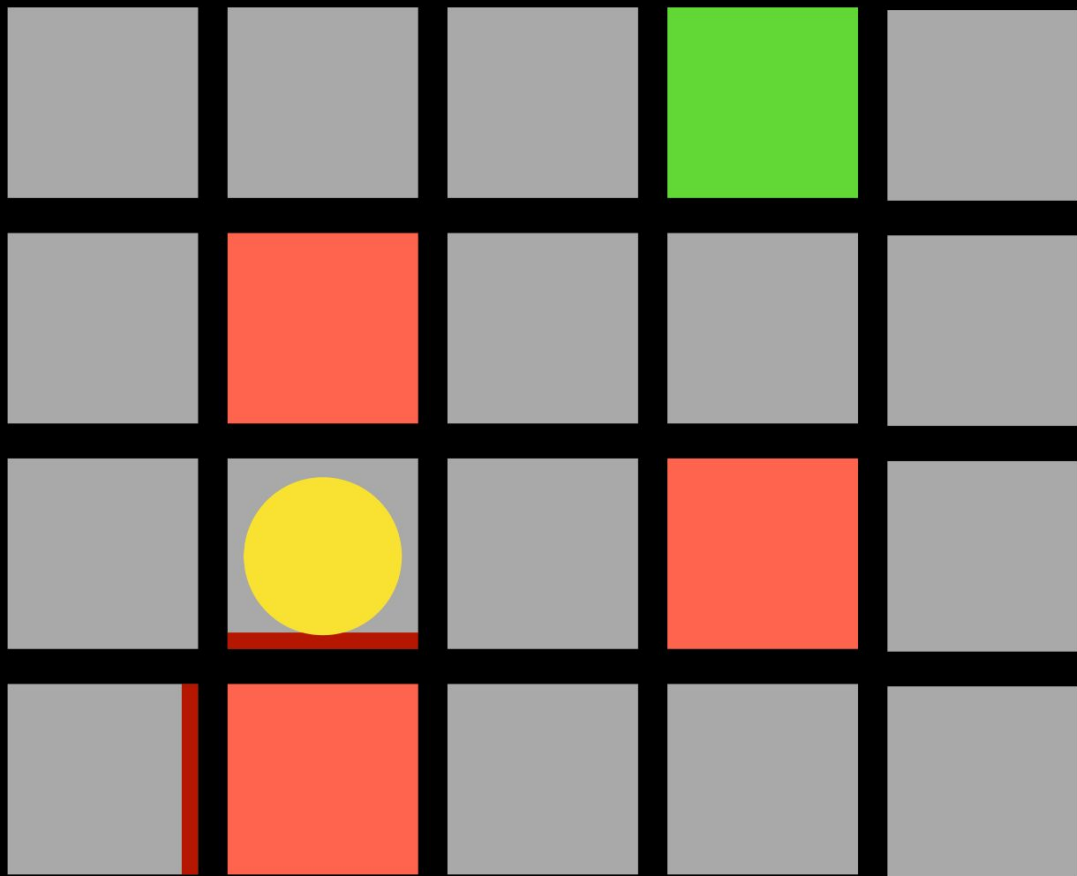


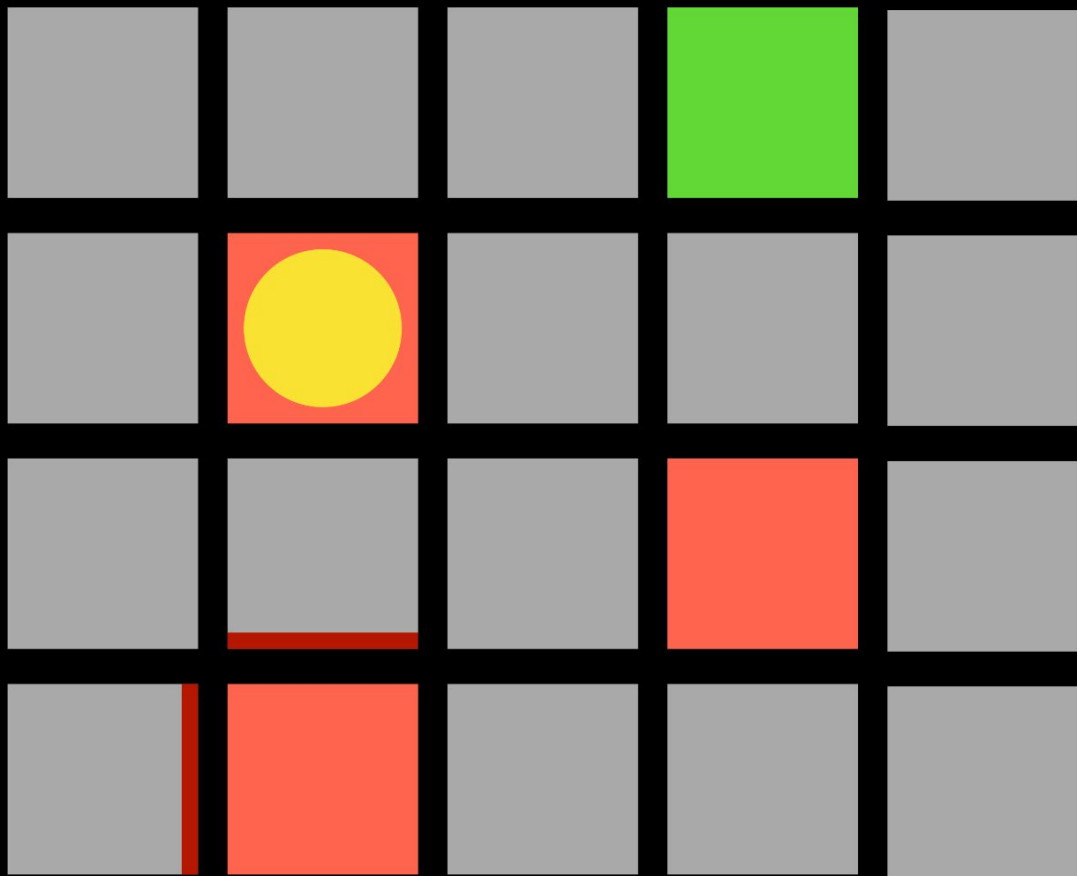


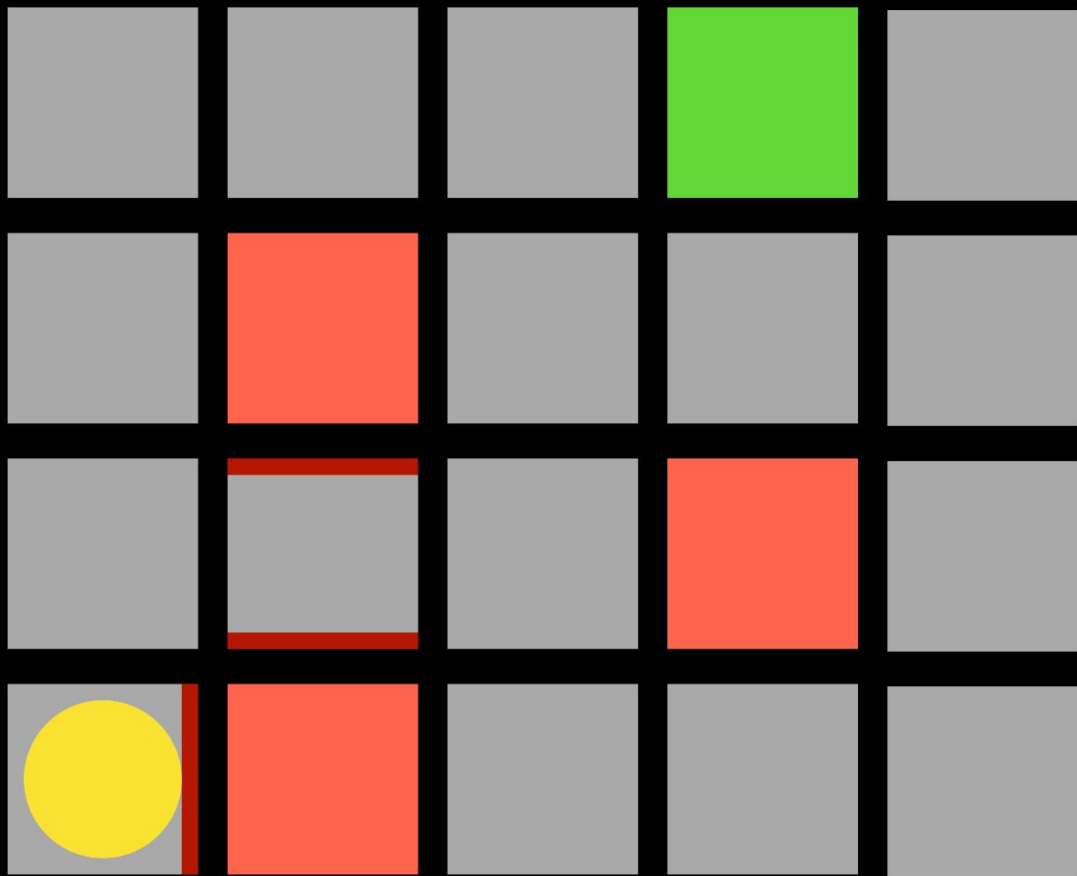


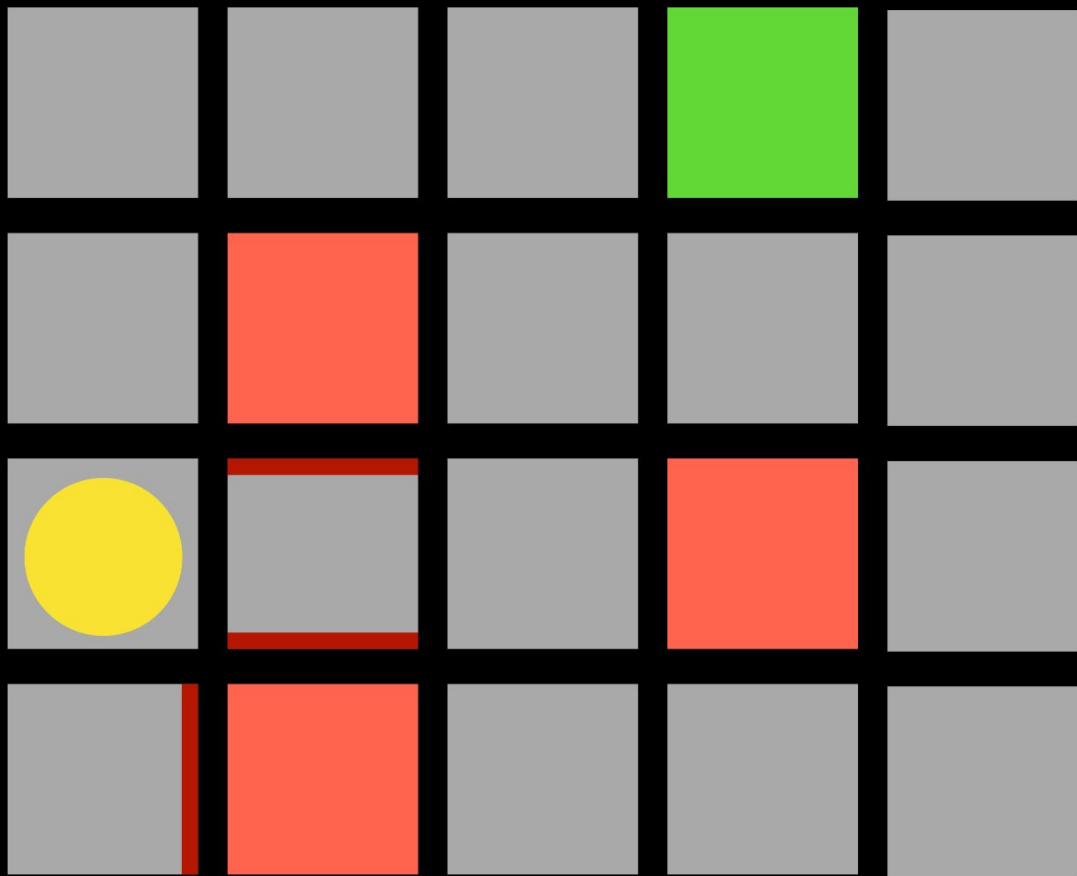


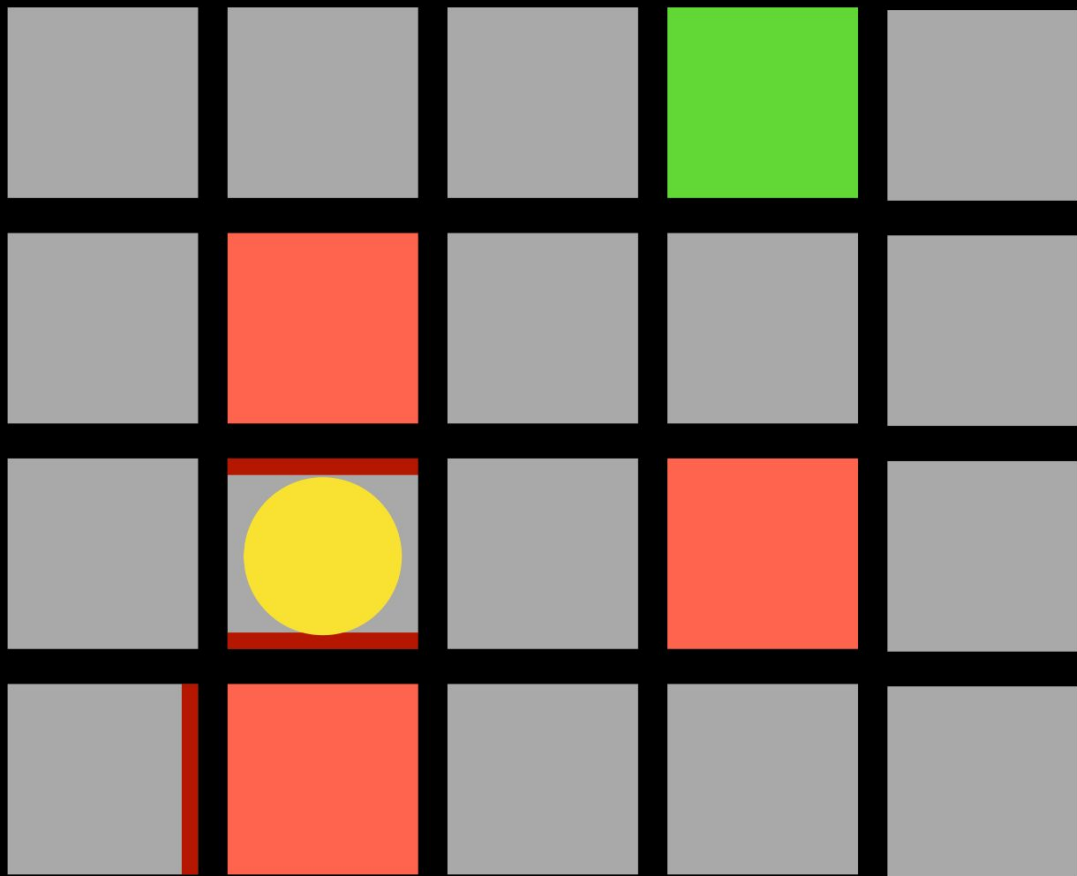


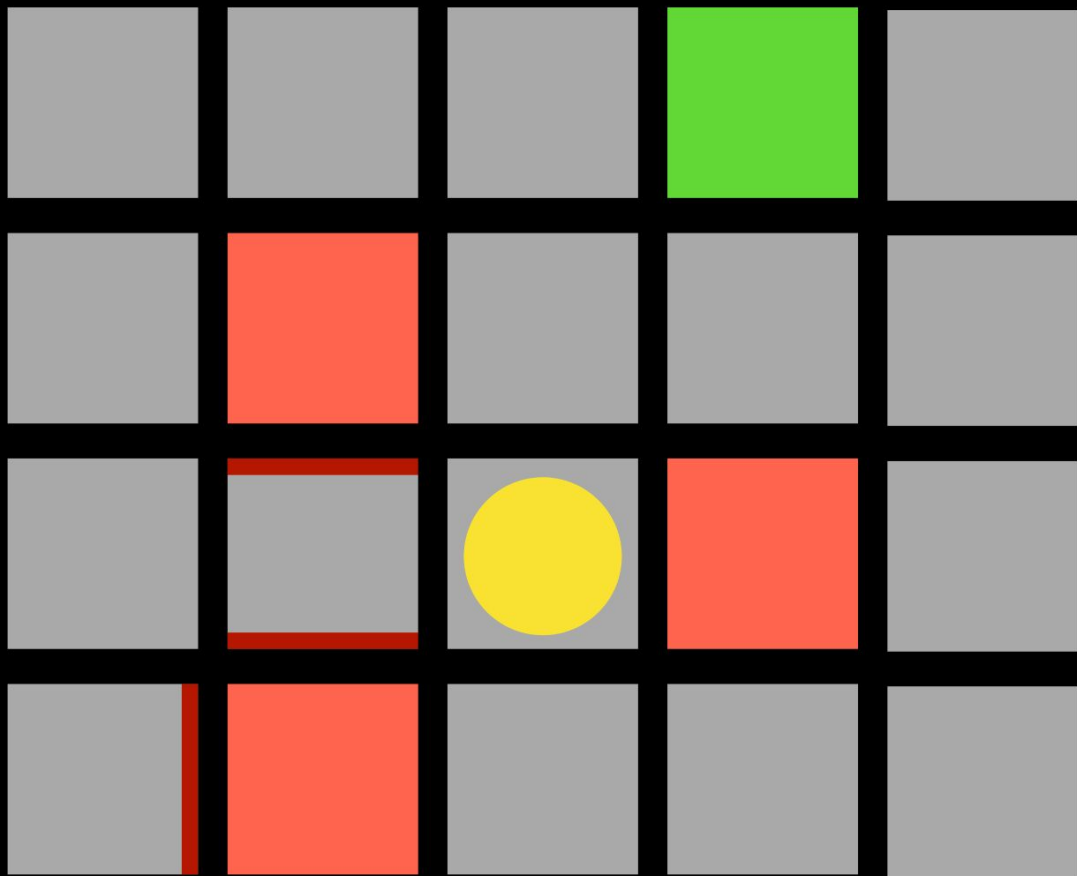


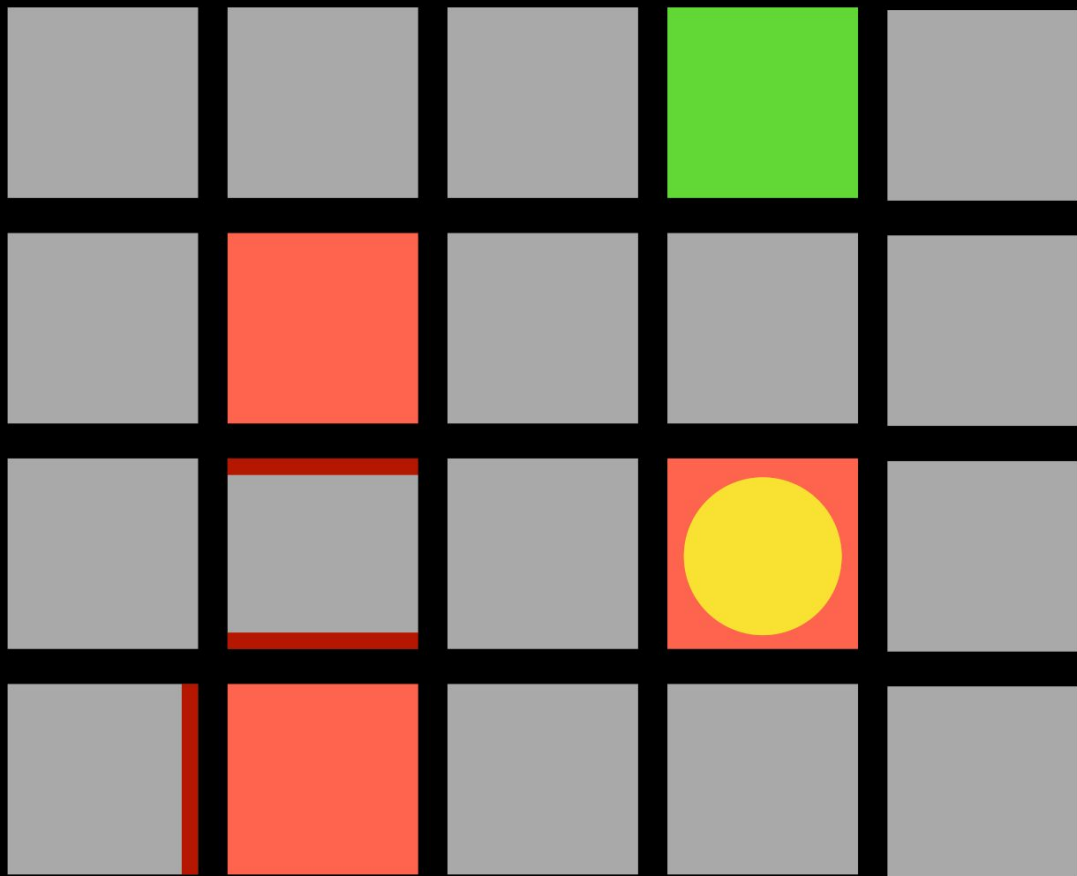




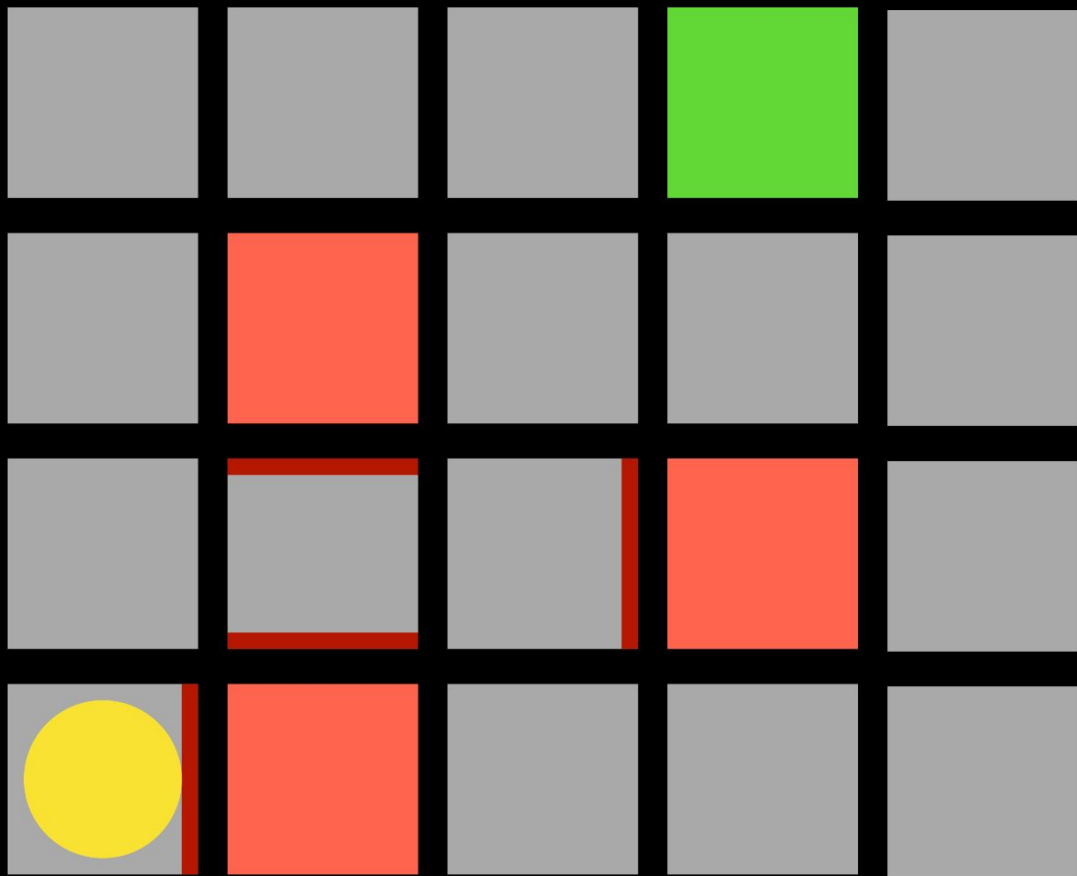


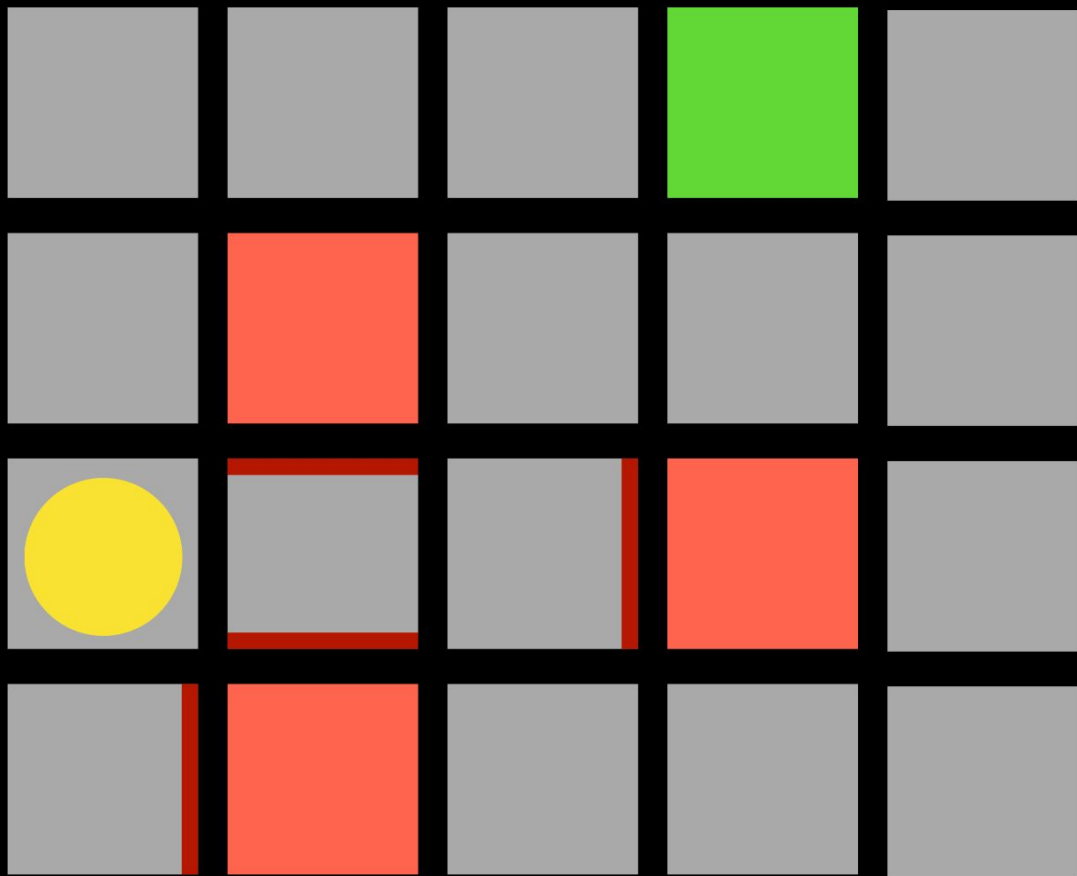


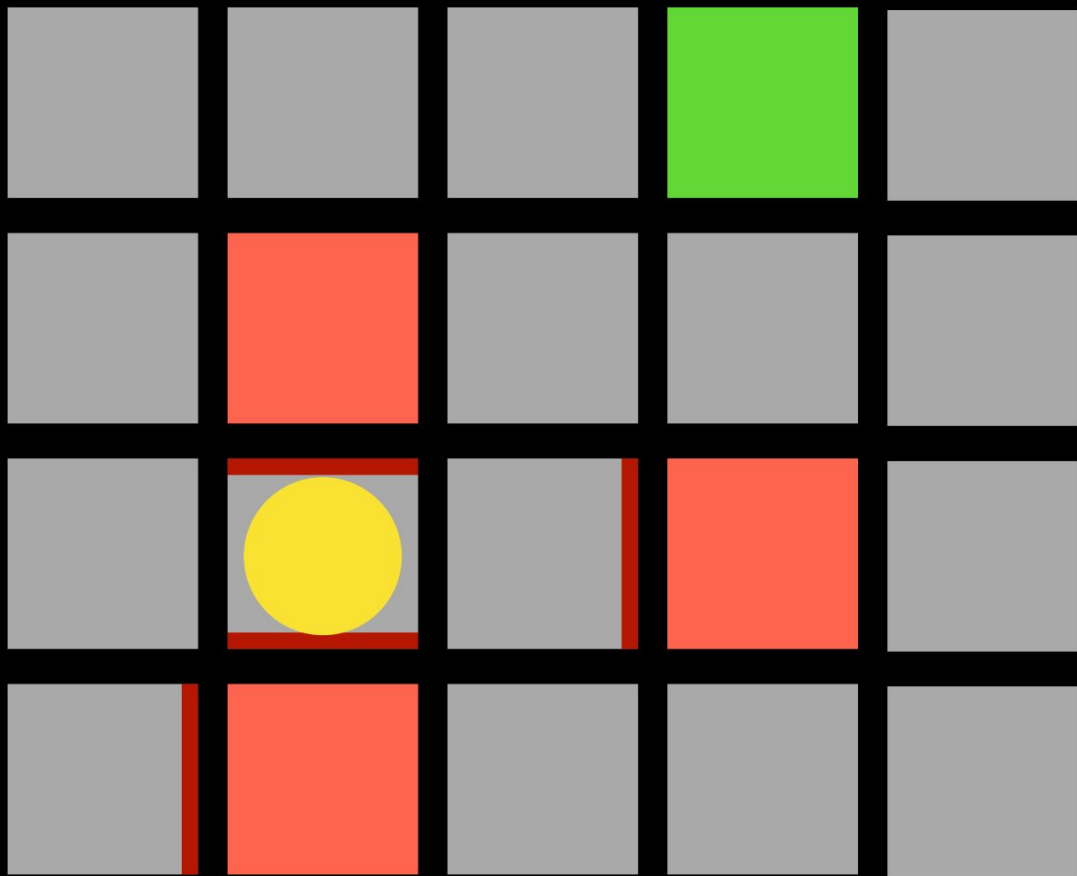


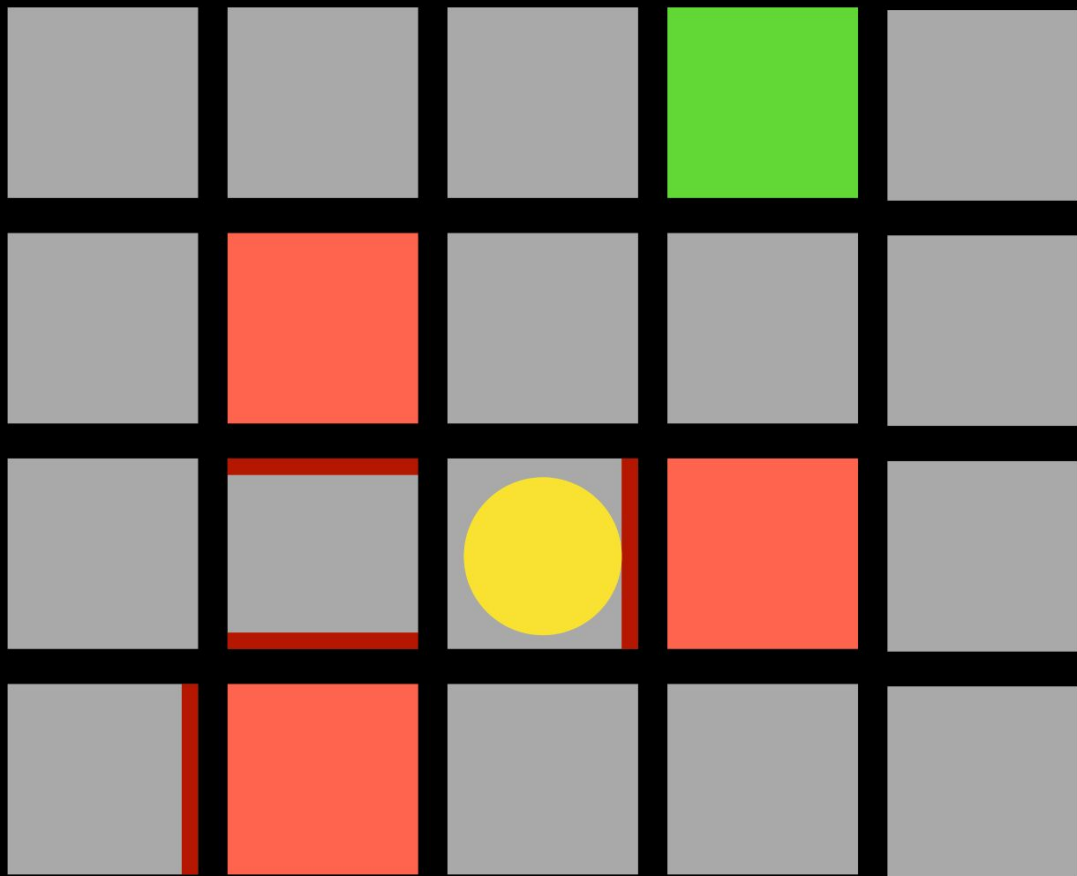


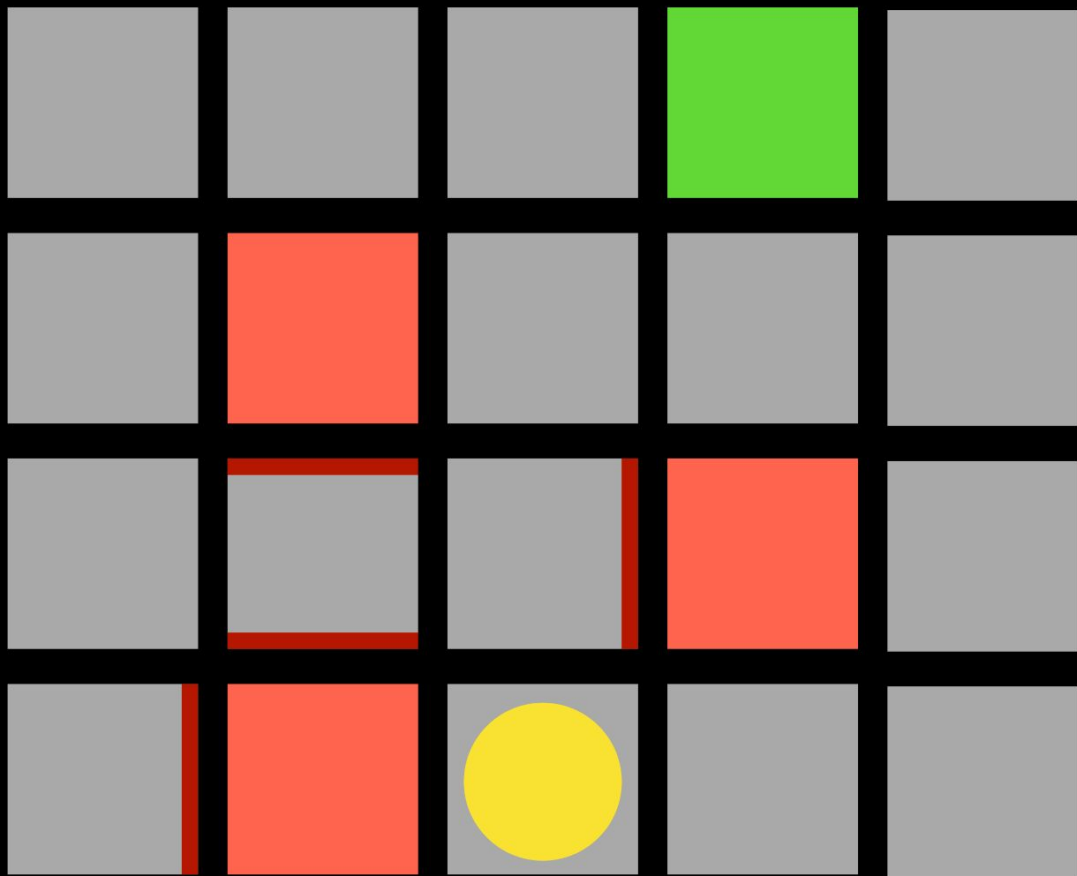


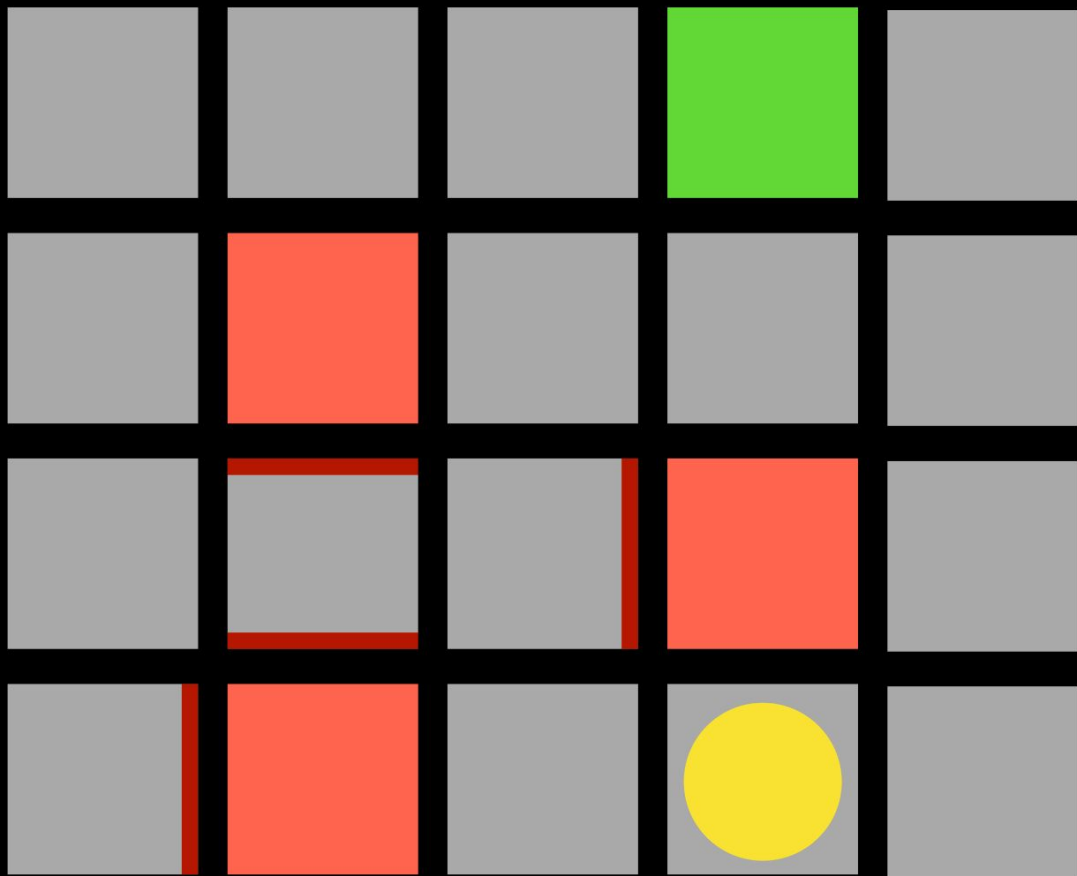


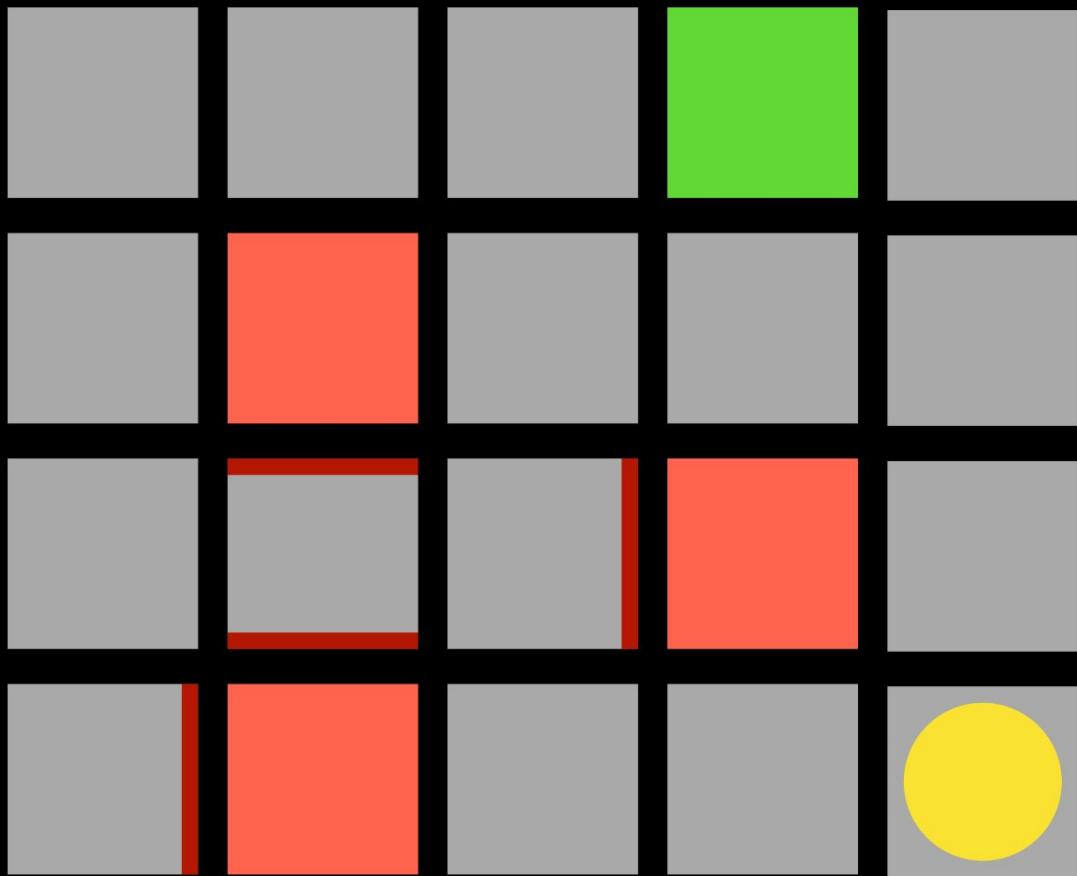


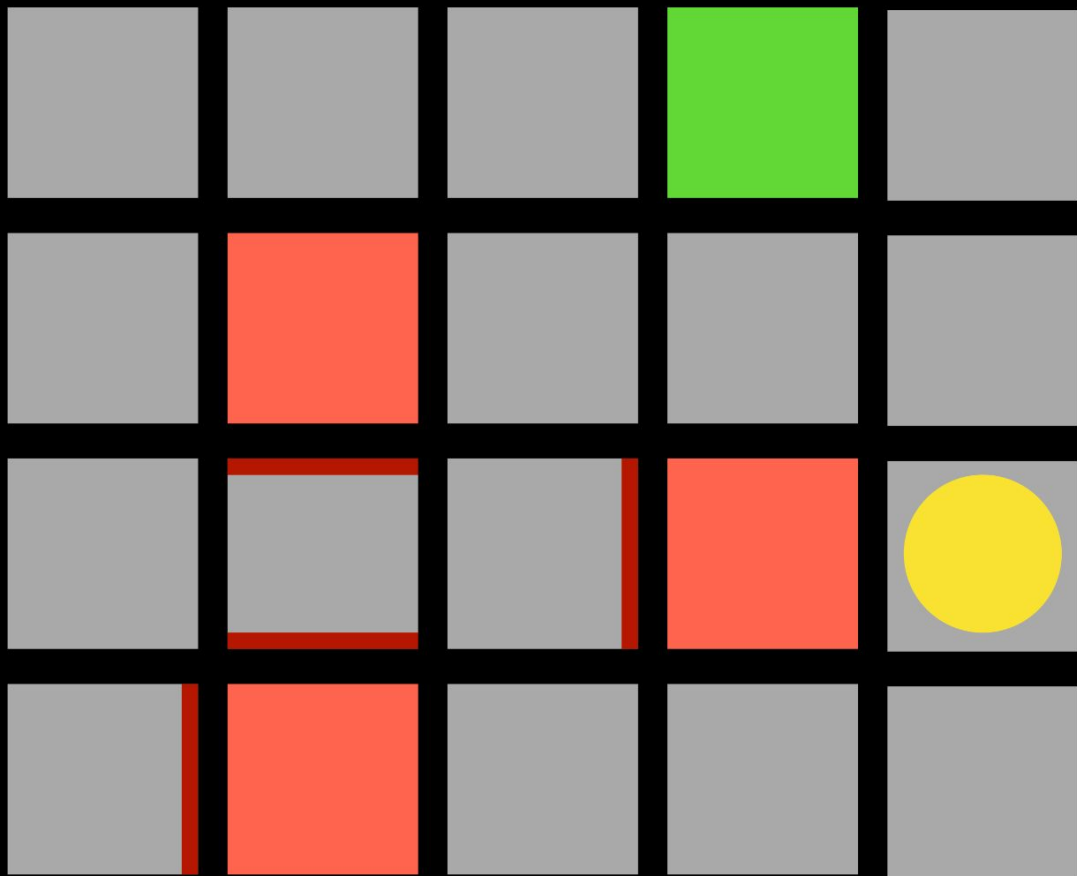




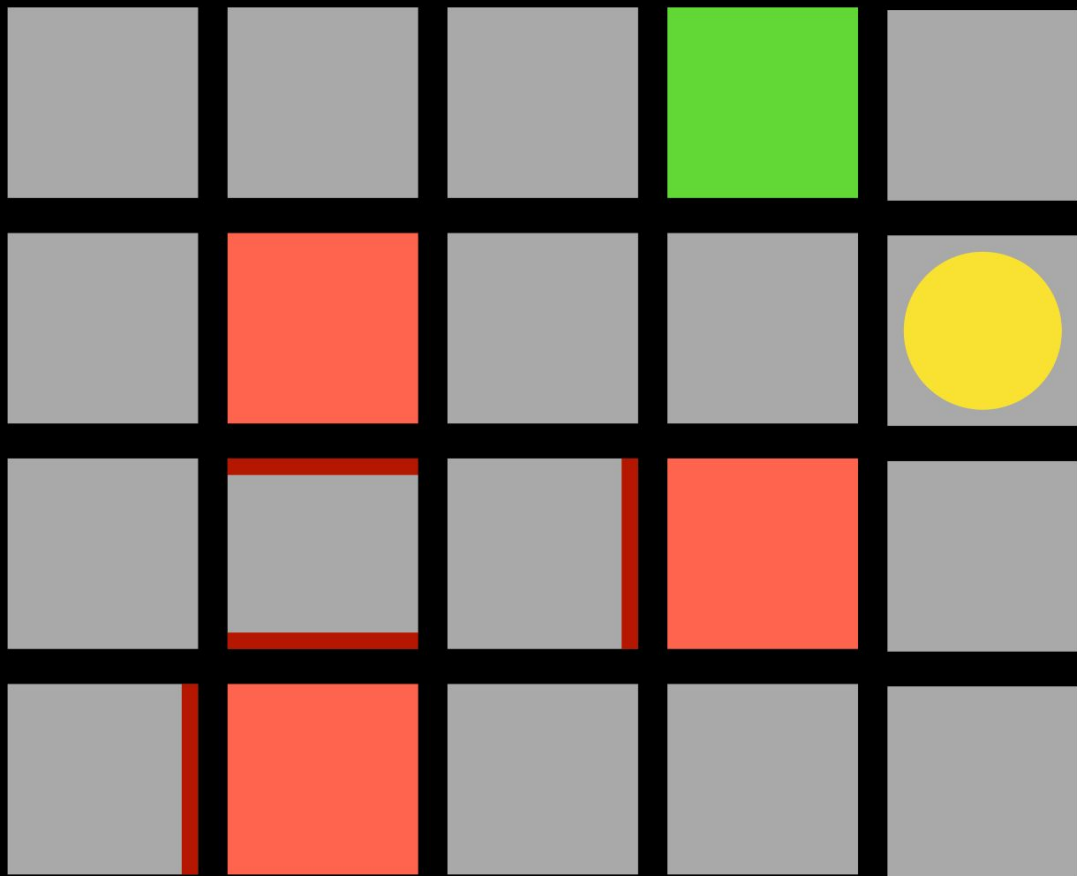


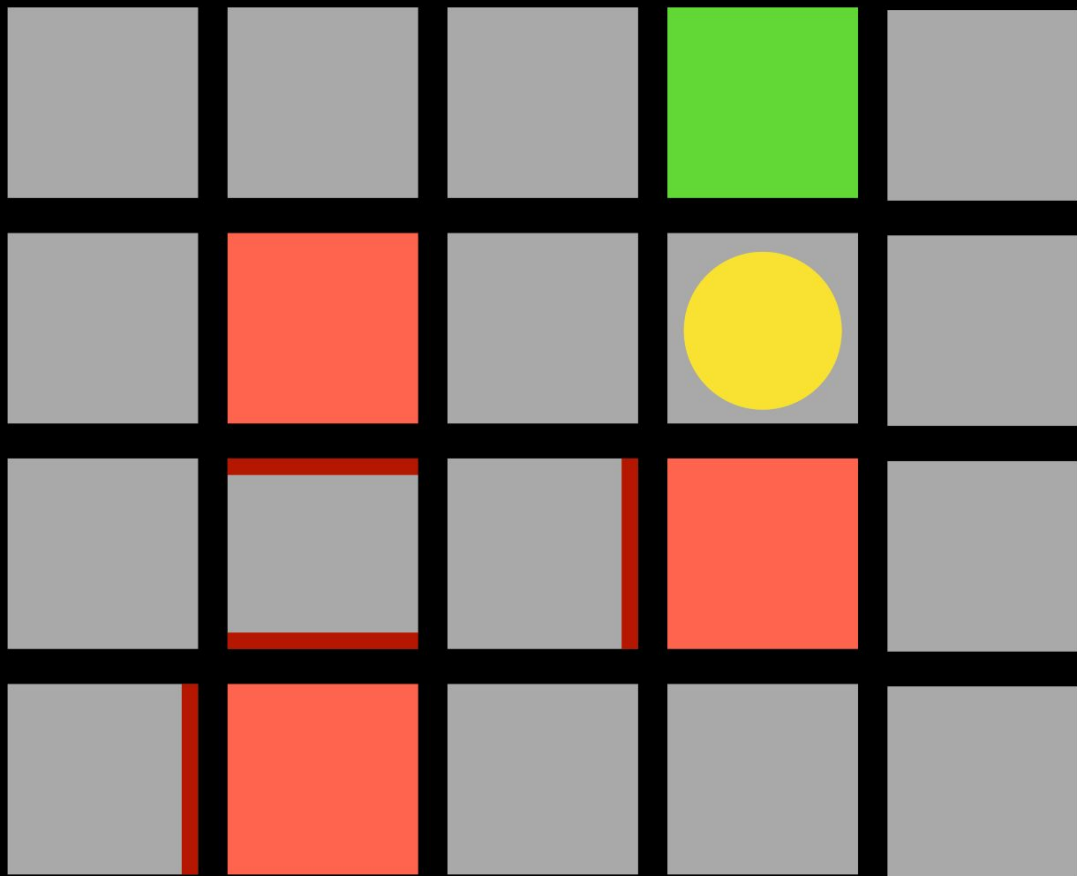


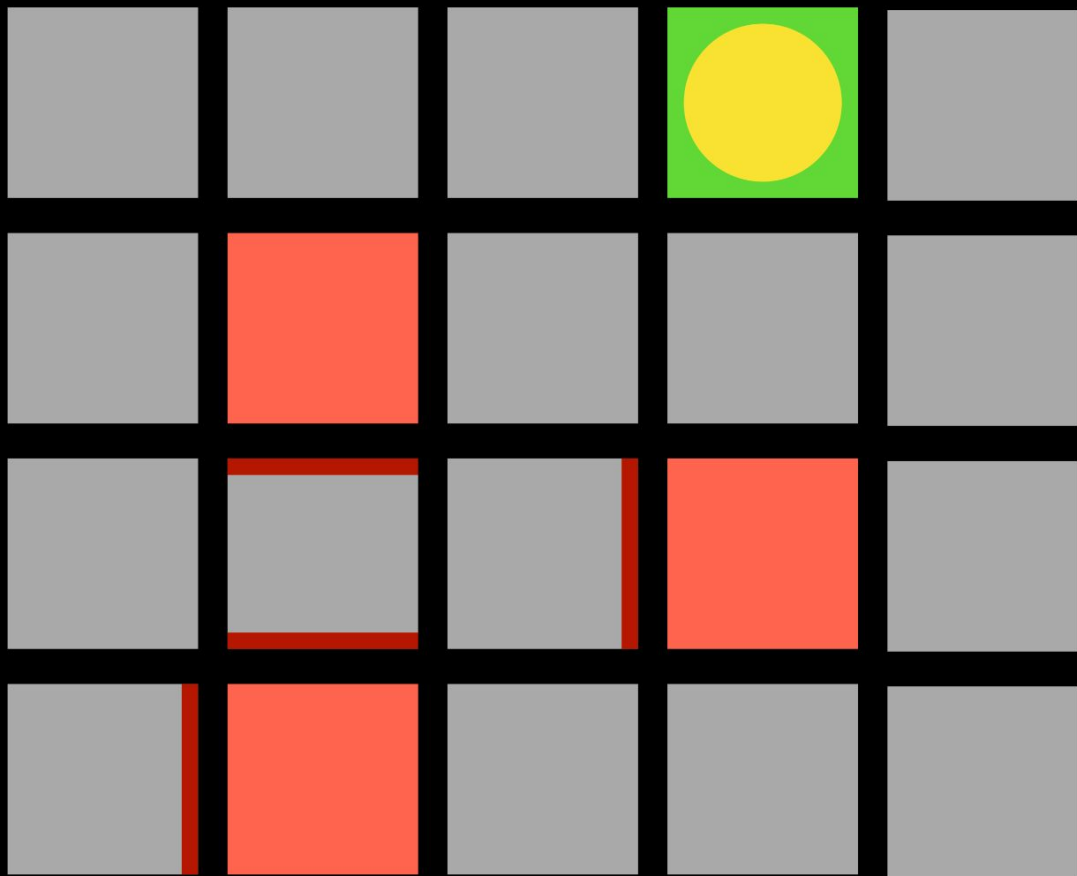


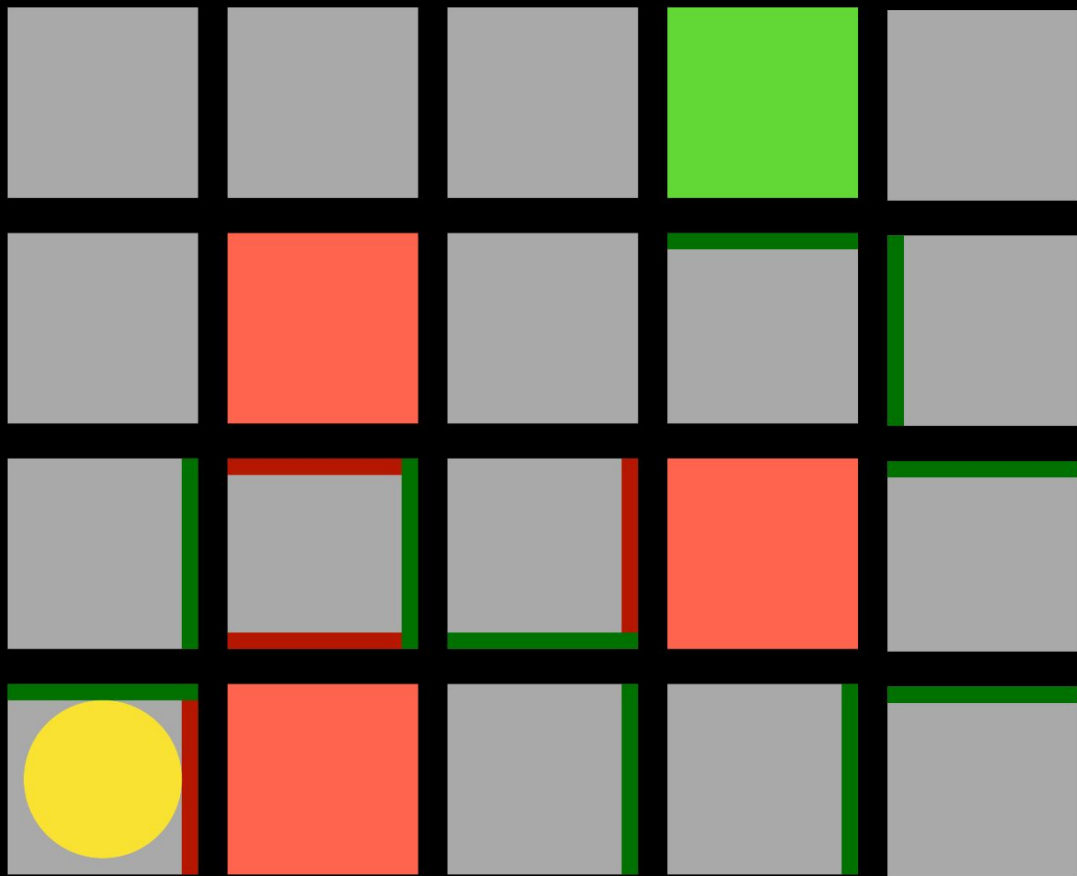


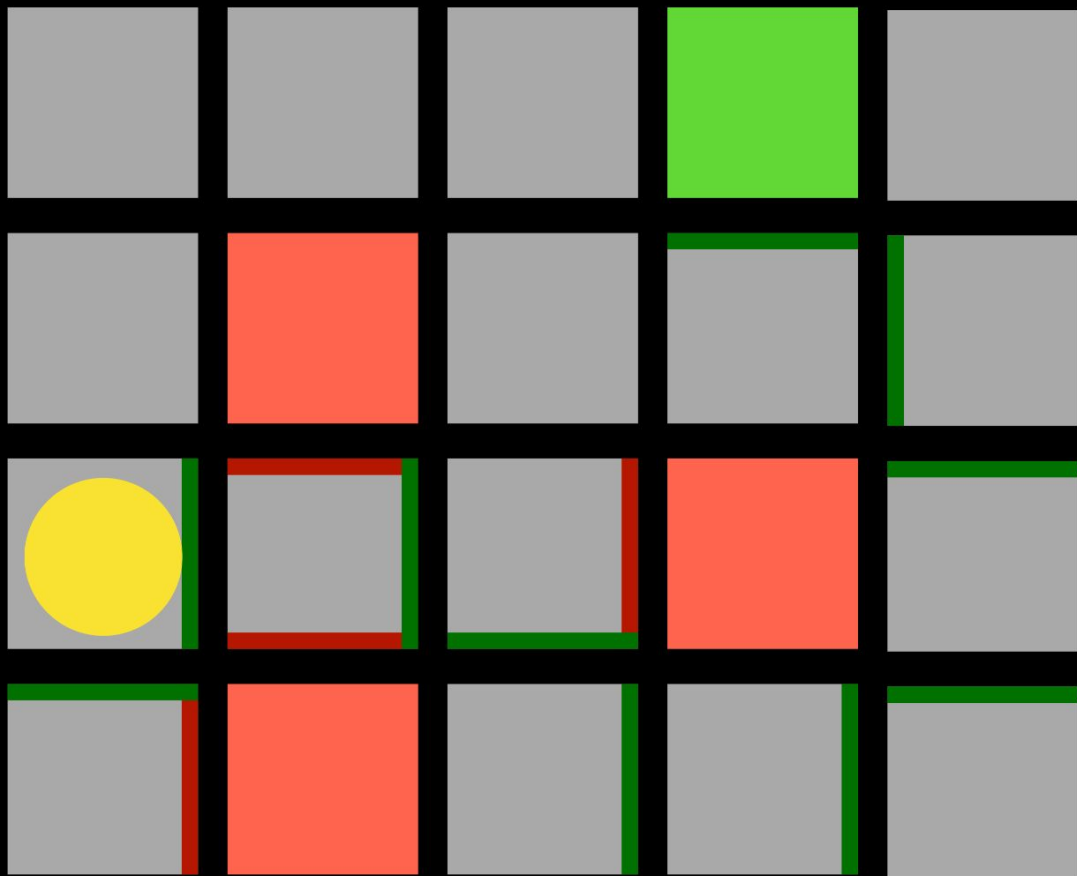


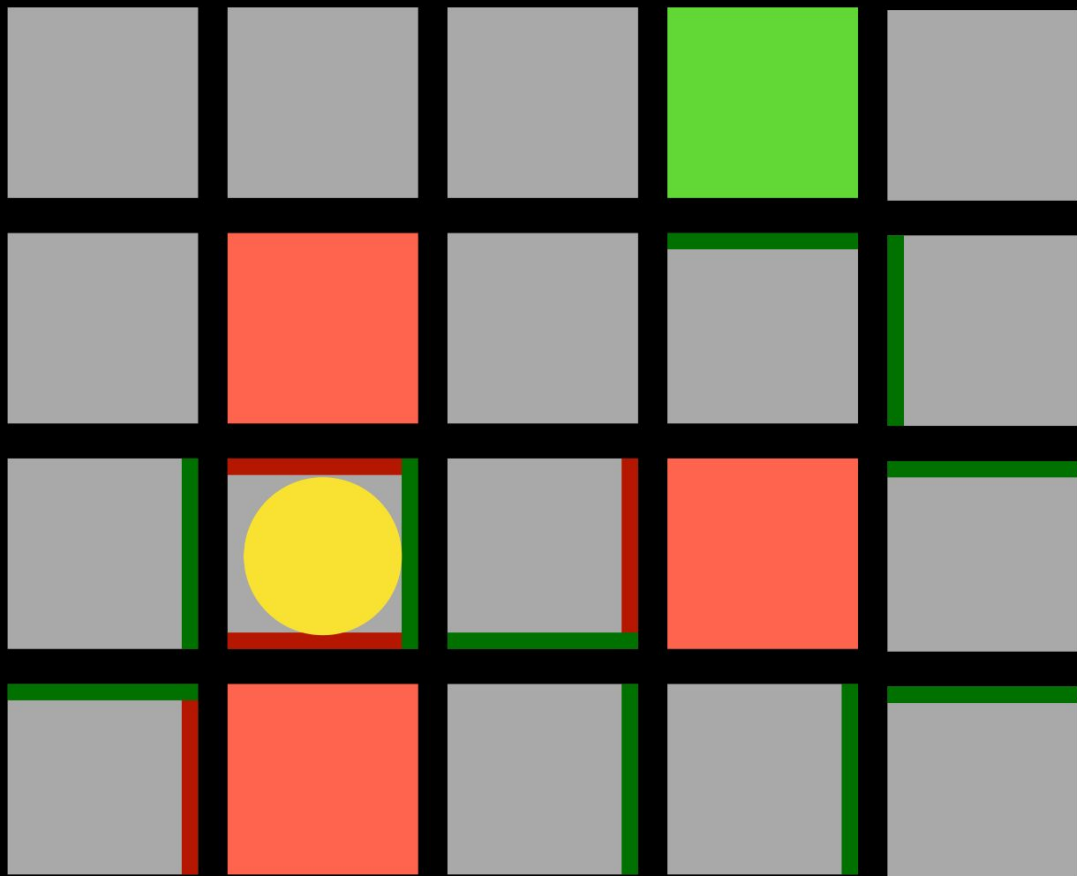


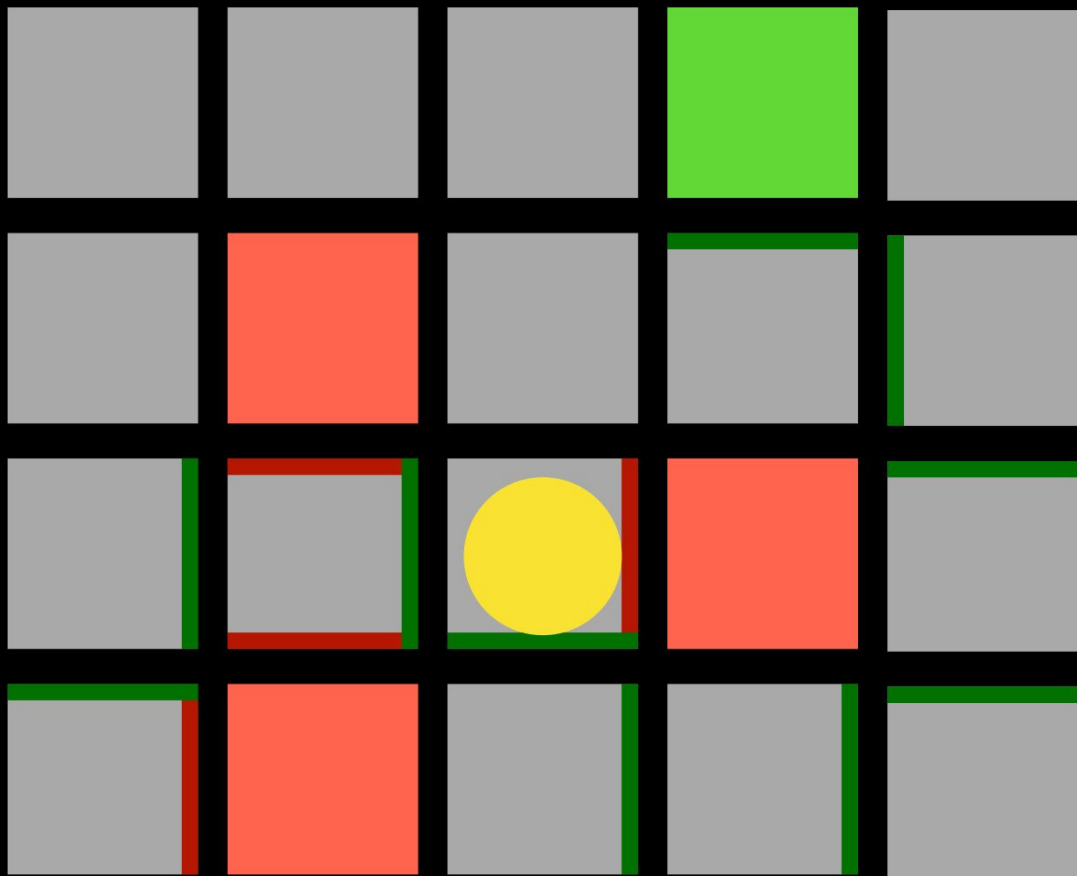


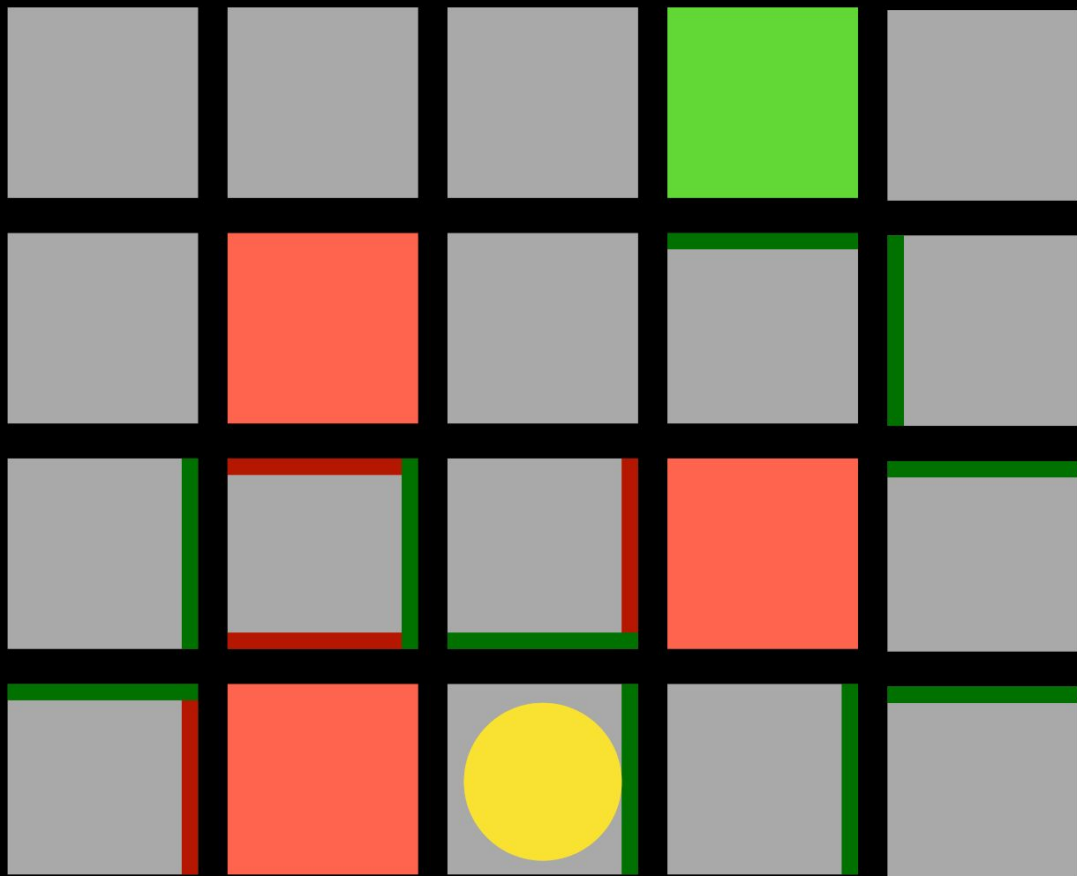




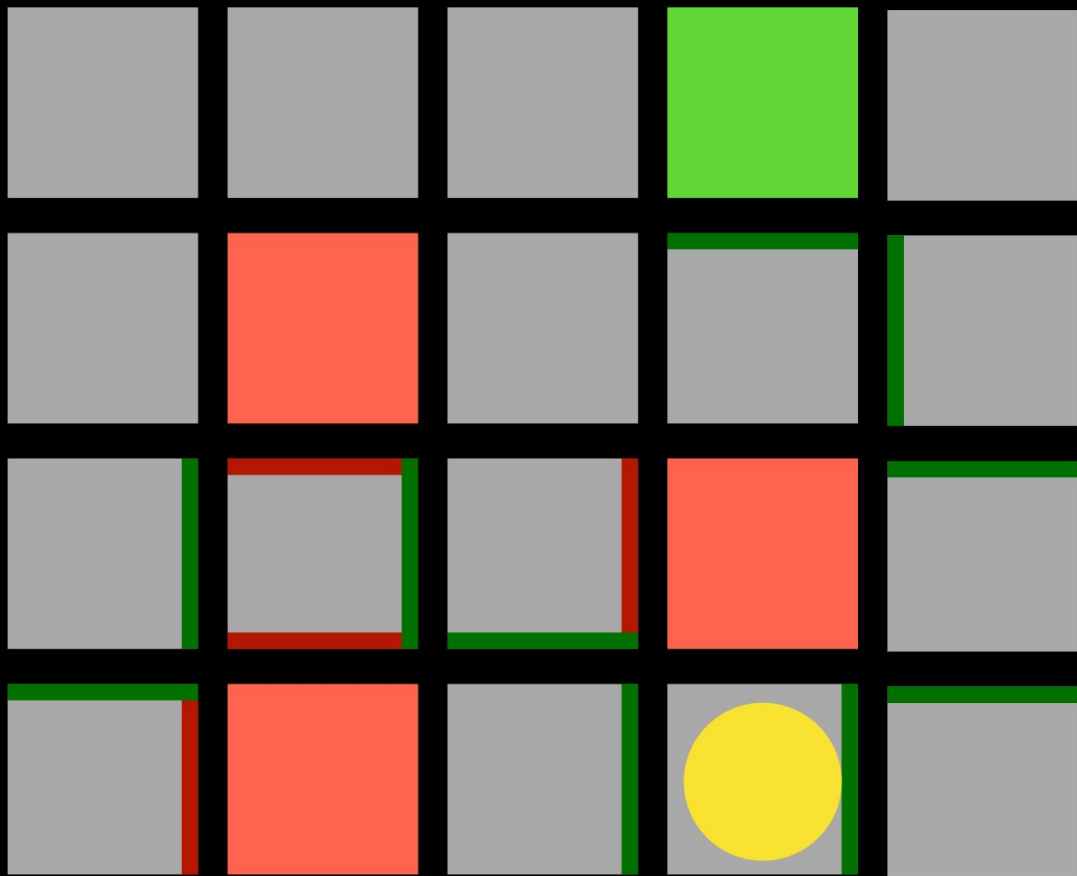


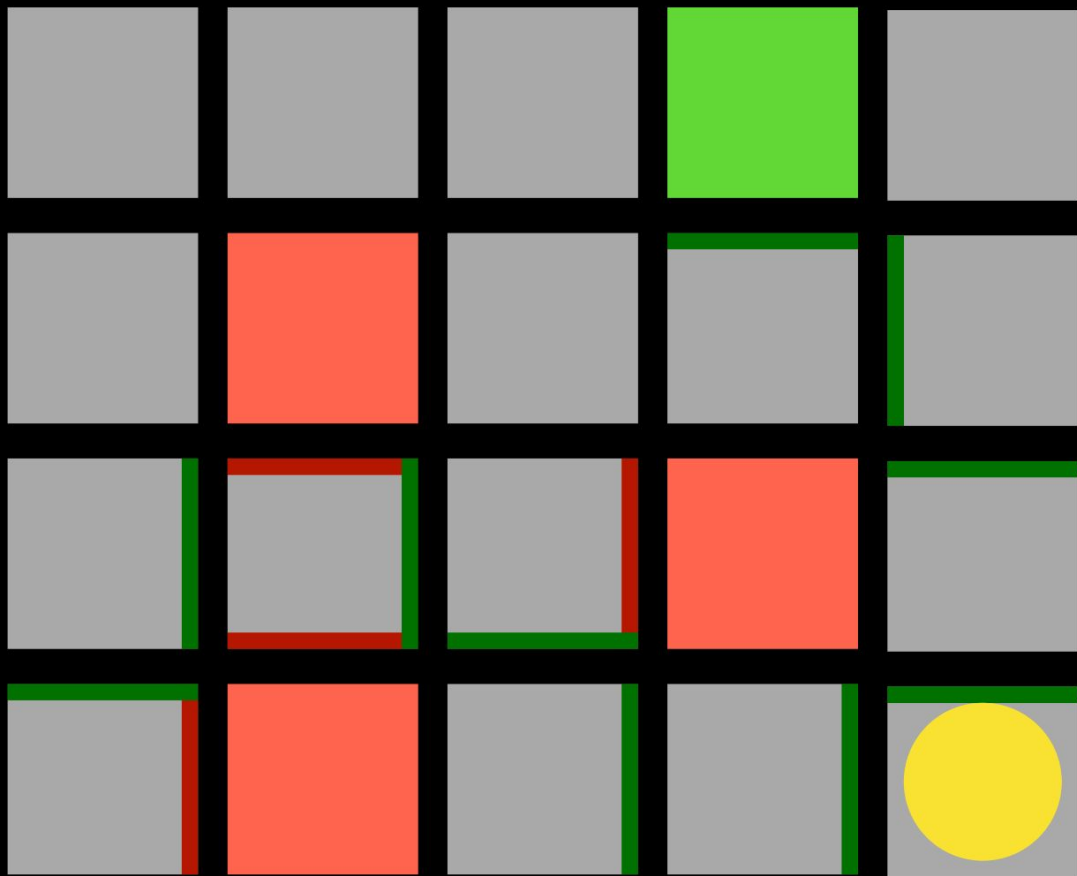


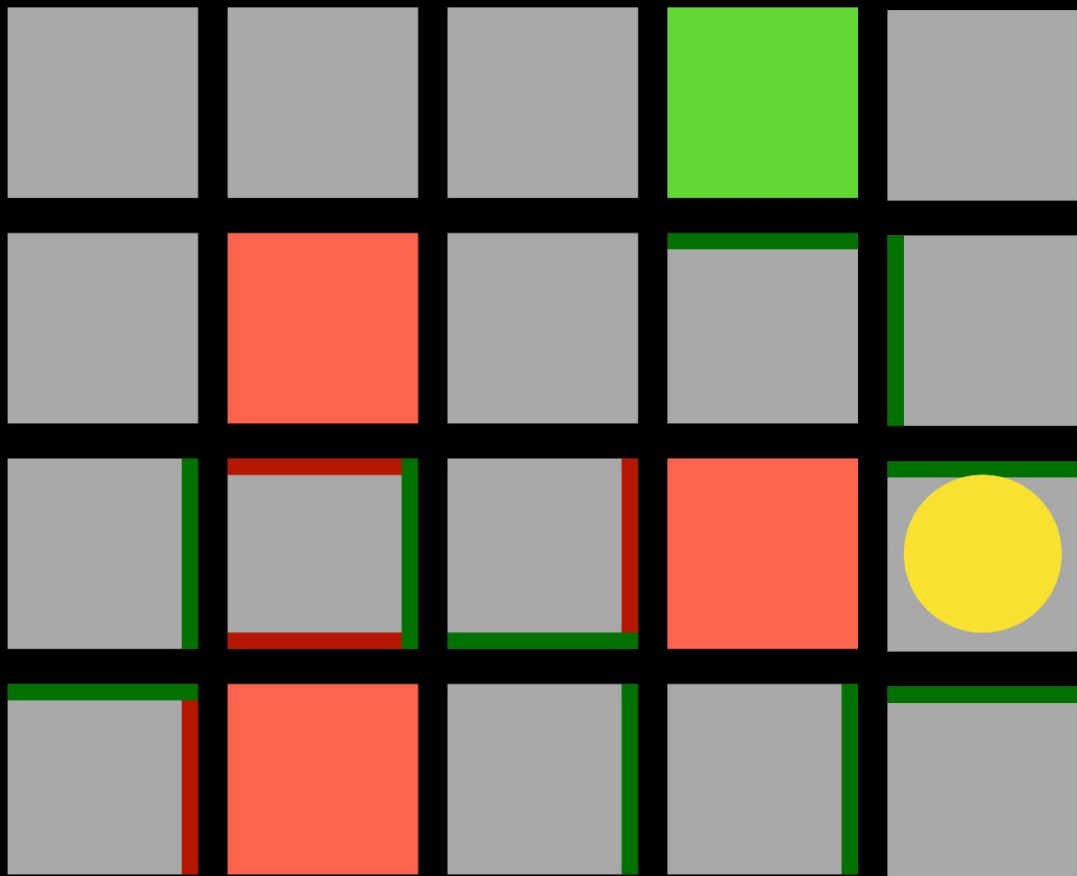


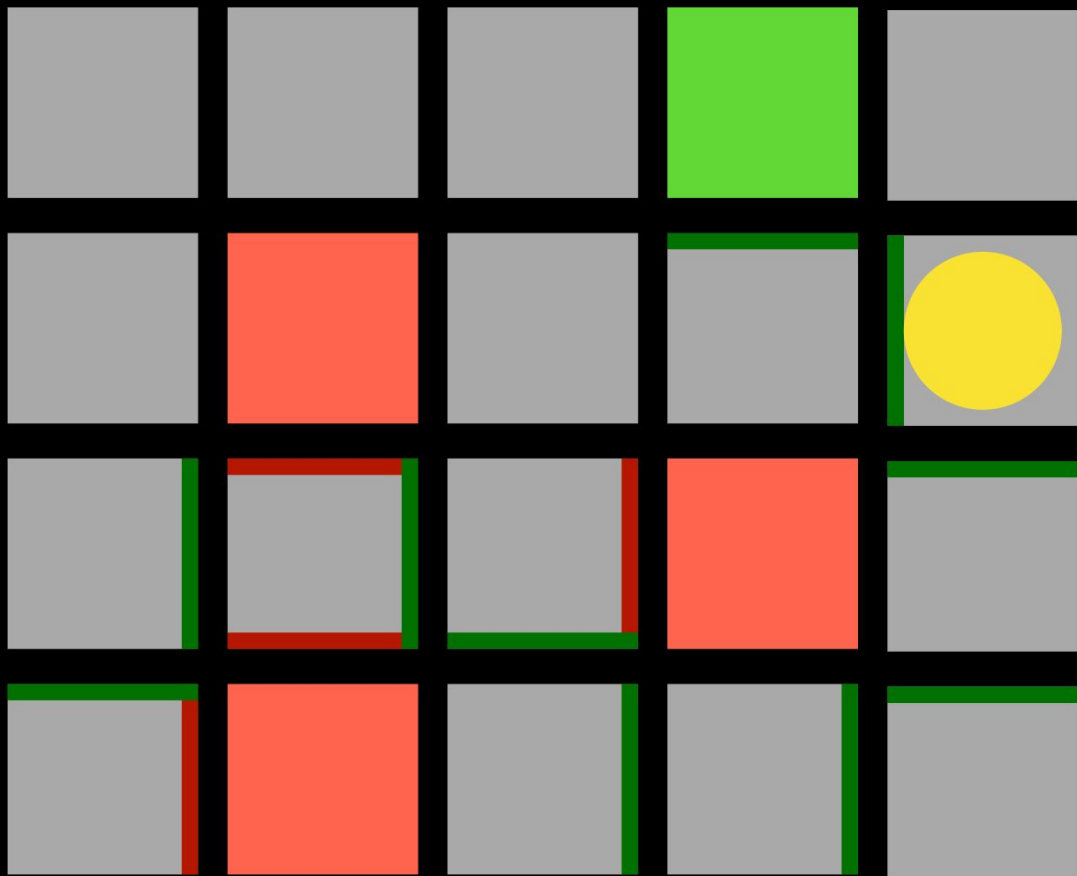


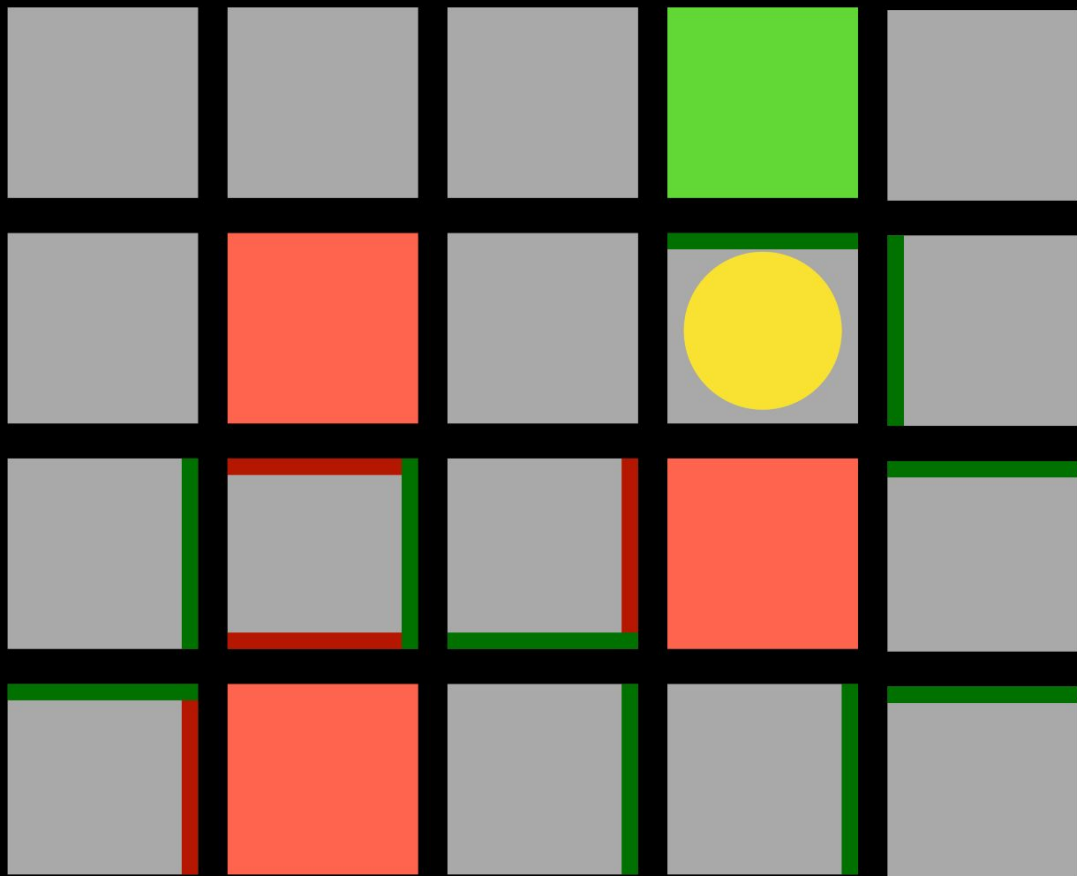


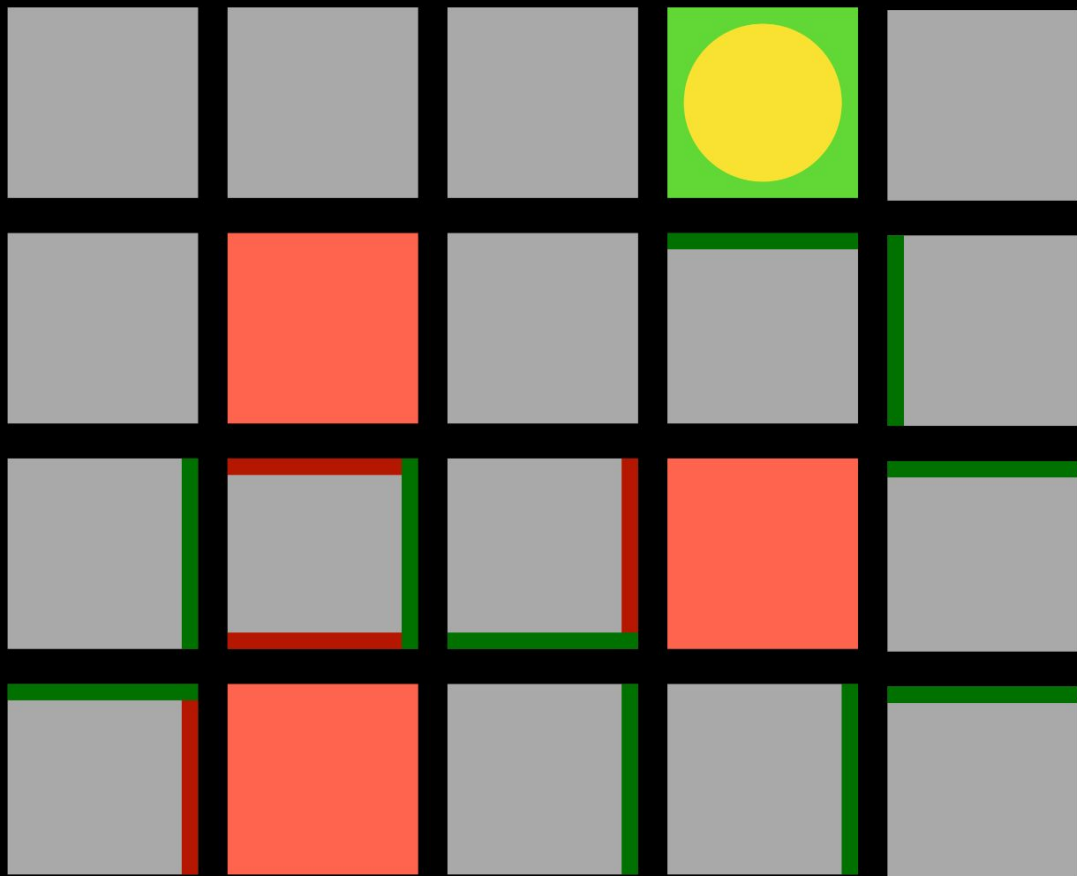












explore vs. exploit

```
epsilon = 0.10
```

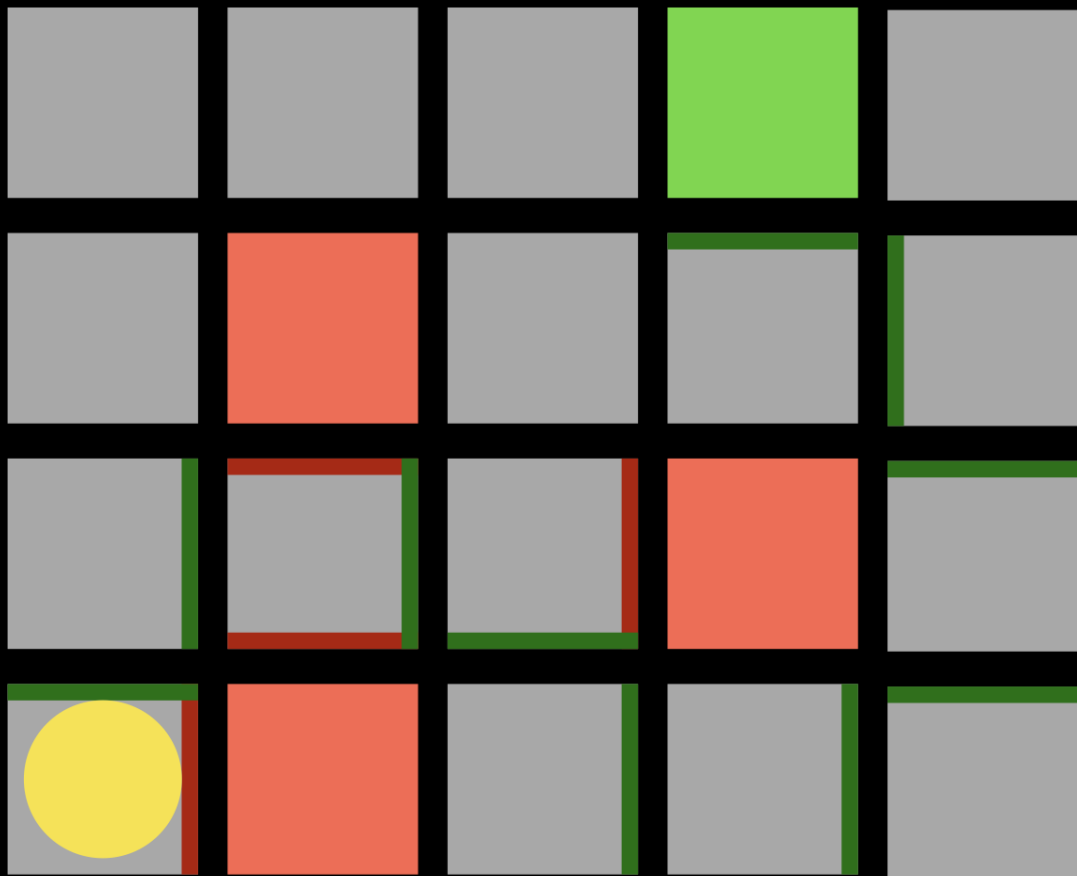
```
If random() < epsilon:
```

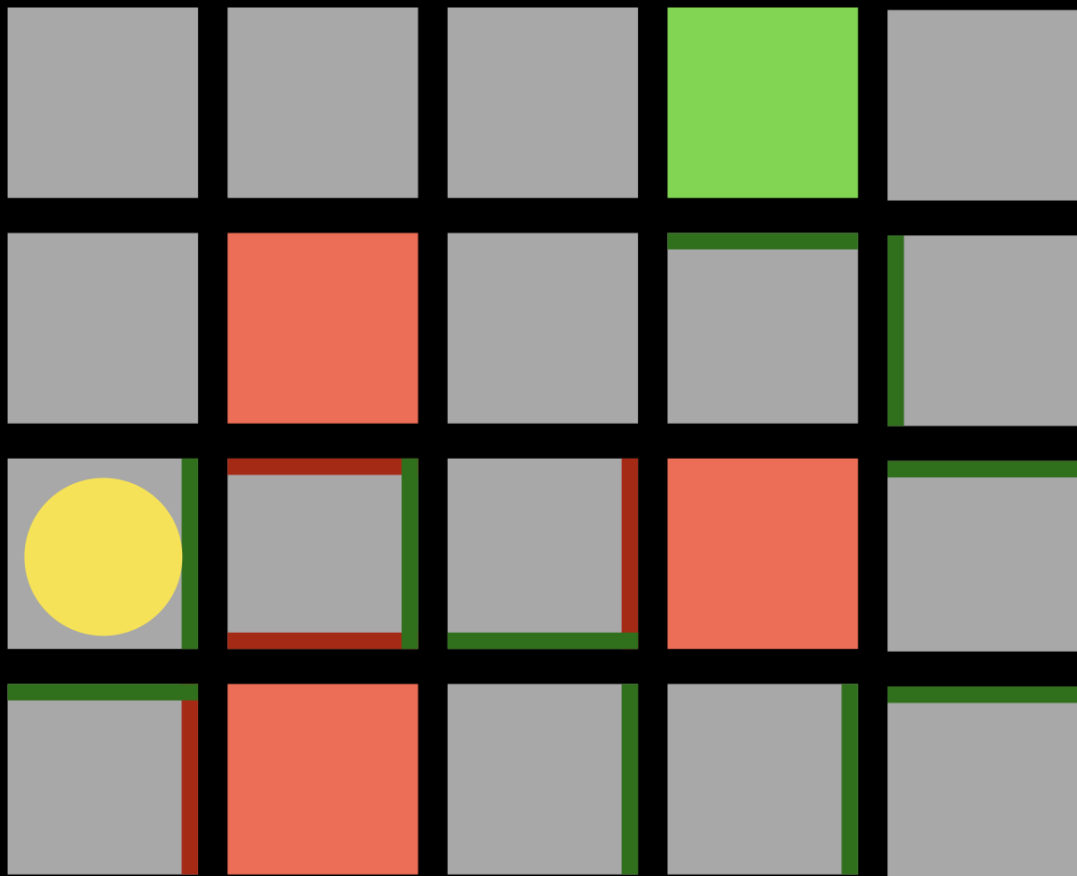
```
    Make a random move
```

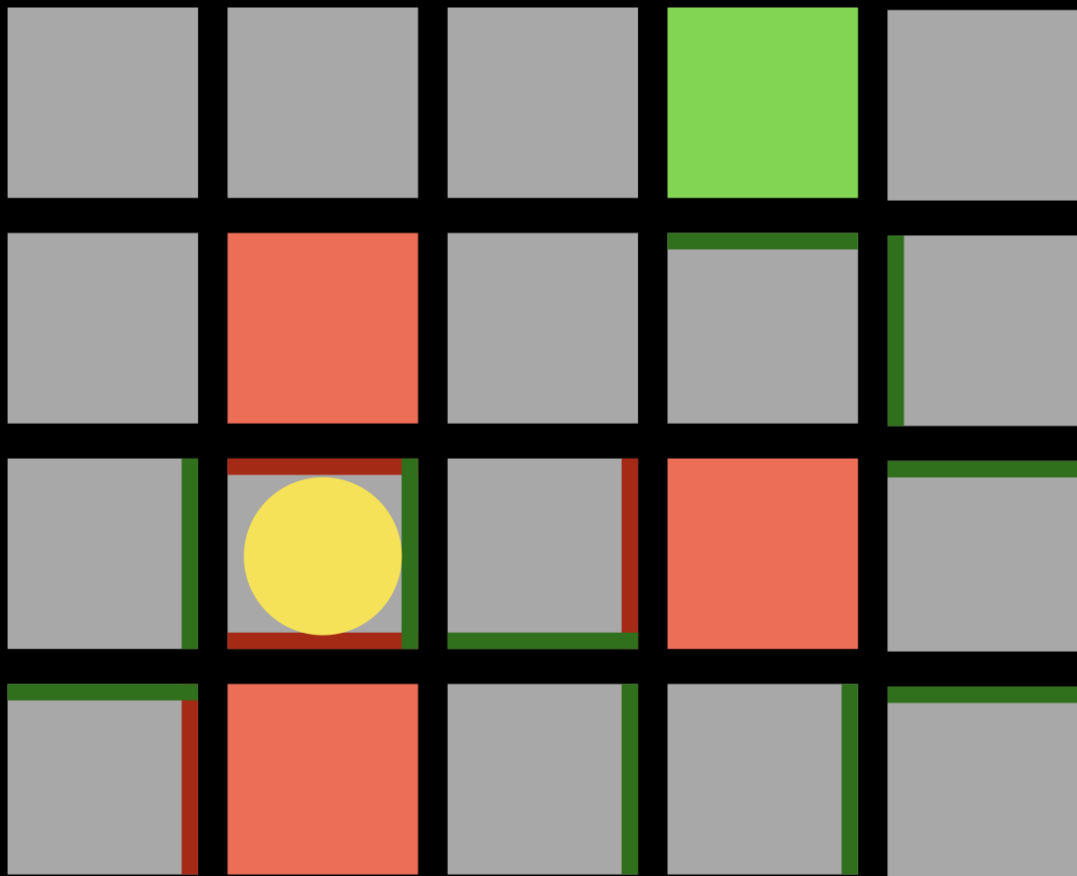
```
Else:
```

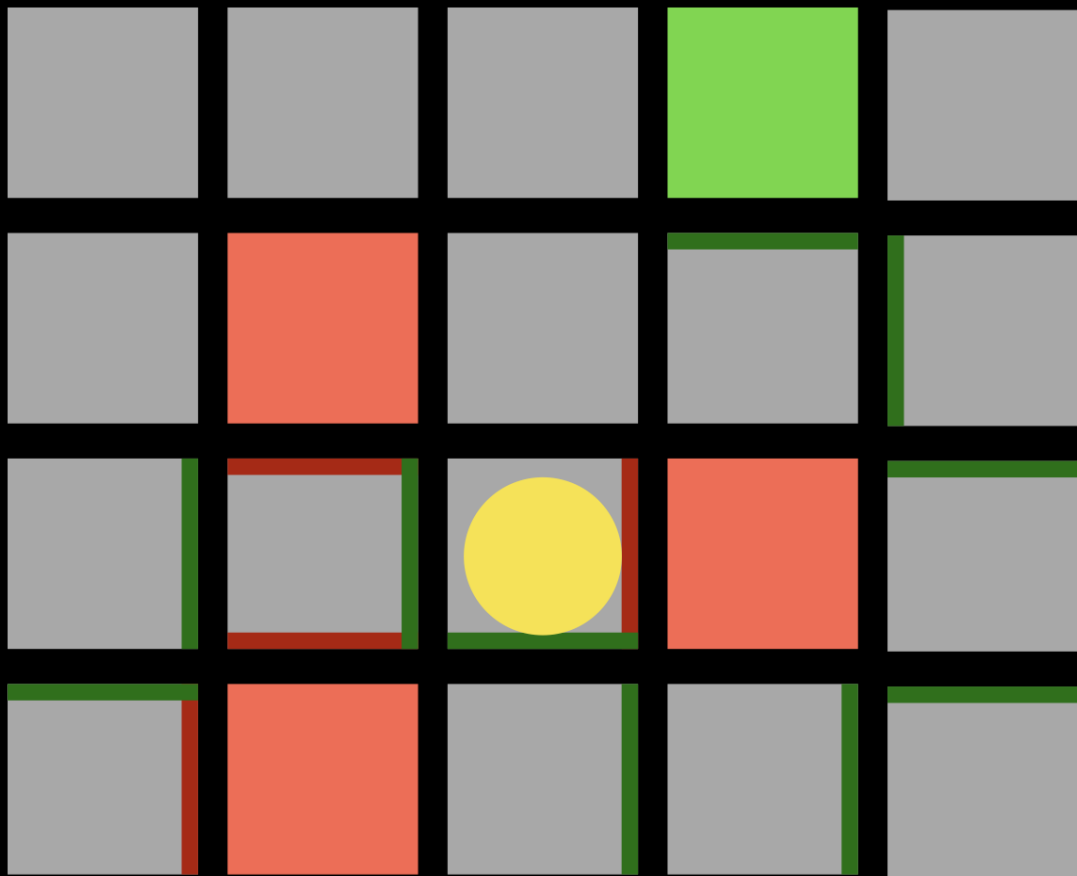
```
    Make the move with the highest value
```

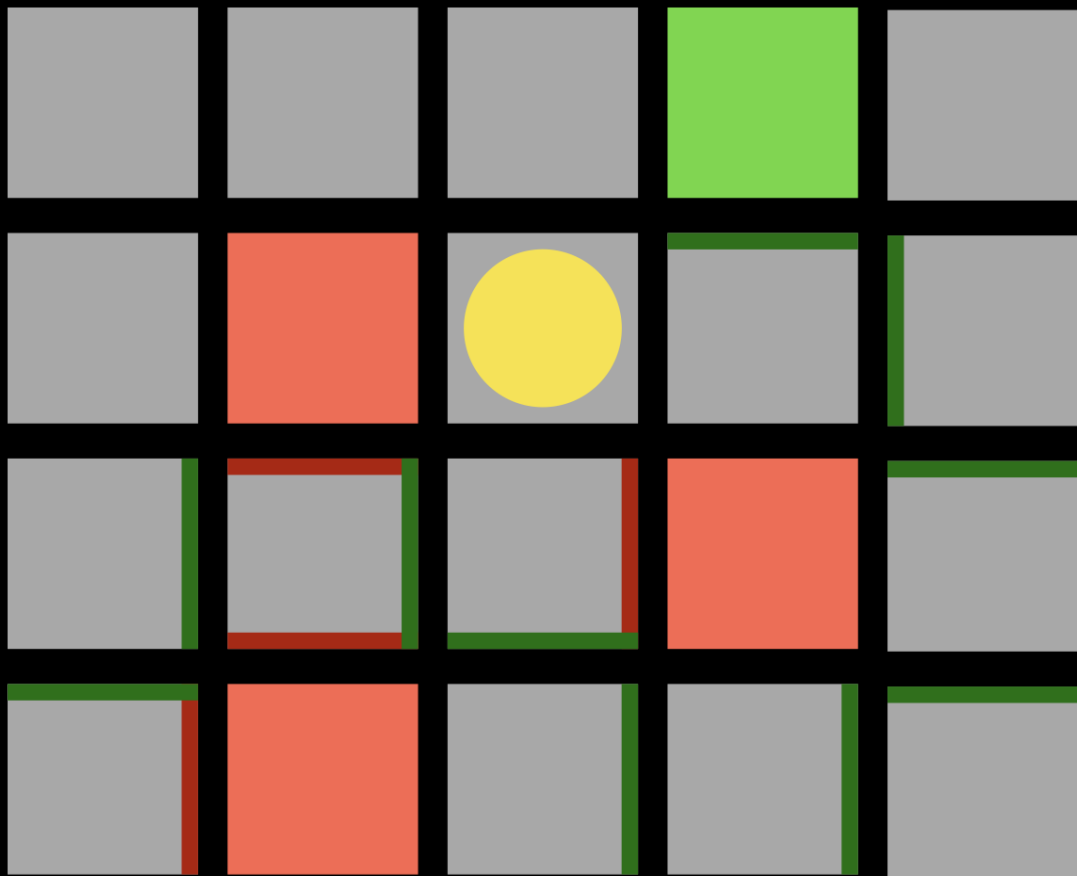


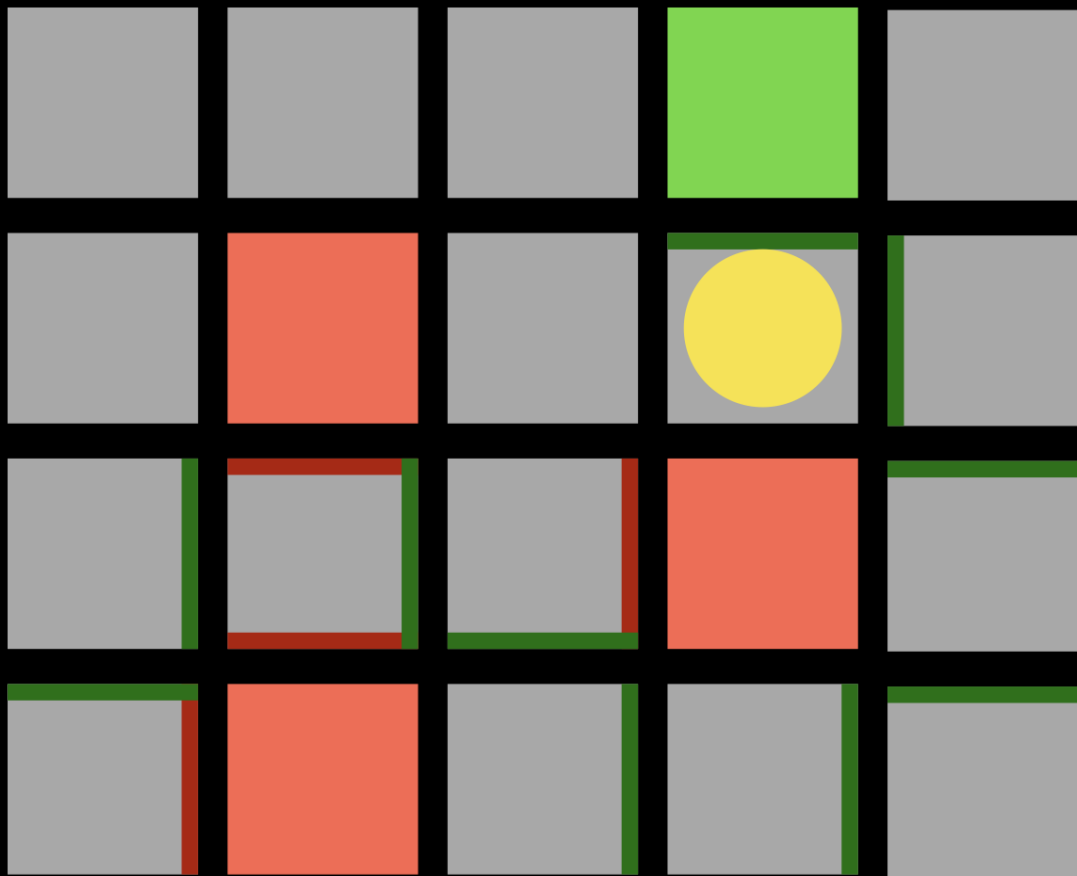


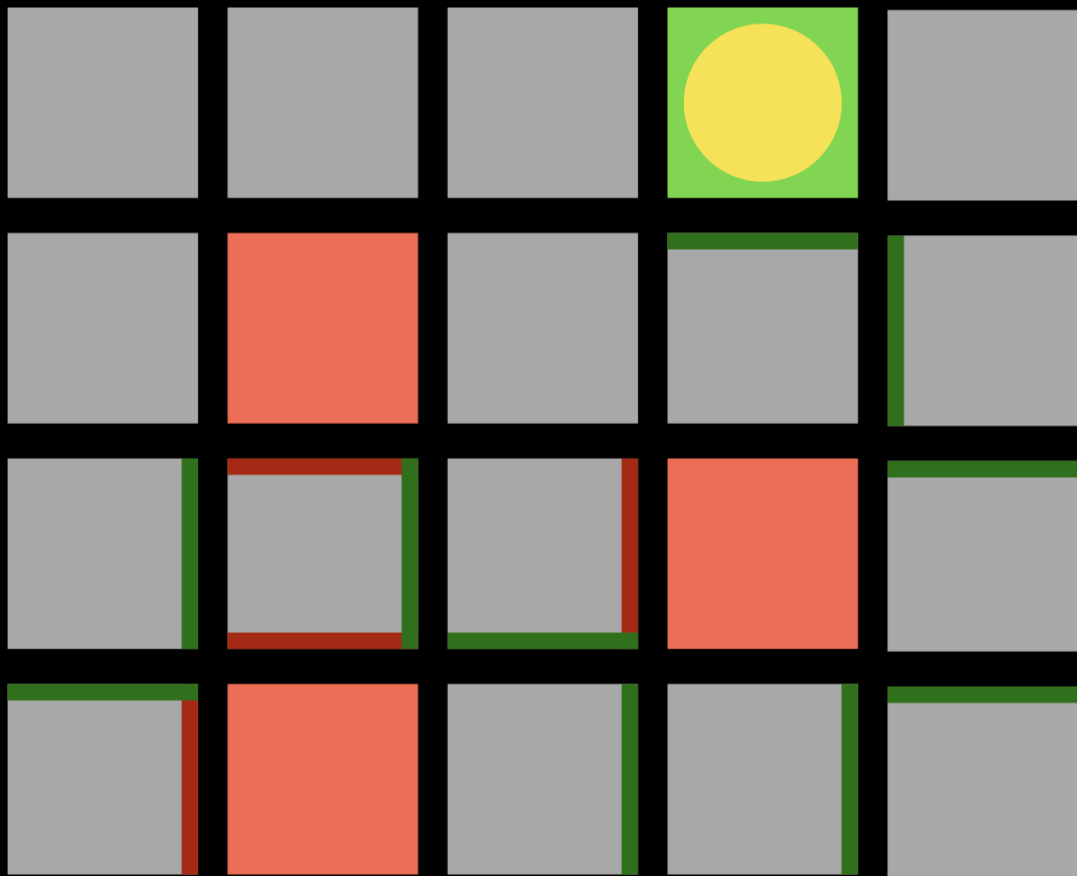










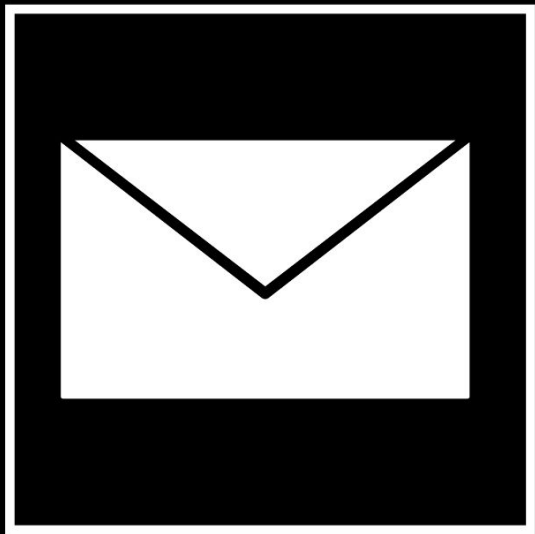


049 2 1





supervised learning

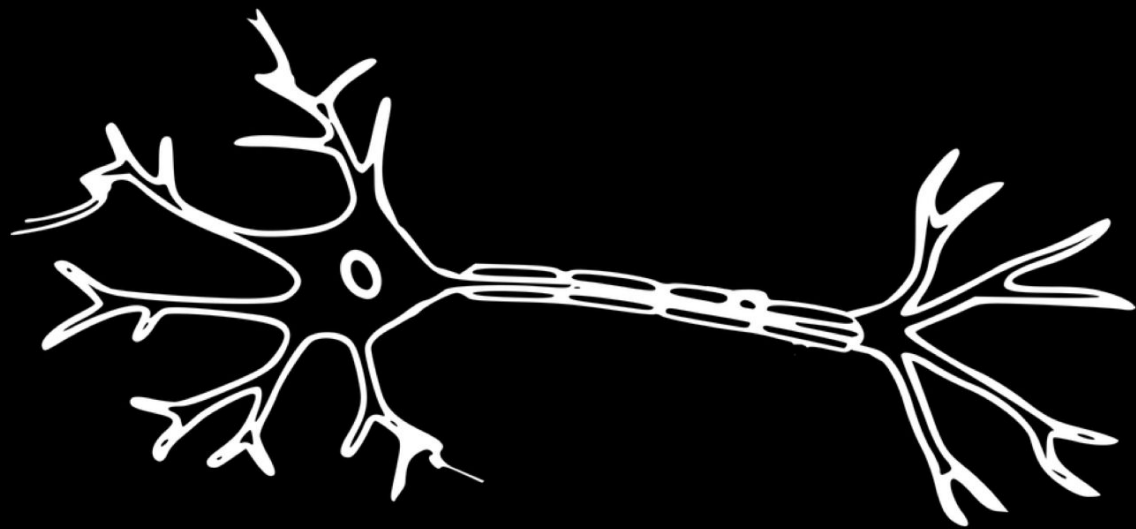


spam

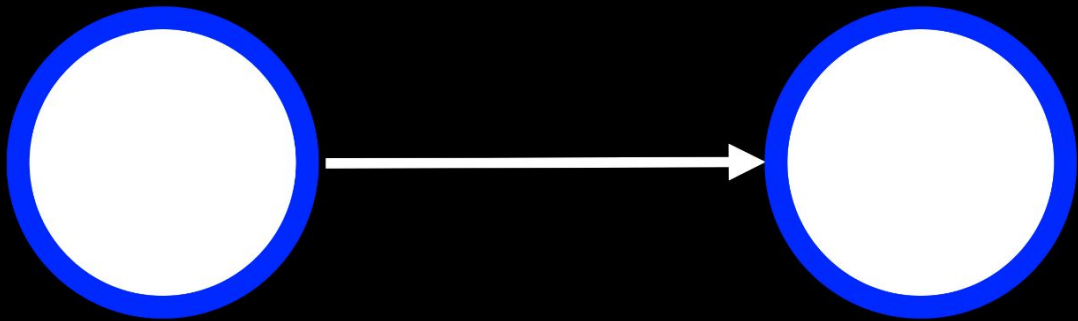
unsupervised learning

deep learning

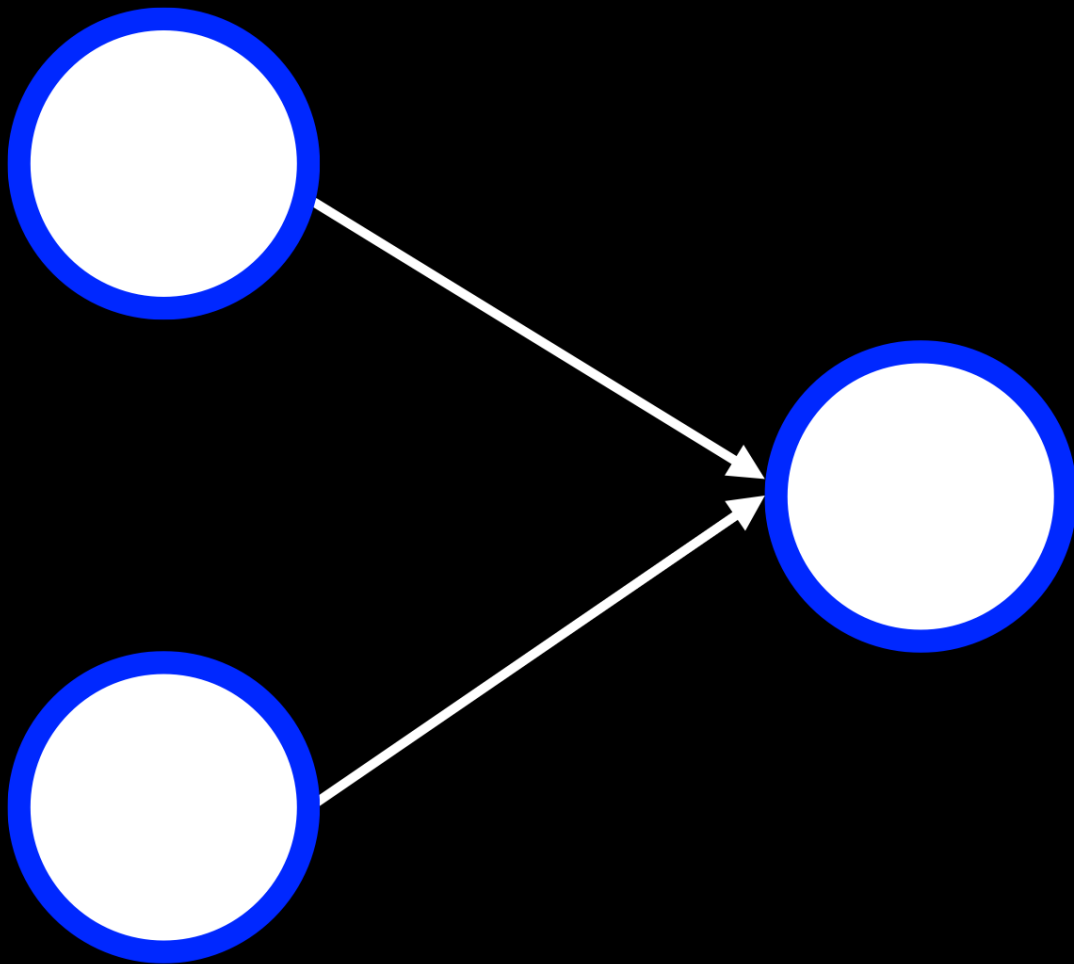
neural networks

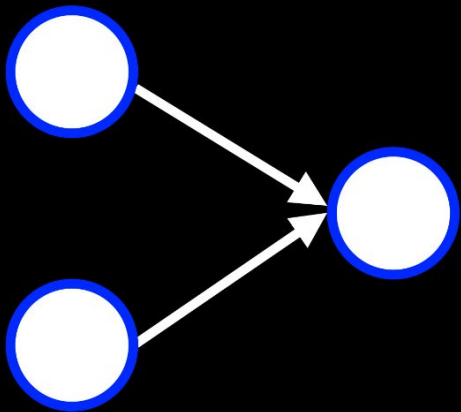


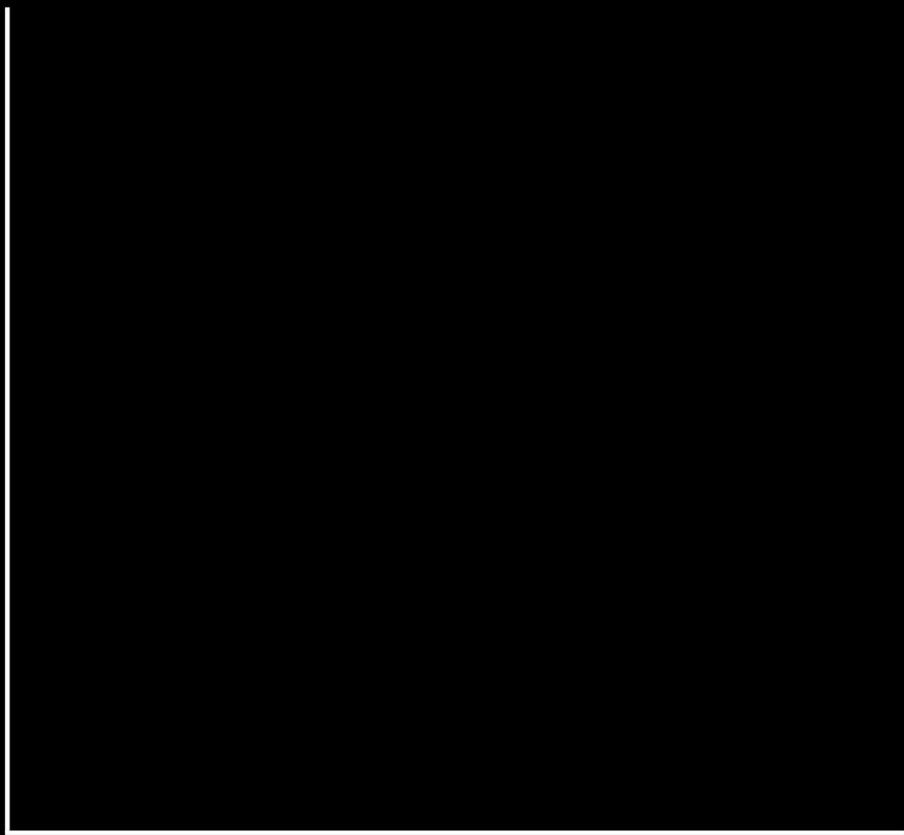
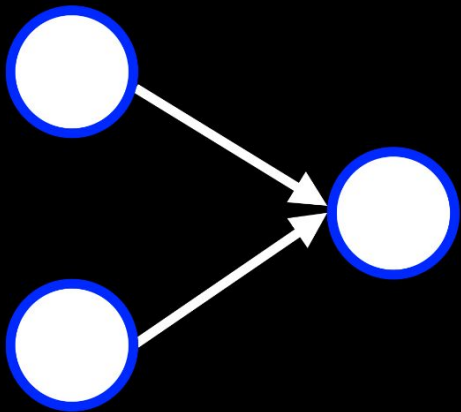


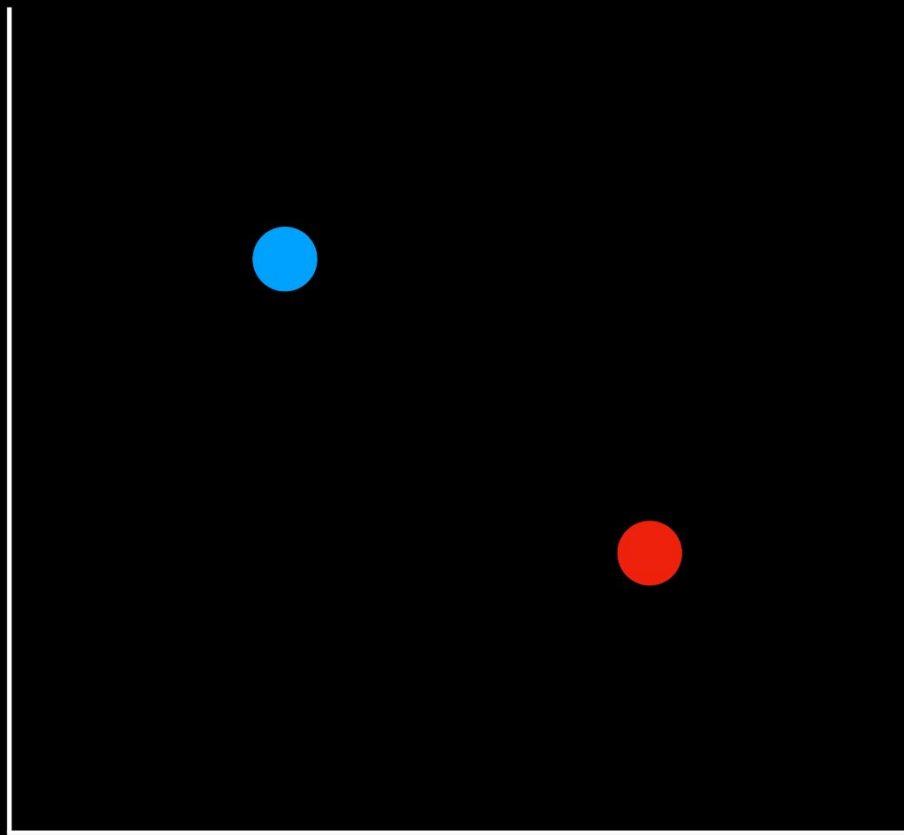
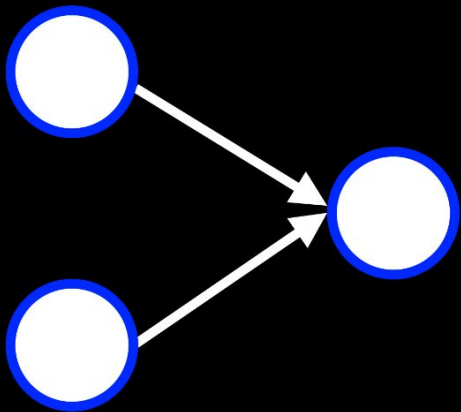


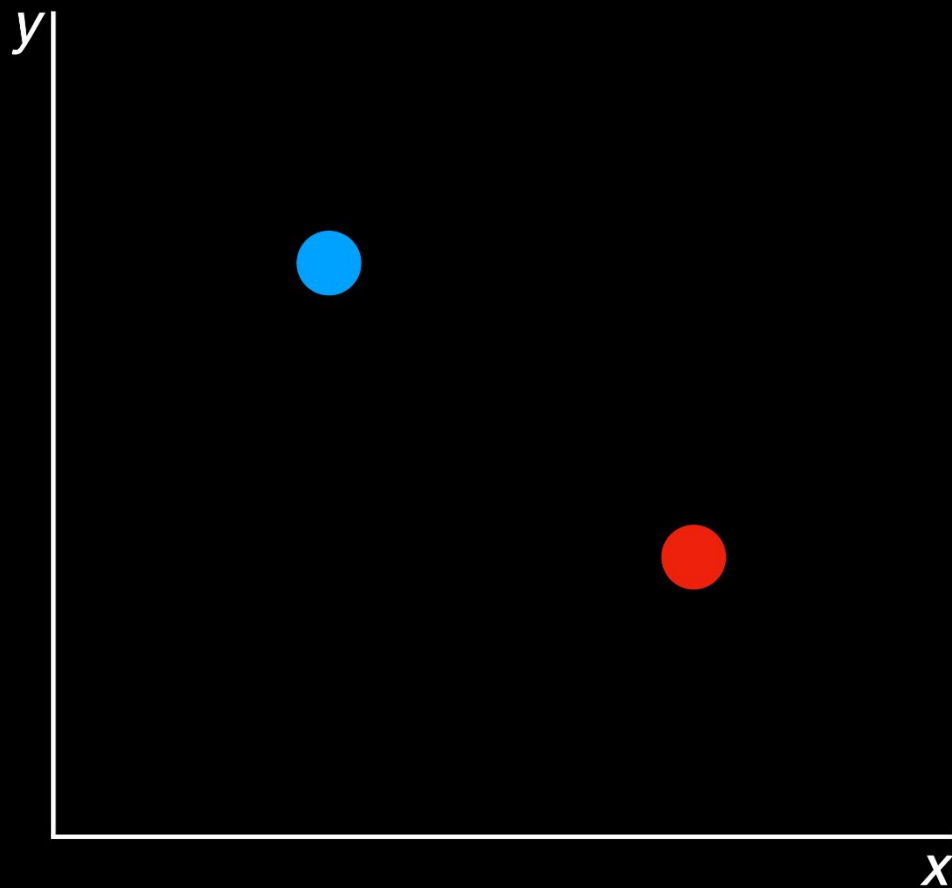
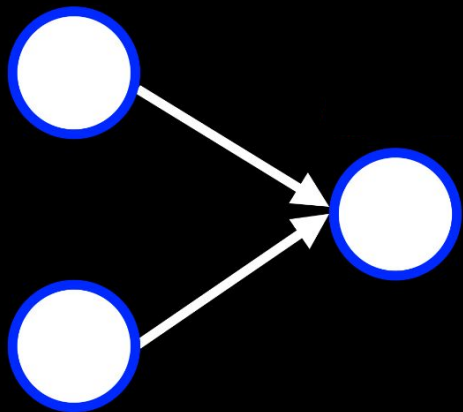


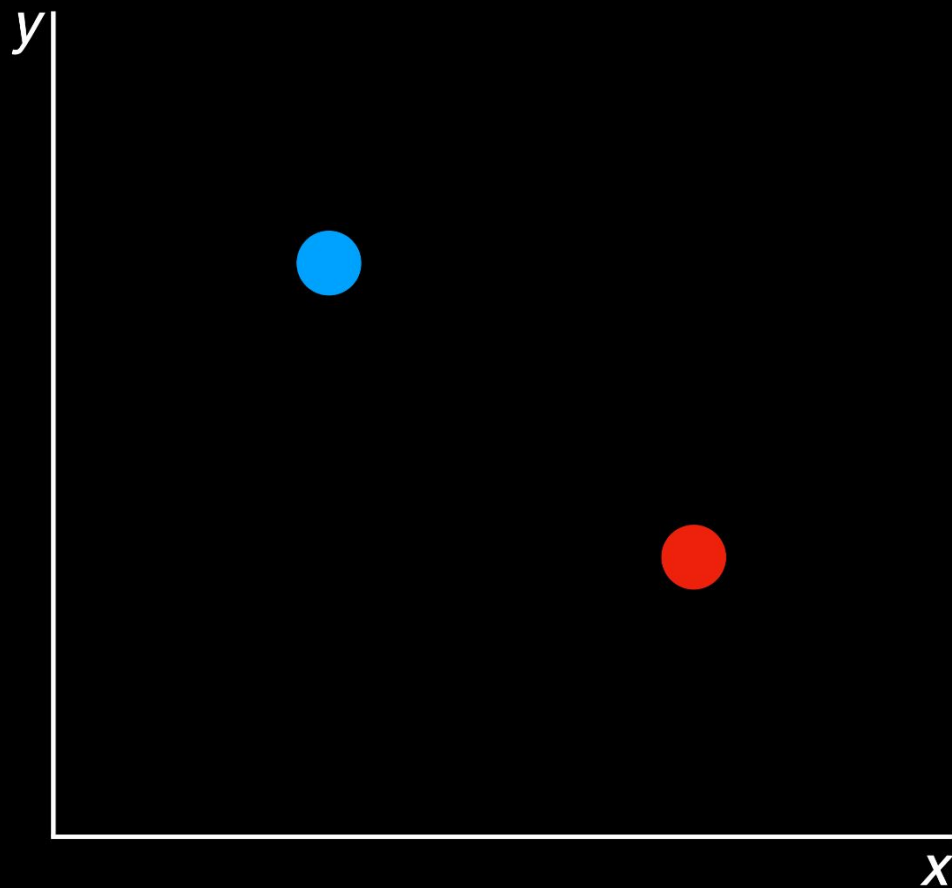
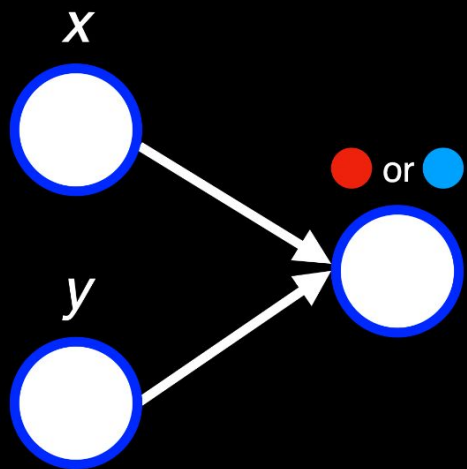


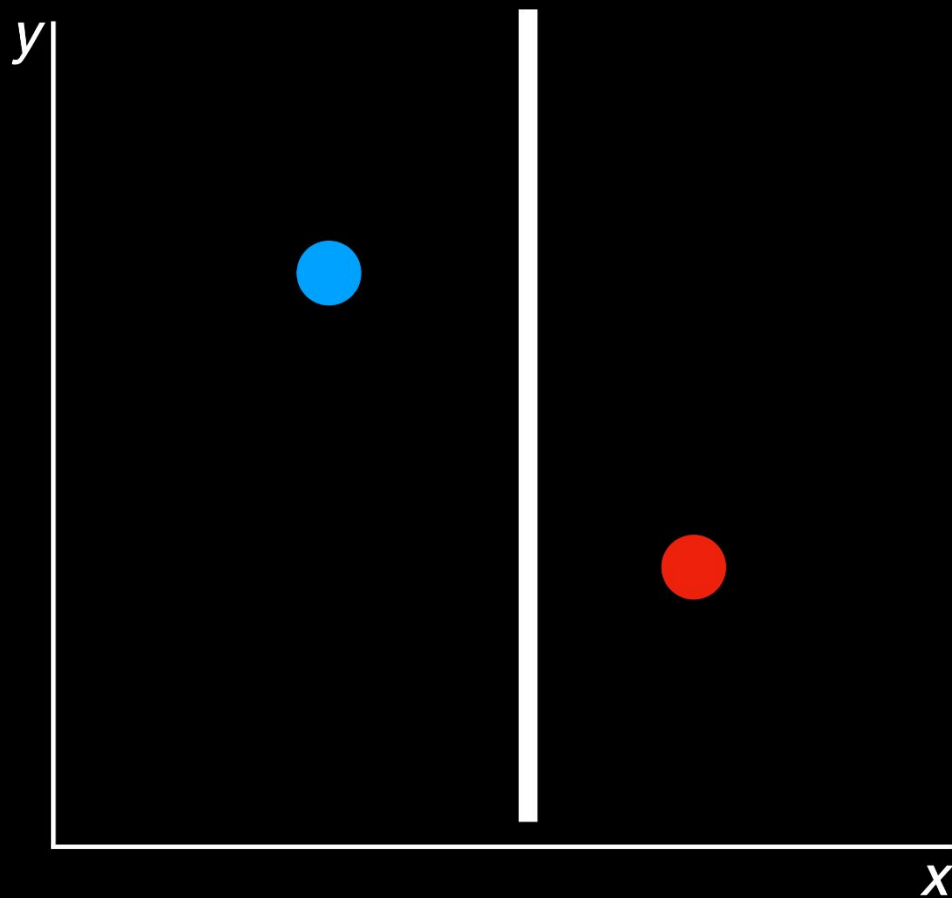
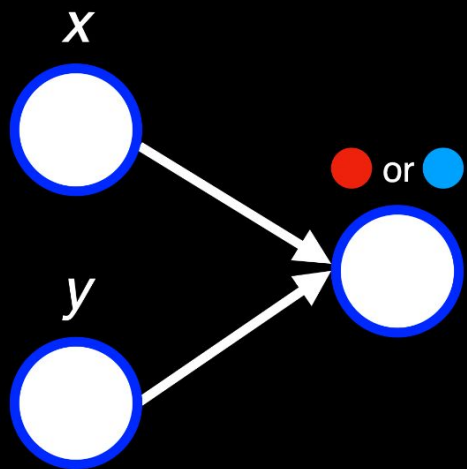


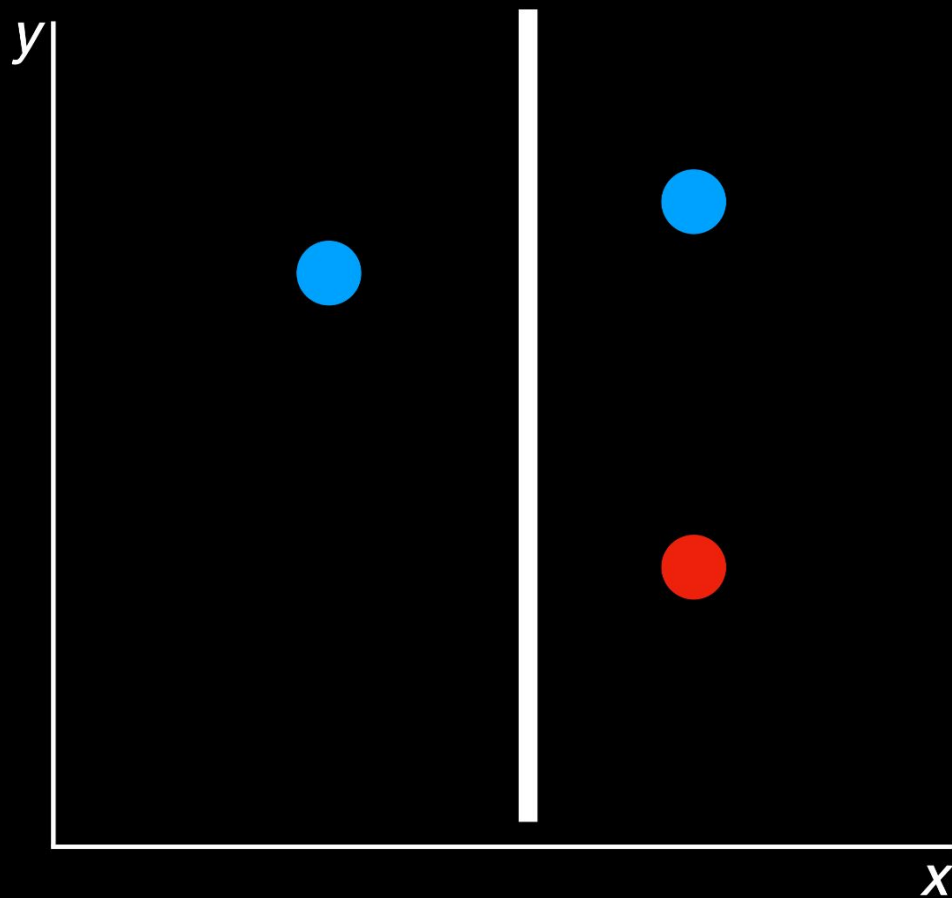
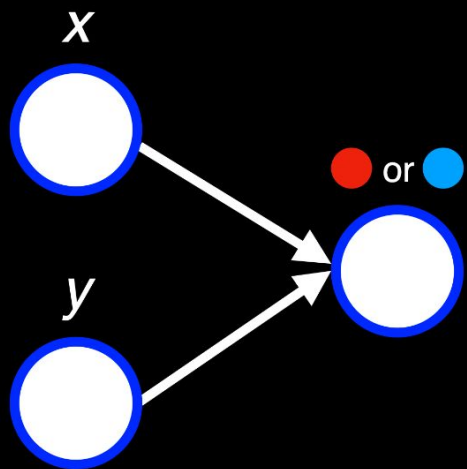




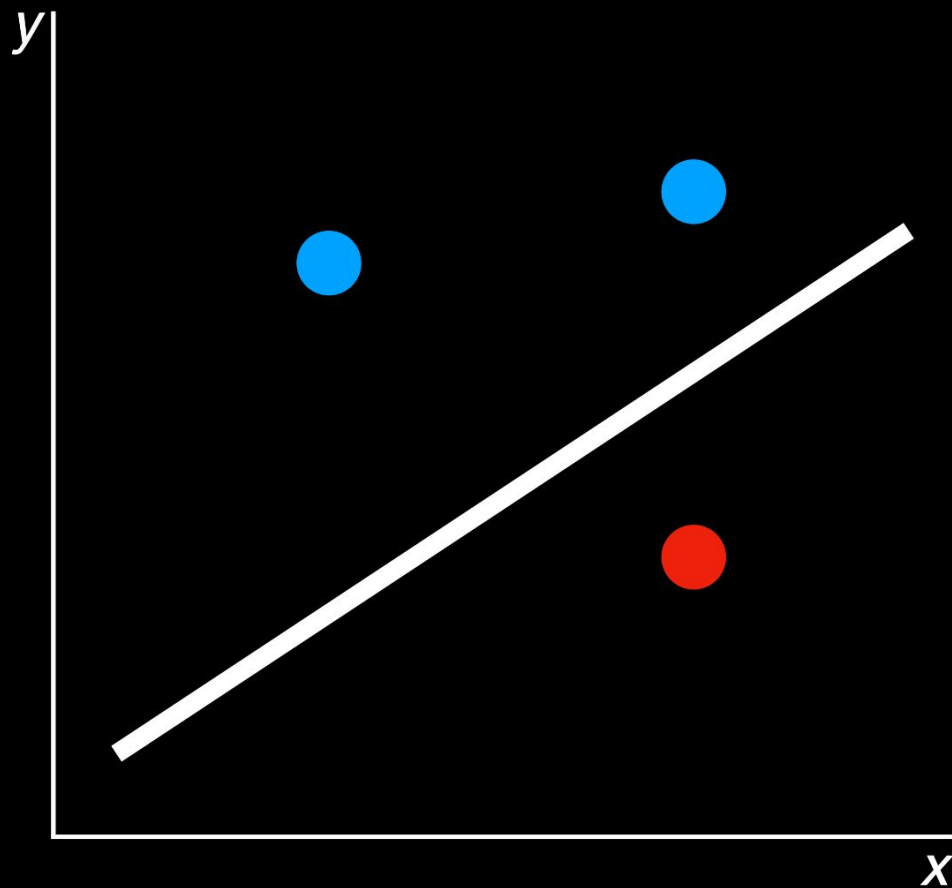
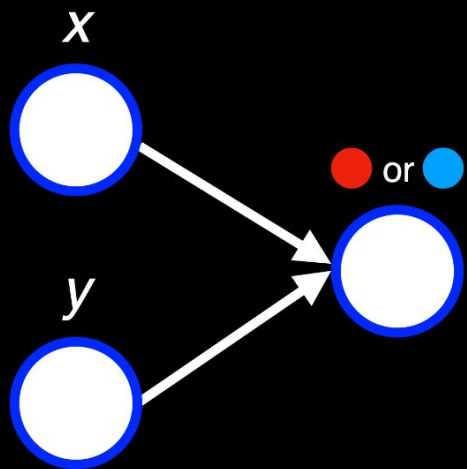


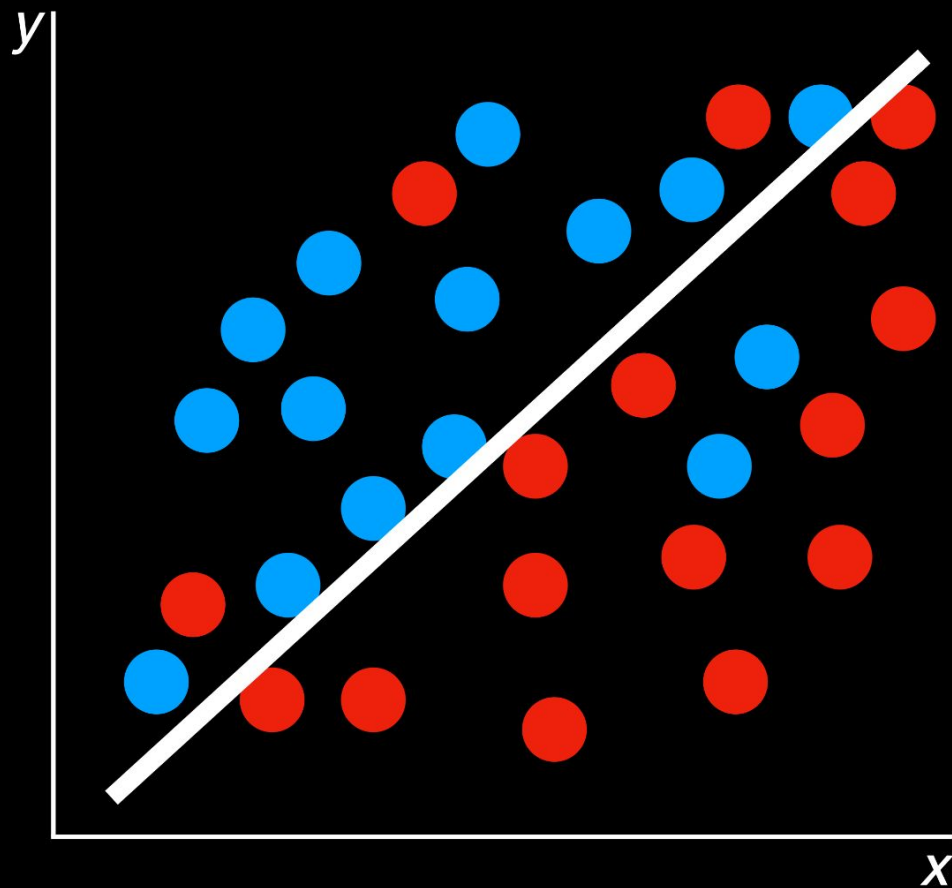
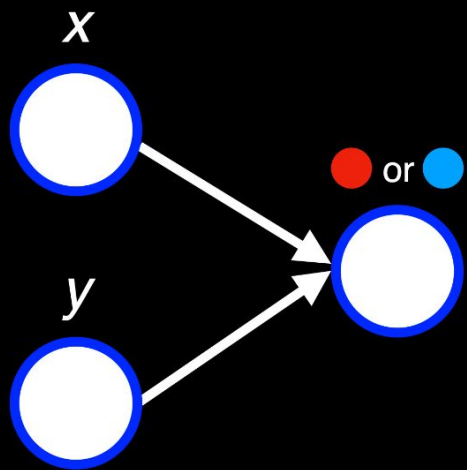


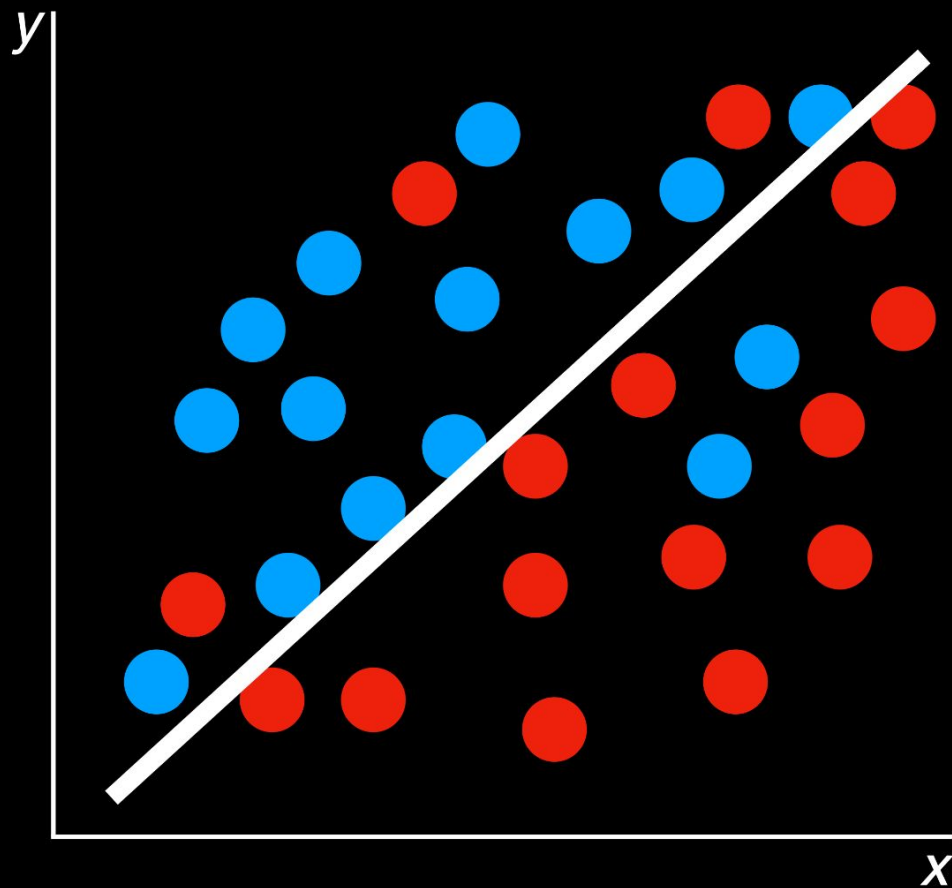
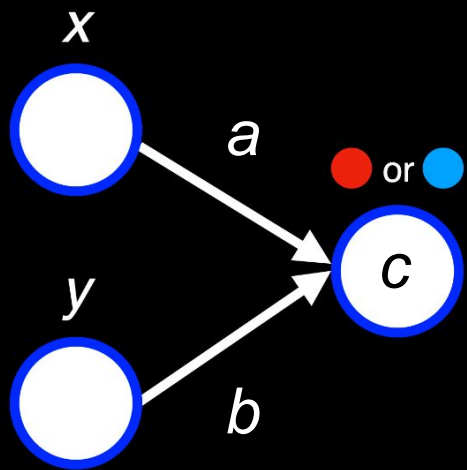




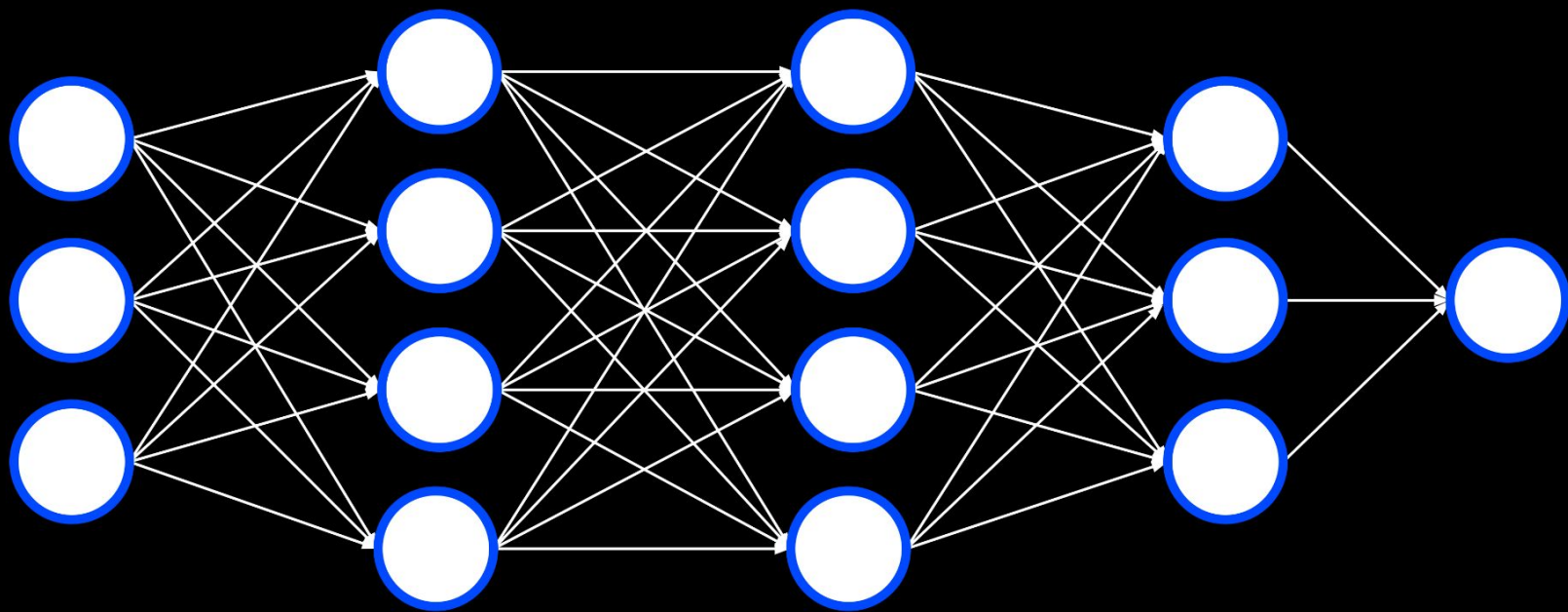


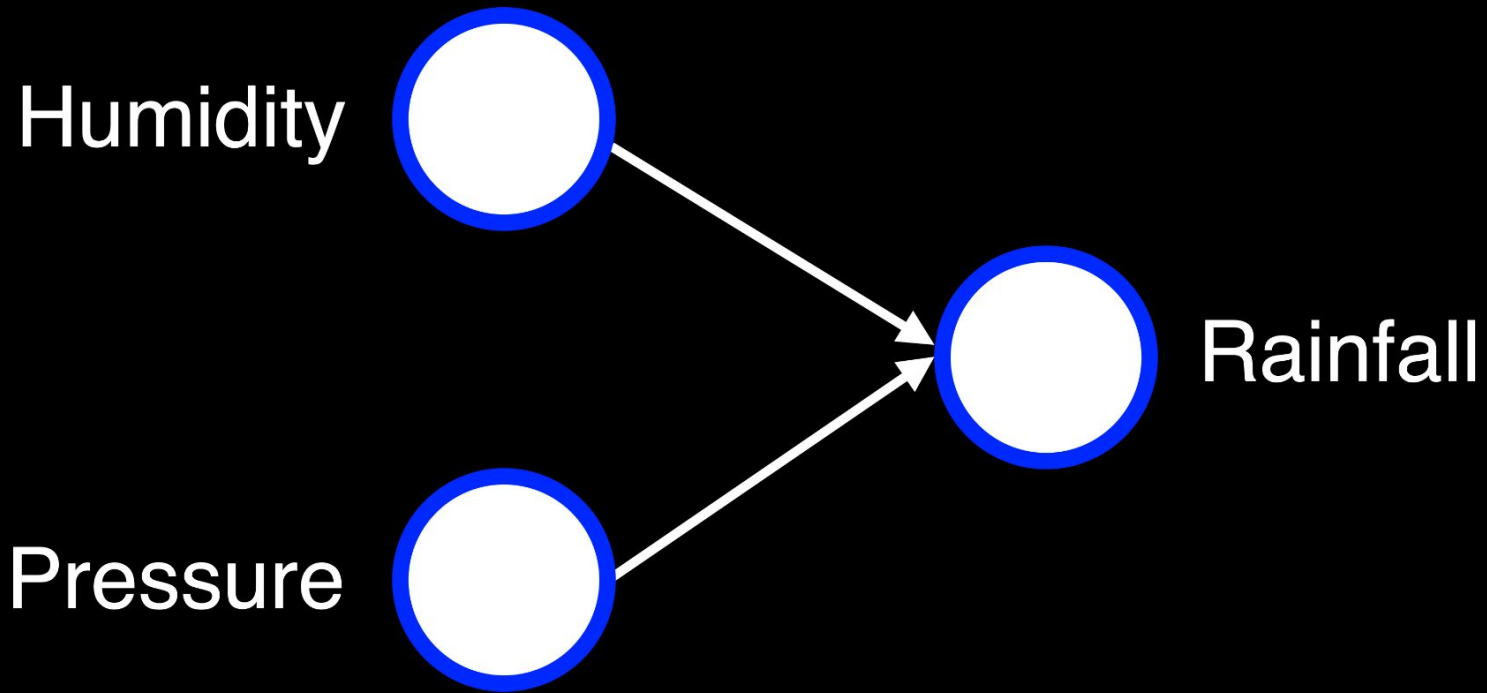


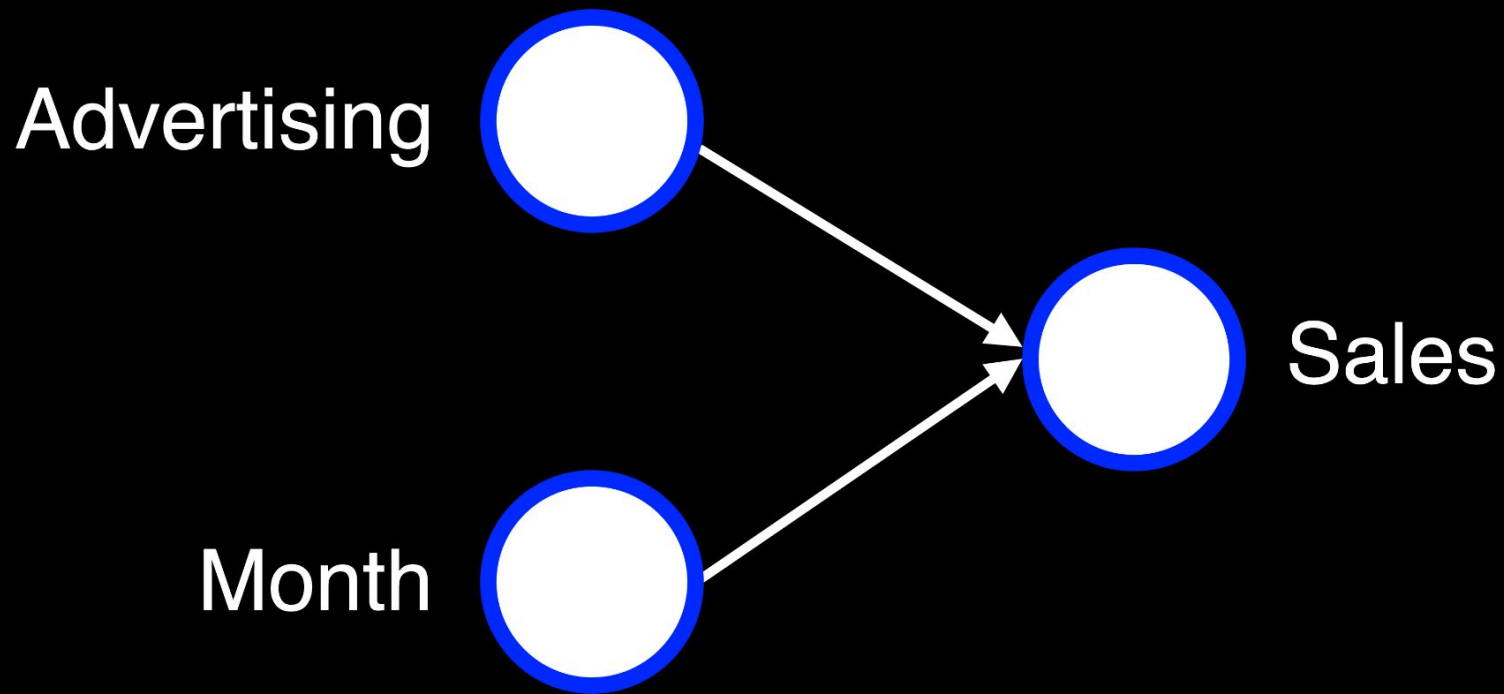




$$ax + by + c > 0$$







large language models





**Transformer Model  
2017**



**GPT-3 Language Model  
2020**



**ChatGPT Model  
2022**

# Generative Pre-trained Transformers

# Generative Pre-trained Transformers

"Massachusetts is a state in the New England region of the Northeastern United States. It borders on the Atlantic Ocean to the east. The state's capital is..."

"Massachusetts is a state in the New England region of the Northeastern United States. It borders on the Atlantic Ocean to the east. The state's capital is..."

*Massachusetts*

*is*

*a*

*state*

...

*the*

*state's*

*capital*

*is*

*Massachusetts* ①

*is* ②

*a* ③

*state* ④

...

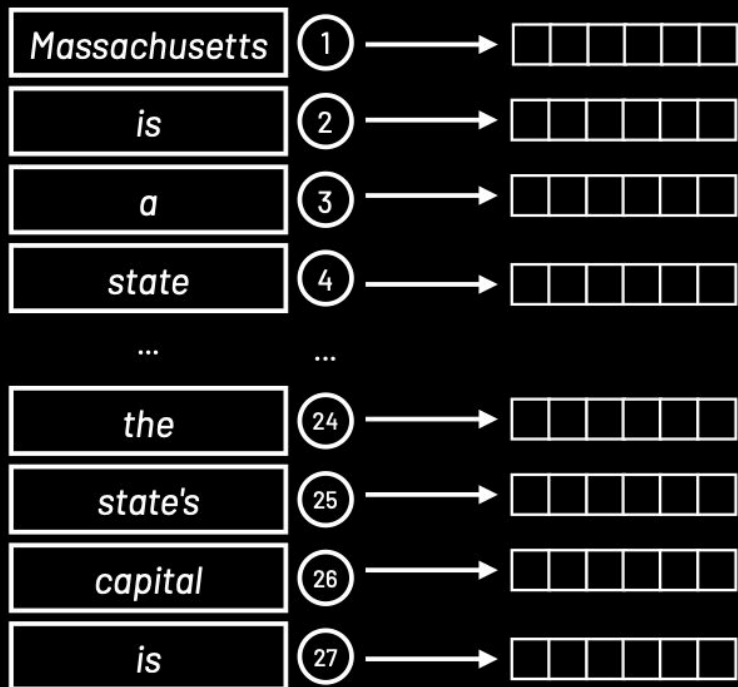
...

*the* ②④

*state's* ②⑤

*capital* ②⑥

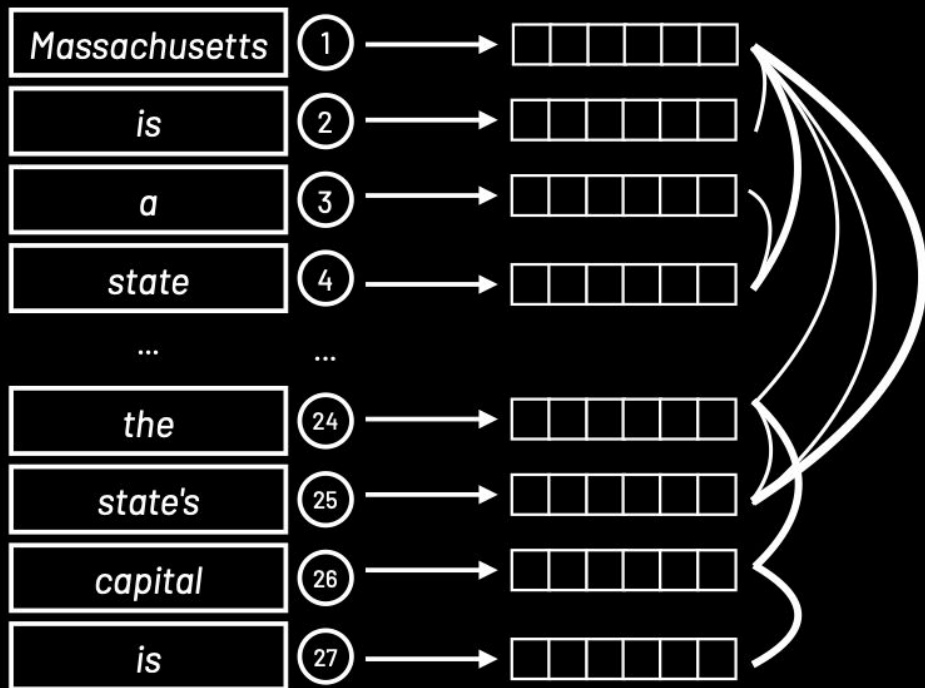
*is* ②⑦

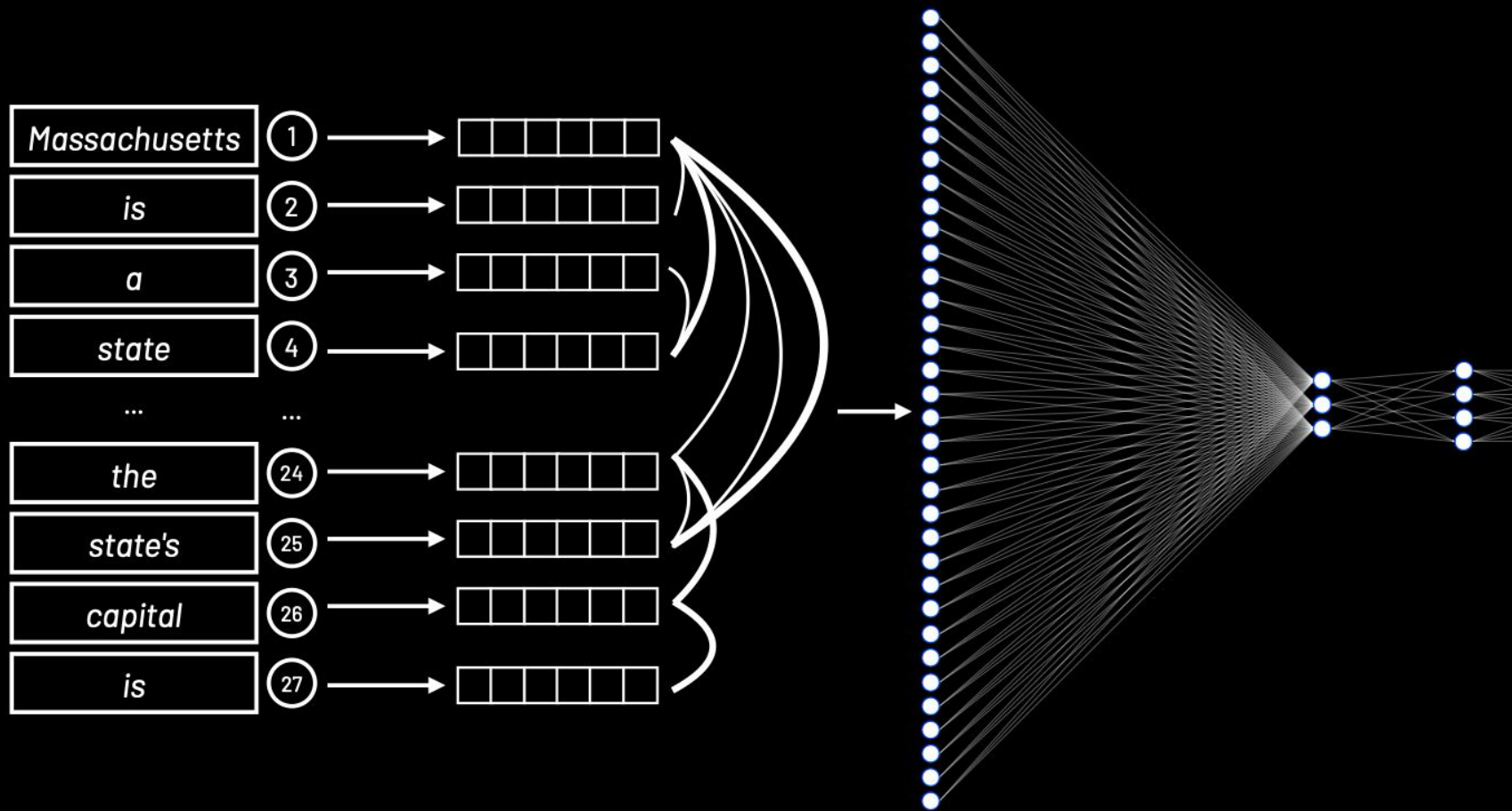




[illegible]

95 -0.014222151 -0.003958608 0.03350398 -0.012164199 -0.01



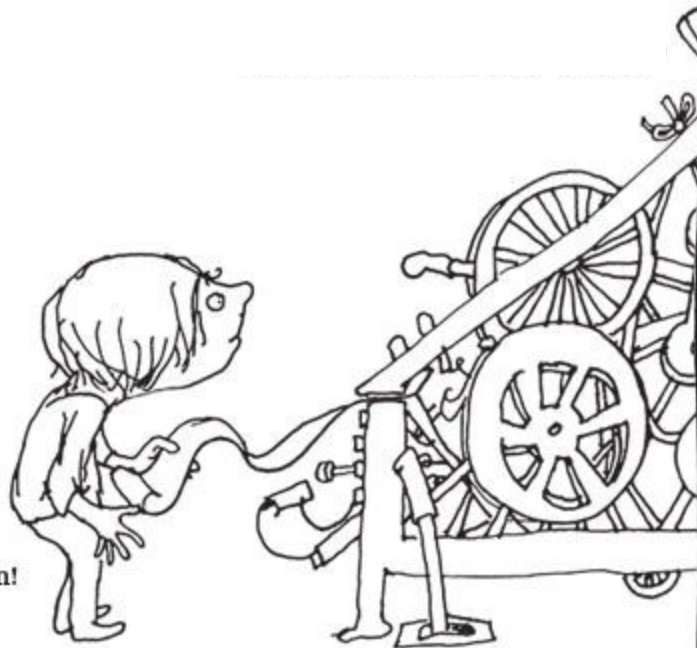




hallucinations

## HOMEWORK MACHINE

Read more  
poems in  
*A Light in  
the Attic*  
by Shel Silverstein!

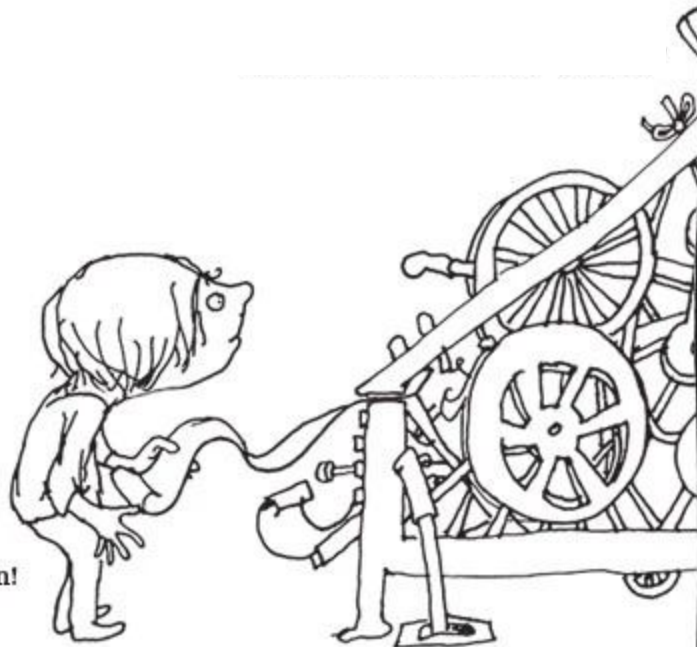


*A Light in the Attic* © 1981 Evil Eye Music, Inc.

## HOMEWORK MACHINE

The Homework Machine, oh the Homework Machine,

Read more  
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the Attic*  
by Shel Silverstein!

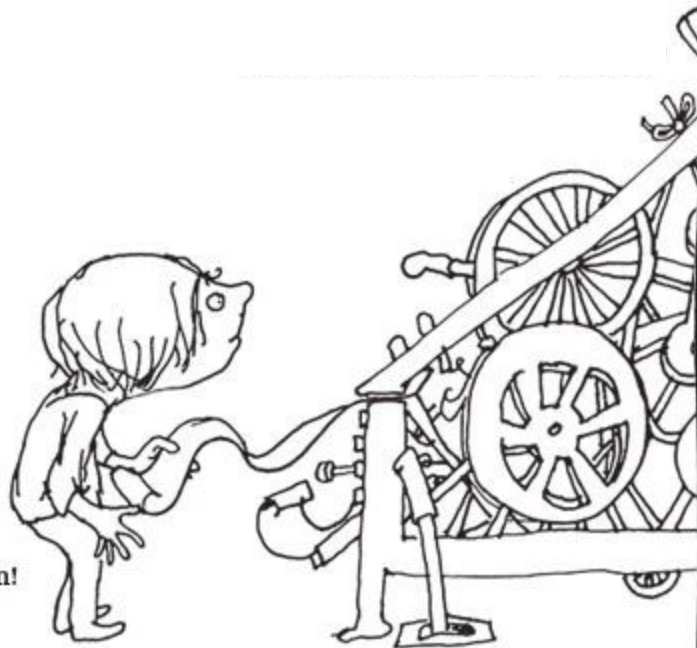


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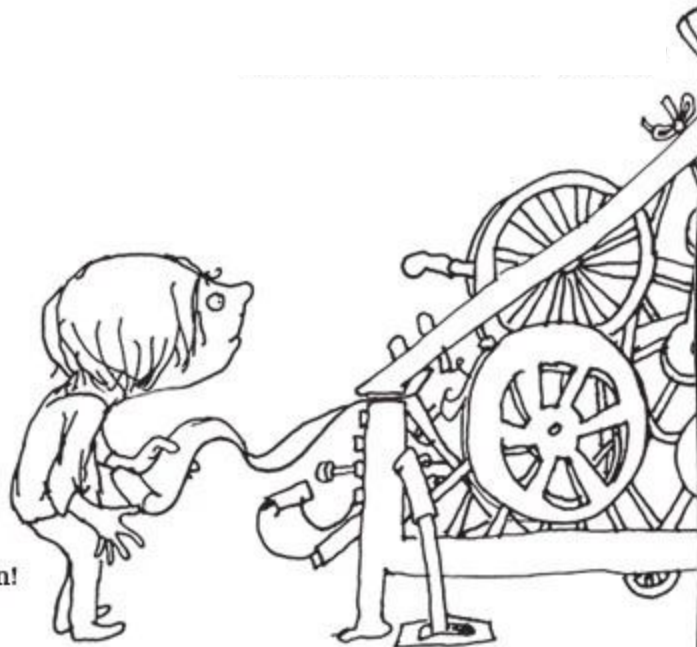
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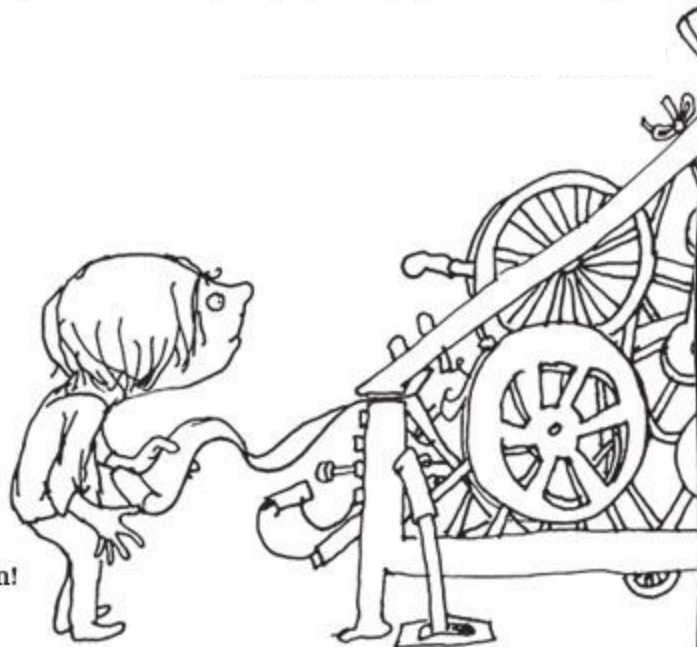


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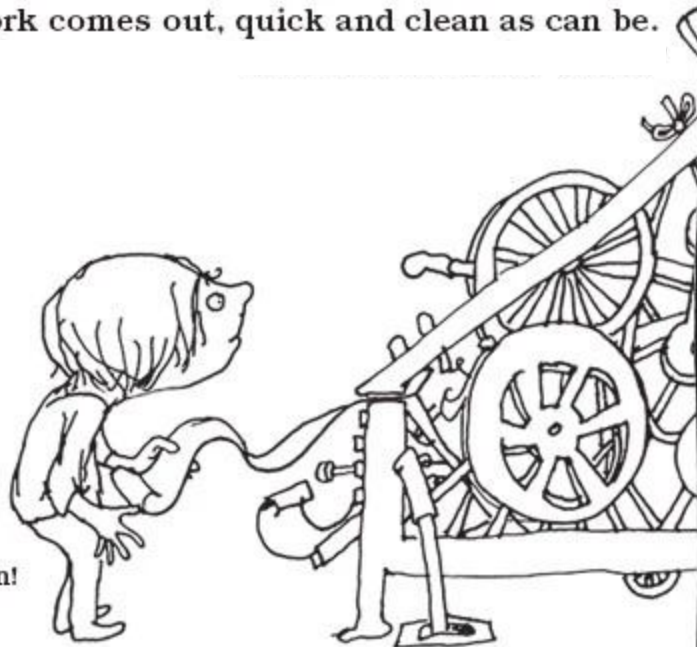


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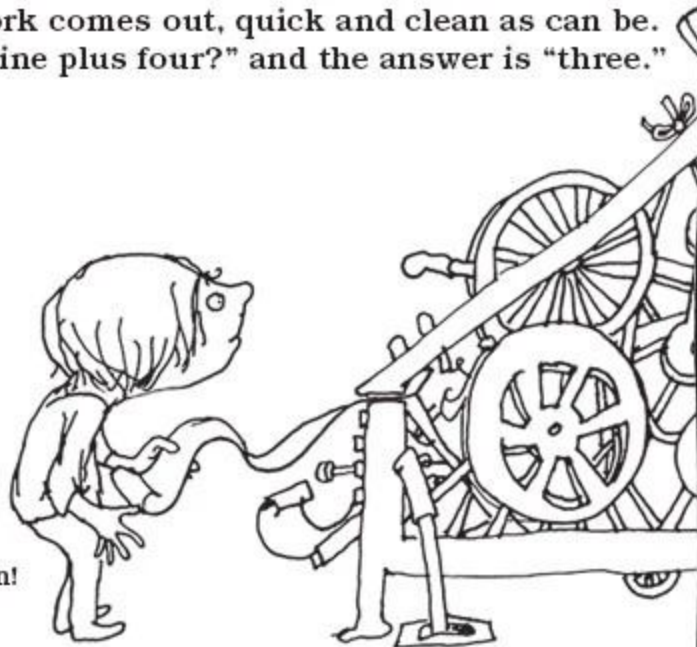


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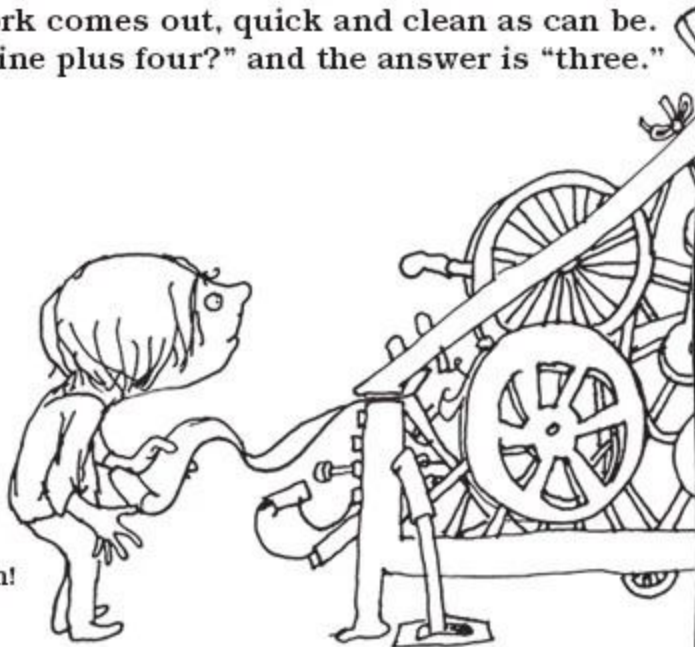


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Three?

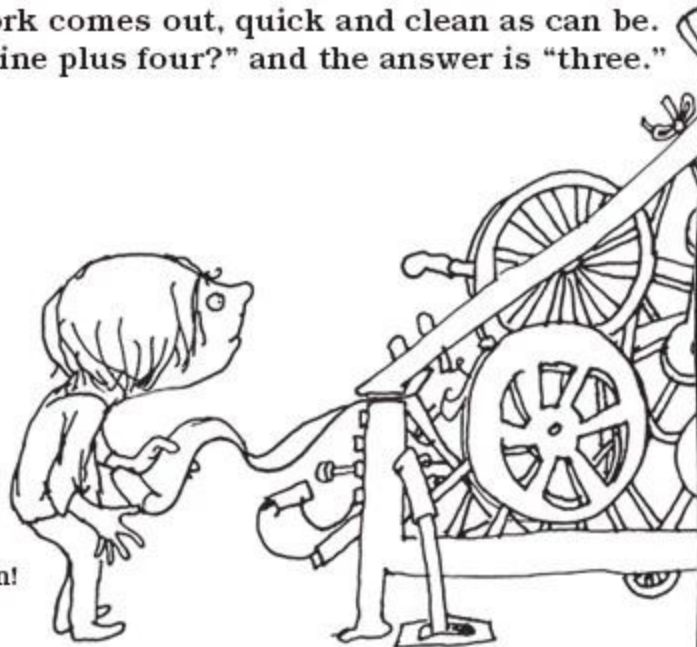
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Oh me . . .

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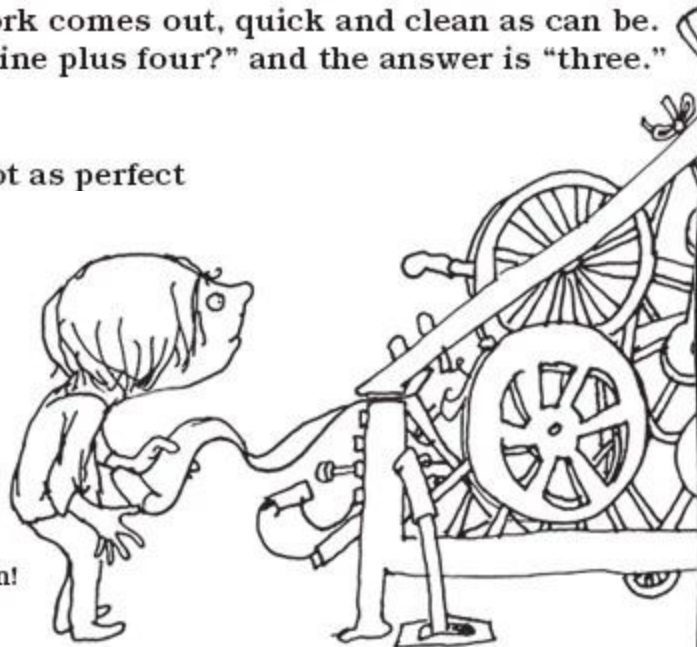




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Three?  
Oh me . . .  
I guess it's not as perfect

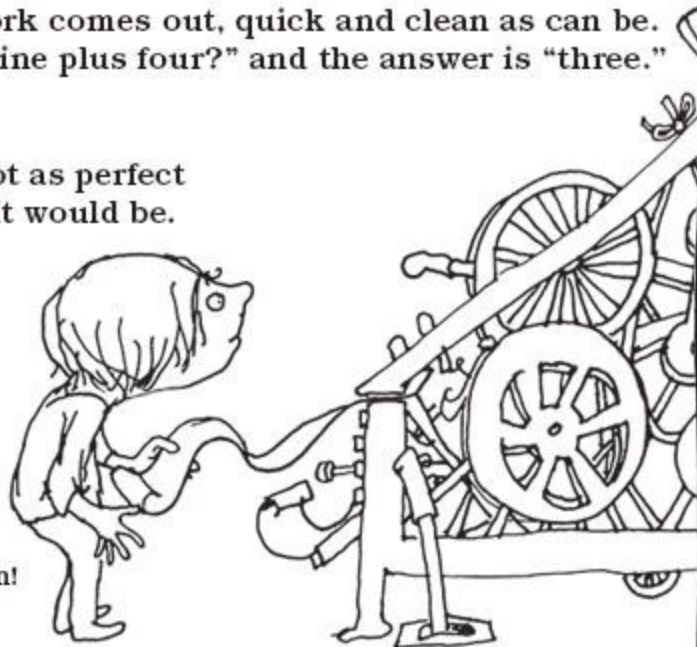
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poems in  
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Your homework comes out, quick and clean as can be.  
Here it is—"nine plus four?" and the answer is "three."  
Three?  
Oh me . . .  
I guess it's not as perfect  
As I thought it would be.

Read more  
poems in  
*A Light in  
the Attic*  
by Shel Silverstein!





**This is CS50**